



In-depth survey: deep learning in recommender systems—exploring prediction and ranking models, datasets, feature analysis, and emerging trends

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Abstract

Due to the exponential growth of online information, users are often welcomed with a huge range of products and services along with descriptions, reviews, and comments. Although this information available to users is valuable, at the same time, massive data sources confuse them to retrieve desired content, which is known as information overload. Recommender systems are examined as effective tools that play a vital role in filtering information and ultimately addressing the information overload problem. Various online platforms use recommendation systems to provide users with more relevant and personalized content. With the remarkable success of deep learning in the field of artificial intelligence, it procures much attention in the recommendation research area in recent years. The exiting literature on recommender system research commonly distinguishes between two main directions: rating prediction and top-N ranking. In this survey paper, we examine deep learning methodologies in the context of both rating prediction and top-N ranking recommendation approaches. Additionally, we investigate pre- and post-modeling critiques of recommender systems and provide insights into exiting benchmark datasets, feature learning analysis, and evaluation measuring techniques. In the end, we highlight the new generation recommender system trend with respective future research directions.

Keywords Information systems · Recommender systems · Deep learning survey · Rating prediction · Top-N ranking

1 Introduction

Recommender systems seek individual interest by reducing their workload and selecting the exact product fast and accurately. These systems show usefulness by assisting end users in the right direction, as sometimes end users do not know what they are looking for; here, the suggestions from

recommender systems become very helpful in understanding their needs. These systems keep the customer engaged, increase website traffic, and lead to the company's profit [302]. Presently, many companies like Amazon, Spotify, Facebook, Netflix, and Google have created history in building recommendation models and using them on their respective platforms.

In recent years, deep artificial neural networks have shown preeminent achievements in multiple research areas such as image processing, speech recognition, computer vision, and natural language processing (NLP) [31]. Deep learning is also gaining influence in the field of recommendation systems research. Furthermore, the hidden layer architecture of deep neural networks (DNNs) can extract latent features, capture nonlinear user–item interactions, or be able to make nonlinear mapping and infer relationships among features [156]. Due to these attractive features of DNNs, there is a propensity to employ them in recommender systems to achieve better recommendation accuracy. Several studies indicate the potency of deep learning

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in catching generalizations and generating better recommendations in different applications like e-commerce, e-learning, health informatics, and online social networks [222, 267], and [252].

The expansive landscape of deep learning-based recommender systems highlights the ongoing necessity for enhancements to improve recommendation efficacy, especially in diverse contexts. The literature in recommendation research domain presents two major research lines: rating prediction and top-N ranking [109, 288, 363]. Rating prediction [254] involves predicting a user's ratings for an item using past preferences, while top-N ranking [364] recommends a list of items from a large instance. Despite numerous reviews exploring deep learning-based recommender systems and their challenges, there remains a gap in comprehensive analyses categorically examining literature focused on rating prediction and top-N ranking. Additionally, existing reviews often overlook crucial aspects such as discussions on dataset descriptions, feature learning, and post-modeling evaluation metrics. This study aims to address this gap by systematically reviewing and comparing existing deep learning articles within these research lines, exploring available datasets, feature learning, and evaluation metrics to contribute to future research and advancements in this field.

1.1 Previous state-of-the-art surveys

How does this survey differ from previous state-of-the-art surveys? Research on recommender systems is diverse, with many researchers presenting surveys focusing on specific recommendation areas. The recent attention has been on summarizing the problem and challenges in scholarly paper recommendation [14], exploring text-based recommendation literature [152], conducting a review on reinforcement learning-based recommender systems [4], cross-domain recommender systems [154], and session-based recommender systems [328].

Additionally, some survey studies particularly investigate deep learning-based recommender systems. For example, [19] and [271] provide brief overviews of deep learning techniques in recommender systems. [82] classify deep learning-based recommender approaches based on their integration methods and levels of dependence on deep learning. [281] discuss the contribution of deep learning-integrated recommendation systems across various application domains. Similarly, [225] examines deep learning studies in key classification techniques and explores its application in social network-based and context-aware recommender systems. Another survey study focuses on deep learning-based recommender system, along with a discussion of related challenges such as accuracy, cold start, scalability, and other issues [148]. In a specific scope,

the survey article focuses on general user modeling approaches that explicitly consider learning latent representations for users in scenarios including recommendations [178]. It reviews and discusses shallow matrix factorization (MF), deep collaborative and recurrent neural network studies in related fields. [427] introduce various types of recommendation methods, technological trends, and application domains that leverage deep learning. Furthermore, recent survey studies include review deep learning methods and contemporary issues in recommender systems [17]. The most up-to-date work in [400] provides a descriptive analysis of deep learning methods. [244] and [58] provide an analysis of state-of-the-art deep learning-based recommender systems, along with datasets, and various metrics used to evaluate their performance. [156] focus solely on rating prediction deep learning approaches. The literature work on recommender systems is broadly divided into two research lines: rating prediction and top-N ranking. In the literature, [288] examine the details, highlighting the dominating differences between these research lines in solving recommendation tasks. Despite this, there is a lack of a comprehensive deep learning-based analysis of these research lines within existing surveys. To assist improvements with rating prediction task and top-N ranking task, it becomes important to deeply investigate their methodologies and applicability.

With this objective, this survey paper covers a thorough examination of contemporary deep learning state of the art for both rating prediction and top-N ranking approaches. Unlike the mentioned deep learning-based recommender system surveys, which either offer limited or no pre- and post-modeling analysis, this survey examines critical evaluations of recommender system. It provides an analysis on benchmark datasets, feature learning studies, and evaluation measuring techniques to give readers a profound understanding of the evolution of recommender systems. Furthermore, we investigate current research trends in the recommendation system and discuss potential future improvements for each of these trends. A brief comparison between ours and previous deep learning state-of-the-art surveys is presented Table 1.

There are other recent deep learning surveys focus on specific applications or subjects such as health recommender systems [266], e-learning recommender systems [21, 269], and knowledge graph-based recommender systems [95], which differs from our broad investigative scope and are not included in the comparison.

1.2 Paper selection

For this survey, we used ACM Digital Library, Google Scholar, IEEE Xplore, Science Direct, and Web of Sciences search engines to collect representative

Table 1 Comparison between our survey and previous deep learning state-of-the-art surveys

Publication	Survey focus	Deep learning literature	Pre-modeling analysis (Dataset and feature learning)	Pre-modeling Analysis (Evaluation Metrics)	Current trend
[19]	Brief survey of traditional deep learning techniques in recommender system (RS)	Limited focus only on traditional deep learning models	–	–	–
[82]	Analysis of integration methods and dependence levels of deep learning in RS	Limited focus only on traditional deep learning models	Lists dataset but lacks detail	Limited compared to this survey	Limited compared to this survey
[281]	Studies deep learning's impact on RS applications and over conventional methods	Primarily centered around content-based, collaborative filtering (CF), and hybrid systems	–	–	–
[225]	Examines deep learning models by RS classification techniques and applications in social network and context-aware systems	Focus only on traditional deep learning models	–	–	Includes discussion of future research directions
[17]	In-depth review of deep models in RS and in various applications	Focus only on traditional deep learning models	–	–	Identifies general future works
[244]	Brief survey of recent advances in deep learning models along with datasets, and evaluation metrics	Focus only on traditional deep learning models	Summarizes 3 datasets (MovieLens, Epinions, Amazon)	Summarizes some accuracy-based metrics	Limited compared to this survey
[400]	Comprehensive review of recent research efforts	Focus on traditional and attention-based deep learning models	–	–	Includes discussion of future research directions
[58]	Explores deep learning RS, along with datasets and commonly used metrics	Focus on traditional and attention-based deep learning models	Lists datasets with publications but lacks detail	Summarizes only rating prediction and ranking metrics but no formula discussion	Includes discussion of open issues with future research directions
[156]	Review deep learning models for rating prediction tasks	Focus on traditional and attention-based deep learning models	–	–	Limited compared to this survey
[271]	Brief survey of traditional deep learning technique-based RS	Limited focus only on traditional deep learning models	–	–	Limited compared to this survey
[178]	Examines recent advances in user modeling from the perspective of representation learning	Focus only on deep CF and sequential networks	–	–	Narrow scope includes discussion of potential topics on user modeling approaches
[148]	Investigate survey and criticism of numerous state-of-the-art deep RS	Brief analysis of all studies in one section	--	–	Discussion of some existing challenges
[427]	Examines deep learning models specific to content-based, sequential, cross-domain and social recommendation	Focus on traditional and graph-based deep learning models	–	–	Includes summary of challenges and future developments

Table 1 (continued)

Publication	Survey focus	Deep learning literature	Pre-modeling analysis (Dataset and feature learning)	Pre-modeling Analysis (Evaluation Metrics)	Current trend
Our survey	Comprehensive review of deep learning models for rating prediction and top-N ranking tasks along with analysis on benchmark datasets, deep feature learning approaches and evaluation metrics	Focus on both traditional and modern deep learning models, such as graph-based, attention-based, and large language model (LLM). Additionally, it includes quantitative performance analysis of the current optimal model against traditional models	Summarize exiting datasets with relevant details and feature learning categories with their respective deep learning methods	Summarize evaluation metrics including both accuracy and beyond accuracy metrics	Includes discussion on new generation recommender systems, along with insights into future research perspectives

*In this survey, deep learning models are categorized into traditional, modern, and hybrid. This classification is highlighted in Sect. 3 of the article

recommendation literature published during the period 2003–2024. We analyzed the papers from high-profile artificial intelligence (AI) journals and conferences, including AAAI, ICDM, ICML, IJCAI, KDD, SIGIR, and WWW. The primary keywords we used are information systems, recommender systems, deep learning, rating prediction, and top-N ranking.

This survey provides a detailed review of the existing literature on recommender systems with deep learning architecture. The primary contributions of this study are as follows:

- We briefly provide an overview of recommender classification methods, modeling user behavior trends, and explore the general aspects of recommendation tasks: rating prediction and top-N ranking as part of the research background.
- We comprehensively discuss notable deep learning models covering the literature on rating prediction and top-N ranking tasks.
- We investigate pre- and post-modeling critiques of recommender system research and present a brief discussion on benchmark datasets, deep learning-based feature learning analysis, and evaluation metrics to provide a wide range of utilization knowledge in this domain.
- We analyze the recommender system's current trend and the prospects associated with the discussed trend in this research field.

The remainder of this literature survey is arranged as follows: Sect. 2 presents the recommender system research background. Section 3 reviews existing deep learning

models employed in the recommender system, focusing on rating prediction and top-N ranking research areas. Section 4 explores fundamental aspects of recommender system research, including pre-modeling considerations such as publicly available datasets and feature learning analysis, as well as post-modeling evaluation metrics. Section 5 features the new generation recommendation trend and outlines future research directions in each. Lastly, Sect. 6 concludes the article.

2 Research background

To provide a foundation for the study of recommender systems, this section first describes the techniques used in recommender systems, presents an understanding overview of representation learning in modeling user behavior trends, and then discusses the tasks of rating prediction and top-N ranking.

2.1 Overview of recommender system techniques

This section provides an overview of recommender engines and outlines a general classification of recommender systems. Additionally, it introduces a specific recommender technique known as a graph-based recommender system.

2.1.1 General recommender system classification

Recommender systems are typically categorized into three main algorithms: collaborative filtering, content-based, and hybrid:

- Collaborative Filtering (CF): is the most widely used method for creating personalized recommendation systems. The Amazon team proposed CF, an early version of machine learning algorithms [191]. The CF works by offering relevant things to consumers based on their past decisions [25]. It compiles the information provided by users and generates fresh recommendations based on other users' responses. CF models are further classified into two categories. Namely, memory-based CF and model-based CF. The memory-based finds the similarities between the features and makes predictions based on those similarities. The model-based CF utilizes the machine learning concept. The algorithm learns from the data and recognizes patterns to estimate the predictions. The state of the art vary from MF to clustering to deep learning methods.
- Content-Based: Content-based recommender systems extract content features and recommend similar items based on users' behavior or rated history. The algorithm considers users' profiles and item attributes as content features. In the current trend, several content-based algorithms use side information in the form of meta-data [28] or user-generated [395] features. Deep learning approaches offer a flexible framework to discover existing structures in content data to learn better features [208]. For example, visual features combine with deep learning to model an artwork recommender system [223].
- Hybrid: Hybrid system combines two or more recommendation techniques. The basic idea involves amalgamating content-based and CF approaches for improved results. Another research idea defines hybridization as a fusion of two or more DNNs [145]. Another hybridization concept incorporates side information such as tags in CF [218].

2.1.2 Knowledge graph-based recommender system

As the name implies, it acquires knowledge from graph data. For recommender applications, a graph is formed with the help of users, items, and features as nodes, and the connection between users and items, or items and features represent edges, as shown in Fig. 1. This graph structure is further analyzed to find users' preferences for the items. A knowledge-based recommender system is suitable when data is sparse as the method can use the information from different sources to recommend. For example, a

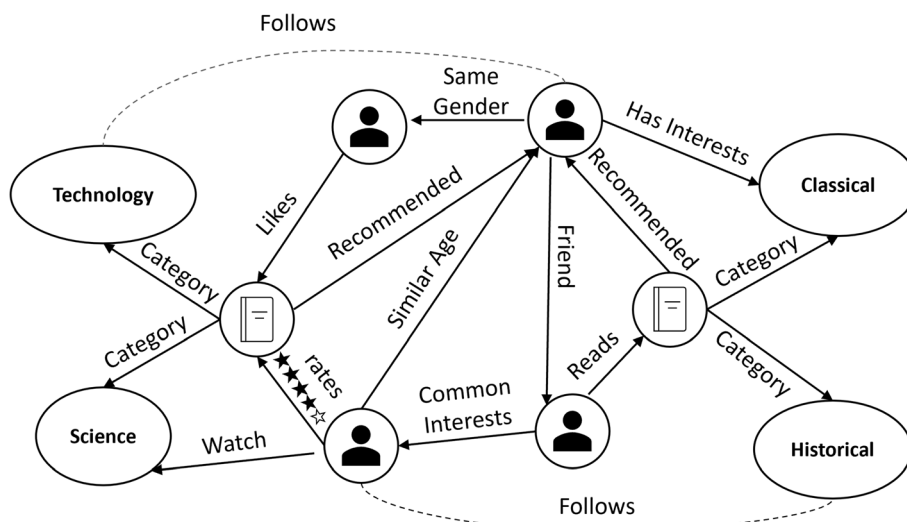
knowledge-based algorithm includes context information (side information extracted from Facebook) along with a user profile to construct the graph [262]. Knowledge-based is one of the robust methods applied to those applications, where traditional CF and content-based methods cannot be used.

2.2 Overview of representation learning in modeling user behavior trends

Numerous strategies and research articles have emerged to address various challenges in recommender systems, such as improving recommendation accuracy, solving the cold start problem, handling data sparsity, enhancing representation learning (especially in user behavior modeling), achieving recommendation scalability, ensuring recommendation diversity, providing personalized recommendations, developing application-based recommendations, and improving recommender system explainability. User behavior modeling (UBM) is particularly relevant to practical recommendations as it is integral to understanding user interests, making it a significant area of research attention in recommender systems. By leveraging interactive patterns between users and items, UBM brings compelling improvements in many recommendation tasks [120]. The fundamental concept of UBM is to map users into a low-dimensional representation space [160], facilitating the learning of latent representations for each user through items, item features, and user–item response matrices. UBM derives these user interest representations from static explicit data (such as click behavior and rating matrices), or unstructured data (like user behavior patterns and diverse heterogeneous features), or sequential data (capturing dynamic user behaviors). This approach is not limited to recommender systems; it also has applications in response prediction and other related tasks. Over recent decades, numerous methodologies have emerged to characterize user behavior and deliver personalized recommendations to users [178]. For instance, [280] uses MF, while [274] employs nonnegative matrix factorization (NMF) to fit relation matrix as the product of low-rank latent factors. Additionally, [217] adopts probabilistic matrix factorization (PMF) method to interpret the rating matrix as the representation, integrating user preferences and user connections to improve model interpretability and recommendation quality.

Recent research has increasingly explored deep learning approaches to enhance the nonlinear representation learning capabilities in recommendation techniques and construct robust user preference model [327]. Key facets of deep representation learning in user modeling pertinent to recommendation systems encompass feature extraction techniques for extracting meaningful features from user

Fig. 1 Knowledge graph-based recommender system



data, embedding methods for creation of low-dimensional user representations of users, application of diverse deep learning networks to capture complex user behaviors, and tailoring personalized recommendations based on learned user representations.

Various deep learning models with different working principles, such as Boltzmann machine [253], autoencoder [177], multilayer perception [68], convolutional neural network [419], recurrent neural network [20], attention mechanism [418], hybrid network [327], and LLM [2, 261], are employed to learn feature representations between users and items [327]. These advancements emphasize ongoing efforts in user modeling and recommendation systems. For example, a literature study proposes a neural network-based context-aware user-item representation learning model by exploiting reviews and interaction data to estimate personalized rating score for new user-item pairs [350, 351]. Addressing privacy concerns related to learning user embeddings, FURL extends neural network personalization to federated learning [27]. FURL can personalize models in federated learning by learning task-specific user representations (i.e., embeddings) [234] or by personalizing model weights [303]. Another article focuses on dynamic user preference modeling and introduces a time-aware attention framework that learns from recently user-interacted items and their interaction time [327] to improve predictions made based on CF techniques to perform high-order feature interactions for making final predictions. Innovative approaches continue to emerge, such as continual user representation learning models that utilize task embeddings to create relation-aware, task-specific masks, thereby retaining learning capabilities and capturing task relationships during training [160]. Additionally, a multiview user representation learning model improves user modeling by integrating

knowledge from different networks [331]. The proposed work leverages information from multiple social platforms to predict user attributes for enhanced recommendations. This evolving research scenario demands a thorough understanding of recent practical methods in user modeling through representation learning. To explore their advancements, advantages, disadvantages, and their application in rating prediction and top-N ranking tasks, in the next section, we include an analysis of several of these deep learning-based methods.

2.3 Rating prediction and top-N ranking

As research in the field of recommender systems continues to grow, a variety of scenarios and tasks have been explored. In this study, we focus on two primary stages of generating recommendations, emphasizing their distinct strengths and weaknesses to establish a general understanding of the differences between them. Table 2 outlines the key distinctions between rating prediction and top-N ranking approaches. The decision to use one method over the other should be guided by the specific goals and constraints of the recommendation scenario.

3 Literature review: state-of-the-art deep learning recommendation models

Historically, recommender systems are generated using strategies like user-based similarity, item-based similarity, user-item past interaction learning, and using additional information like geographical, temporal, and contextual data. Modern recommendation research is inclined toward deep learning because of the method's potency in catching complex user-item relationships and generating better

Table 2 Key distinctions between rating prediction and top-N ranking approaches

General aspect	Recommendation tasks	
	Rating prediction	Top-N ranking
Description	A type of recommendation task that estimates the rating a user is most likely to assign to an item, based on their past behavior and preferences [112]	A type of recommendation task that identifies and recommends the top-N items from a larger pool [112] that a user is most likely to enjoy, based on their preferences and interaction history
Goal	Its goal is to provide an accurate prediction or assigning of the numerical rating to products and uses this rating to filter and sort a list of products to present to the user	It accounts for all products in the collection and aims to create a subset of products that the user may like or likely to interact with
Area of problem	Regression	Classification
Strength	• Personalized Recommendations: Rating prediction algorithms have the ability to generate personalized recommendations [163, 221] tailored to individual users' unique preferences • Enhanced User Experience: By predicting ratings for products, these algorithms can improve user satisfaction by recommending products that match their preferences [224] • Quantitative Evaluation: The accuracy of rating prediction models can be quantitatively assessed using metrics such as mean absolute error (MAE) or root mean squared error (RMSE). These methods are straightforward to implement [105] and provide valuable insights into understanding initial ground performance of recommendation models	• Diverse Recommendations: Top-N ranking algorithms provide a diverse set of recommendations [146], allowing users to explore new and variety of products • User Engagement: By presenting a curated selection of top recommendations, these algorithms can enhance the user experience by promoting exploration and potentially increasing user engagement and interaction with the recommendation platform • Flexibility: Top-N ranking algorithms can be designed to support different recommendation scenarios, such as recommending movies, products, or articles, making them versatile [214, 215] for various applications
Limitations	• Cold Start Problem: Rating prediction models may face difficulties in predicting scores when dealing with new users [229] or products that have minimal historical data [360], a situation commonly referred to as the cold start challenge • Sparsity: In datasets with a large number of users and products, the data matrix is often sparse [229], which can hinder the accurate prediction of ratings for all user–item combinations • Difficulty in Capturing Complex Preferences: Rating prediction algorithms may struggle to capture complex user preferences, especially when users' preferences are diverse or evolving over time	• Lack of Personalization: Top-N ranking algorithms may prioritize popular [226] or trending products at the expense of personalized preferences, resulting in less individualized recommendations for the user • Difficulty in Handling Long-Tail Products: Top-N algorithms may struggle to recommend niche or long-tail products that have limited popularity or user interactions [110] • Evaluation Challenges: Evaluating the effectiveness of top-N ranking algorithms can be challenging [5], as conventional metrics such as precision and recall may not fully capture the quality of recommendations, particularly when user preferences are dynamic or subjective

recommendations. Deep learning is an emerging subdivision of machine learning, composed of several deep layers to process supervised and unsupervised features and analyze input data patterns. Over the past decade, deep learning has expeditiously evolved in the recommendation research discipline, demonstrating prodigious improvement over conventional approaches in terms of user behavior representation learning, recommendation accuracy, system robustness, and algorithm generalizability. Also, the involved deep models have attained considerable success in determining a dynamic array of users' preferences and uncovering complex relationships within user–item data. Besides this, the deep hidden neural architecture attests prowess in achieving better recommendation tasks, with the following capabilities:

- The capability of extracting underlying hidden users and items features.
- Ability to understand nonlinear user–item interactions.
- Scalability in terms of recommending products or items.
- Ability to predict the next item in a sequence for sequential-based recommender applications.
- Ability to incorporate rich resources from external domains.

Several deep learning algorithms are adopted in building recommender systems; **traditional models** like restricted Boltzmann machine (RBM), deep belief network (DBN), autoencoder (AE), generative adversarial network (GAN), multilayer perceptron (MLP), convolutional neural network (CNN), and recurrent neural network (RNN);

modern categorization of models such as deep attention network (DAN), large language model (LLM), graph neural network (GNN), and **hybrid models**. It becomes important to explore their contributions in the field of recommender systems. Both rating prediction and top-N ranking are fundamental tasks in recommender system research, and there is ongoing exploration and development of methods to improve the accuracy and effectiveness of these approaches. To this end, this section covers the most prominent deep state-of-the-art literature review pertaining to these two principal recommendation tasks: rating prediction and top-N ranking. Relevant to these tasks, Table 3 provides a compilation list of deep learning publications with model summary for rating prediction and top-N ranking recommendation tasks.

3.1 Traditional deep learning models

Traditional deep learning models include a variety of architectures that have greatly advanced the field of recommendation systems. This section provides a literature analysis of traditional deep learning models (RBM, DBN, AE, MLP, CNN, and RNN) for recommendation tasks and presents detailed summaries of these approaches in Table 4.

3.1.1 Restricted Boltzmann machine (RBM)

RBMs are undirected, shallow neural networks that consist of two layers. The first layer is called the visible layer, and the second layer is called the hidden layer. The neurons in RBMs can be connected to each other across layers, but connections between neurons within the same layer are restricted. One feature that distinguishes RBMs from other neural networks is that they use two biases: the hidden bias for the forward pass and the visible bias for the backward pass. An RBM-based recommender system can predict item ratings for users by analyzing their past ratings and those of similar users.

RBM approaches for rating prediction. RBMs are a type of stochastic network. The stochastic nature of inner neuron units implies that they rely on probabilities rather than deterministic values. The concept of probability is effectively demonstrated in a recommendation method by [268], where the authors used users' movie ratings as input for the visible layer, while the hidden layer represents binary user features. During training, the authors applied conditional probabilities to find connections (latent factors) between ratings (visible layer v) and users features (hidden layer h). In their proposed model, RBM also incorporates an additional information vector r into account, which represents a binary vector of rated and unrated movies by the user. The

derived equations for conditional probabilities over $(v|h)$ and r are given as:

$$p(v_i^k = 1|h) = \frac{e^{(b_i^k + \sum_{j=1}^F h_j w_{ij}^k)}}{\sum_{l=1}^K e^{(b_i^l + \sum_{j=1}^F h_j w_{ij}^l)}} \quad (1)$$

$$p(h_j = 1|V, r) = \sigma \left(\sum_{i=1}^m \sum_{k=1}^K v_i^k w_{ij}^k + b_j + \sum_{i=1}^M r_i D_{ij} \right) \quad (2)$$

Here, w and b represents weights and bias. D represents learned matrix that shows the effect of r on h . They used marginal likelihood to optimize weights and gradient $\Delta W_{ij}^k = \in (\langle V_i^k h_j \rangle_{\text{data}} - \langle V_i^k h_j \rangle_T)$, where $\langle V_i^k h_j \rangle_{\text{data}}$ denotes data observed from the training set and $\langle V_i^k h_j \rangle_T$ denotes expected data from the model.

Unlike traditional RBM that models pair-wise collaborative effects directly, another study constrained the RBM parameters using content information to discover the most reliable content information for the results [104]. A different approach proposed neighborhood conditional RBM (N-CRBM) model that integrated neighborhood features into the CF method to address the cold start problem. [374]. [203] considered item category as an additional layer. This layer comprised of two parts: a user-based model (to build a vector with each user's ratings on items and the conditional layer of item categories) and an item-based model (to build a vector of item ratings and the conditional layer of item features). Next article proposed a deep multilayer RBM architecture for item-based CF [75]. Each RBM layer is trained separately and treated as a single item. The multilayer RBM model achieved success in tailoring recommendation systems.

RBM approaches for top-N ranking. Most RBM implementations are designed for rating prediction tasks, with only a few methods tailored for top-N ranking. Because incorporating negative sampling into RBM training is technically challenging [354]. However, a recent method not only included negative sampling but also demonstrated the efficiency of conditional RBM (CRBM) for top-N tasks [44]. They created positive and negative samples from users' feedback matrices and passed both into the defined conditional layer. Another research showed the effectiveness of incorporating side information, such as item details and preference relations (the order of preference for items), into deep recommendation models [253]. The authors integrated side data into a CRBM that generates item rankings for any user. Additionally, the approach retrieved local and global user-item interactions to improve ranking quality. Another method introduced an explainable RBM-CF technique to achieve both accuracy and interpretability in top-N ranking [1].

Table 3 List of deep learning publications with model summary for rating prediction and Top-N ranking

Model category	Deep learning models	Rating prediction publications	Top-N ranking publications	Model summary
Traditional	RBM	[1, 67, 75, 101, 104, 133, 170, 203, 205, 268, 307, 374, 384]	[1, 44, 147, 233, 253, 332, 354]	RBM perform well in item rating tasks by learning the probability distribution of item ratings for users based on their previous ratings. Their ability to learn representations of user-item latent factors is one of their key strengths, though they may not scale as well as newer methods
	DBN	[38, 39, 127, 133, 246, 335, 385, 392, 399]	[242, 377, 404, 415]	DBNs effectively reduce dimensionality and extract features from sparse data, offering good handling of supervised learning over RBMs [38] for complex feature extraction and classification tasks, though their training can be computationally intensive
	AE	[16, 31, 67, 164, 177, 240, 248, 250, 260, 270, 290, 291, 298, 299, 318, 323, 339, 401, 434]	[54, 71, 167, 179, 249, 276, 298, 315, 318, 323, 354, 370, 378, 428, 435]	AEs make recommendation models more adaptable to multimedia scenarios and better at handling input noise. Their ability to process heterogeneous data provides significant opportunities to recommend diverse items using unstructured data and demonstrates excellence in learning underlying features ([342, 354]). However, AEs require careful hyperparameter tuning, and larger models may struggle with disentanglement and interpretability, limiting their generalizability across recommendation research domains
Modern	MLP	[37, 46, 84, 88, 185, 255, 309, 325, 337, 389]	[10, 11, 40, 41, 45, 53, 68, 118, 119, 142, 365, 426]	In recommender systems, MLPs can model features or serve as MF to model latent factors. They are popular for their adaptability and ease of use but require careful regularization to prevent overfitting [391]
	CNN	[48, 155, 157, 158, 171, 196, 237, 245, 272, 346, 375, 419, 431]	[13, 51, 77, 78, 116, 231, 232, 303, 329, 357, 382, 387]	CNN-based recommenders are particularly good at extracting spatial features and sequences, making them strong for personalized recommendation. By leveraging contextual and visual data, they provide nuanced recommendations [305] tailored to the user's current situation. Despite their strengths, CNNs can be resource-intensive, often demanding substantial data and computational power for effective training
	RNN	[124, 153, 192, 210, 247, 336, 341, 393, 432]	[15, 47, 72, 92, 128, 228, 243, 285, 342, 417, 433]	RNNs are useful for sequential or session-based recommendations. They can effectively model user preferences over time, making them well suited for dynamic and real-time applications. They may create false dependencies by assuming adjacent interactions are related and can overlook collective dependencies ([327, 330])
	DAN	[34, 36, 43, 55, 123, 196, 210, 255, 272, 273, 287, 304, 375]	[7, 29, 35, 42, 77, 78, 80, 93, 115, 130, 198, 204, 251, 284, 411, 418, 425]	Deep attention models perform well in rating prediction and top-N ranking tasks by focusing on key aspects of input data and capturing subtle relationships. They offer high interpretability for explaining recommendations [52], but their success depends on the quality and diversity of the input data
	LLM	[96, 149, 172, 183, 186, 211, 212, 338]	[60, 65, 66, 70, 76, 86, 98, 103, 129, 150, 159, 181, 182, 190, 199, 213, 235, 236, 279, 294, 340, 344, 348, 349, 350, 351, 361, 366, 424]	LLMs demonstrate exceptional capabilities in language comprehension and generation, enhancing recommender systems by drawing insights from large data volumes for personalized recommendations. However, they can be sensitive to input prompts [129], sometimes misinterpret them, and favor popular items too much. Additionally, incorporating LLMs directly into recommendation tasks can be complex [211, 212]

Table 3 continued

Model category	Deep learning models	Rating prediction publications	Top-N ranking publications	Model summary
Hybrid	GNN	[18, 84, 85, 94, 107, 138, 139, 141, 189, 206, 256, 343, 348, 349, 352, 356, 397, 405]	[6, 32, 106, 111, 117, 137, 138, 139, 140, 168, 181, 182, 184, 193, 194, 216, 292, 321, 333, 334, 347, 353, 372, 379, 386, 388, 398, 420, 421]	GNNs have gained prominence in recommender systems for their ability to analyze complex relationships in graph data structures. They excel in CF tasks, particularly in high-connectivity scenarios. However, GNNs can be difficult to interpret when features are correlated and may be memory intensive. Sparse data can also hinder performance by causing weak relationships and inaccurate embeddings [32]
	Hybrid models	[12, 73, 122, 237, 267, 297, 313, 317, 322, 325, 345, 381, 422, 423]	[40, 41, 54, 63, 64, 73, 136, 195, 241, 257, 316, 326, 345, 381, 390, 396, 412, 413, 414, 416, 429, 430]	Hybrid mechanisms create more robust and flexible recommendation systems that can handle various data types and complex interactions. They often achieve superior performance but may also require careful model selection and tuning to optimize performance

3.1.2 Deep belief network (DBN)

DBNs are generative models that excel at shallow unsupervised training. Each RBM representation in the DBN design learns the full input and produces output vectors to be used as input for the next RBM representation in the network. Each hidden layer in the network serves as the visible layer for the next network in sequence. Like RBMs, DBNs are pre-trained using contrastive divergence to update weights. DBNs have several advantages, including easy network expandability, the ability to work in both supervised and unsupervised environments, and fast classification training, making them feasible for recommendation approaches.

DBN approaches for rating prediction. A method was employed to obtain a discriminative classifier for predicting ratings from a set of user comments [38]. The proposed recommendation system first obtained word vectors from user comments using WordNet sentiment analysis and then generated matrix of word vectors as $\sum_{u=1}^n \sum_{p=1}^m [V_{u,p}]^l$, where n represents the total number of user comments, m indicates the total number of products or businesses with user comments, and l is the dimension of the word vector $V_{u,p}$. Next the system utilized these word vectors as inputs to RBM-DBN models. In the first part, RBM performed feature extraction and produced a new vector using the learning process at the hidden layer h as shown in (3), followed by the reconstruction process at the visible layer v as in (4).

$$p(h_{j=1}|v) = \sigma \left(\sum_{i=1}^n w_{ij}^{(h)} v_i + b_j^{(h)} \right) \quad (3)$$

$$p(v_i|h) = \sigma \left(\sum_{j=1}^m w_{ij}^{(v)} + b_i^{(v)} \right) \quad (4)$$

In the second part, DBN used map features extraction as inputs to classify user ratings. The computation at each hidden layer H_j of DBN is shown in (5) and the computation of each output layer Y_k is presented in (6). Both computations at the hidden and output layers are repeated until the target output is reached.

$$H_j = \sigma \left(\sum_{i=1}^n w_{ij} X_i + \theta^{(h)} \right) \quad (5)$$

$$Y_k = \sigma \left(\sum_{j=1}^m w_{jk} H_j + \theta^{(o)} \right) \quad (6)$$

$\theta^{(h)}$ and $\theta^{(o)}$ are biases at hidden and output layers. The complete recommendation system performed outreach in comparison to baseline support vector machine (SVM) and MLP-based recommendation models. Also, to improve

deep DBN performances, the system implemented three preprocessing noise reduction steps (short comments, comments with no opinion words and sarcasm or false comments) from the extracted user comments.

Another approach focused on audio content to predict user behavior [335]. The proposed music recommendation method combined CF (user rating data) and content-based (audio features) using PMF and a DBN. The model computed rating prediction as $r_{ui} \sim \mathcal{N}(\beta_u x_i + \gamma_u y_i, \sigma_R^2)$, which represents normal distribution with mean $(\beta_u x_i + \gamma_u y_i)$ and variance (σ_R^2) . The first term in the mean is the inner product of the item (song i) feature vector (x_i) and user (u) preference over the item vector (β_u) . The (x_i) vector is obtained using a DBN. The second term represents the inner product of user and item latent feature vectors (γ_u) and y_i , respectively) obtained from the collaborative PMF technique. The model defined its cost function using negative log-likelihood, as shown in (7), and optimized it using the alternating least-squares (ALS) algorithm. K represents training user–song pairs set and Reg means regularization term. This model successfully recommended new music to consumers and alleviated the cold start issue.

$$L = \sum_{u,i \in K} (r_{ui} - \beta_u x_i - \gamma_u y_i, \sigma_R^2) + \lambda \cdot Reg \quad (7)$$

In another article, a video recommender system was built using DBN and user-based CF approaches [127]. The proposed approach divided the rating matrix into 0–1 matrices and then passed it to the DBN network to determine neighborhood relationships. To represent vectors from documents, [385] used the bag-of-words approach. These vectors were then passed into parallel DBNs to extract latent factors. The authors implemented a shared layer on top of the DBNs that uses latent characteristics for the rating prediction task.

Instead of MF, a different article applied a Bayesian ensemble classifier to train each class of user ratings in combination with DBN that extracts sentiment features from user comments [39]. Another method trained DBN networks in two steps: unsupervised pre-training and supervised parameter adjustment [392]. This method also used the weight update algorithm to optimize the error function during parameter tweaking. Finally, in the recommendation task, the approach predicted course scores in a Massive Open Online Courses (MOOC) environment, allowing users to filter out low-quality courses and replace them with new ones.

The next approach improved recommendation performance by extracting high-level representations [246]. The method first utilized DBN as a feature extractor to remove the content of items and produce similarities between them. Second, it used traditional item-based CF to calculate the

similarities between items and combined the two similarity matrices to predict ratings. In another study, researchers designed a collective DBN to disentangle high-level features and devised a dual-wing RBM at the top level that learns the representation of group preferences using collective features and group choices [133].

DBN approaches for top-N ranking. An approach was employed that utilized deep feature representation learning and incorporated spatial dynamics of customized preferences to provide recommendations for the top-N Points of Interest (POI) [377]. The methodology employed a multi-modal DBN to merge heterogeneous information (including category, description, and comments) into a unified POI semantic representation. Further it applied PMF to combine geographical data such as geolocation, popularity, and neighborhood features with a semantic representation of POIs based on individual preferences. The method obtained a probabilistic generative process (g) for each check-in C_{ui} of user u at location i as $C_{u,i,k} \sim \mathcal{N}(g(p_{u,r} x_i), \sigma_C^2)$, where $p_{u,r}$ and x_i denote user latent vectors and POI features latent vectors, respectively. The vectors $p_{u,r}$ represent region-aware collective preferences $\theta_{k,r}$ at region r . Each user is assigned a role (either local or tourist) and based on this role; k is assigned. For tourist preferences, k equals zero and for native preferences k equals 1. The model extracted the latent vector x_i from the auxiliary information (a_i) and the DBN $(a_i; \psi)$, with ψ representing the weight parameters in the network. Finally, the model inferred using following negative log-likelihood method:

$$\begin{aligned} L = & \sum_{u \in \text{user}} \sum_{i \in \text{POI}} \sum_{k \in \{0,1\}} I_{u,i,k} (C_{u,i,k} - p'_{u,r} x_i)^2 \\ & + \lambda_u \sum_{u \in \text{user}} \sum_{r \in R} \sum_{k \in \{0,1\}} I_{u,r,k} \| (p_{u,r} - \theta_{k,r}) \|^2 \\ & + \lambda_R \sum_{k \in \{0,1\}} \sum_{r \in R} \| \theta_{k,r} \|^2 + \lambda_I \sum_{i \in \text{POI}} \| x_i - \text{DBN}(a_i; \psi) \|^2 \end{aligned} \quad (8)$$

λ_u, λ_R and λ_I represents hyperparameters. $I_{u,i,k}$ is an indicator function defined as $I_{u,i,k} = \text{values}$, where $C_{u,i,k} > 0$. $I_{u,r,k}$ is another indicator function defined as $I_{u,r,k} = \text{values}$, where $[(l_u \in r) \odot k]$. Here l_u indicates users' home location or local region. The model followed the DBN training process and was trained in both an unsupervised and a supervised manner. In the unsupervised part, the RBM model processed input from the visible layer v to obtain first hidden layer representation $h^0 = v$. Similarly, the consecutive hidden layer obtained representation learning as $p(h^{i-1} | h^i)$. This process repeated each time propagating upward the mean activation values. In the last stage, the model obtained supervised learning by converting DBN into MLP. This step performed chain-rule backpropagation

derivation with respect to parameters ψ and latent factors $p_{u,r}$, $\theta_{k,r}$ and x_i . Finally, in the recommendation step, the system computed the ranking score $S(q, i) = g(p_{u_q, r_q}^T x_i)$ for each unvisited POI i in region r w.r.t query $q = (u_q, r_q, k_q)$, which provides top-N recommendations. The ranking score for cold start POIs was approximately calculated as $S(q, i) = g(p_{u_q, r_q}^T \text{DBN}(a_i, \psi))$. Similarly for cold start users, it was calculated as $S(q, i) = g(\theta_{k_q, r_q}^T x_i)$.

Next, the paper performed text analysis to provide personalized recommendations [242, 415]. The authors extracted interest keywords to characterize the user's preference from the set of news articles read by that user in the past [242]. Another method captured the word semantics similarity for similar cases recommendation using a novel word embedding-based retrieval model [415]. A novel method called discrete deep learning (DDL) implemented a hashing-based deep learning framework to tackle sparsity and cold start difficulties [404]. It is a content-aware model that combines content (hash codes) with a DBN network to extract useful binary representations. It added balance and unnecessary limitations to hash codes to generate informative binary codes. DDL's unified integration approach helped maintain a balance between recommendation efficiency and accuracy. It improved recommendations in sparse situations using hash codes.

3.1.3 Autoencoder (AE)

Unsupervised neural networks, or AEs, are beneficial for tasks like dimensionality reduction and feature learning. The concept behind AEs is to compress large amounts of data into smaller representations and then learn from these compressed forms. AEs are capable of learning nonlinear transformations, and this property makes AE algorithms useful for applications involving efficient image and video data transformations in recommender systems.

AE approaches for rating prediction. Although MF method has become dominant in recommender systems, these systems largely rely on initializing the user and item latent feature vectors [67]. Thus, to improve the quality of these initialization vectors, Deng and team utilized the deep AE method that abstracts high-dimensional rating data into latent features through multiple layers of RBMs [67]. They further revamped the MF model by incorporating trust relations to make recommendations in social networks. In another study, the researchers developed an enhanced deep AE model to handle data sparse and imbalance issues in social networks [250]. A different work applied the concept of social influence to develop a movie recommender system [299]. This method combined CF and content-based filtering by considering the movie properties using content-based filtering and then predicting unrated

movies for target users using CF and a deep AE network. The deep network aimed to calculate the social influence of each user and the users with stronger social influence impose a more significant impact on movie recommendations for target users. Kuchaiev and Ginsburg designed a deep AE model for rating prediction task [164]. This model followed an iterative approach as shown in Fig. 2, where the output y was re-fed into the network to avoid the need for model pre-training. During the training process, the model used nonlinear activation functions in all layers that assisted the model in training with negative values, except the last layer, which used linear function. The model was optimized using the masking approach, which considers only nonzero user representations, and the optimization equation can be represented as the masking squared error loss $MMSE = \frac{\sum_{i=1}^N m_i (r_i - \hat{r}_i)^2}{\sum_{i=1}^N m_i}$. Here r_i and $\hat{r}_i$ denotes actual and reconstructed rating. m_i is mask function. All these features (iterative, nonlinear activation and mask function) contributed to the model effectiveness resulting in high-speed training and comparable results.

In contrast, an earlier approach by [270] utilized single-hidden-layer concept for rating prediction. Their model employed a vanilla version of AE, and minimized the objective function as follows:

$$\min_{\theta} \sum_{i=1}^N \|r^{(i)} - \hat{r}^{(i)}\|_o^2 + \frac{\lambda}{2} (\|W_F^2 + V_F^2\|) \quad (9)$$

The term $\frac{\lambda}{2} (\|W\|_F^2 + \|V\|_F^2)$ represents the regularization term over the weights (W and V). $r^{(i)}$ and $\hat{r}^{(i)}$ denote respective partially observed and predicted ratings obtained during backpropagation. $\|\cdot\|_o^2$ represents consideration of only observed ratings and $\|\cdot\|_F^2$ signifies Frobenius norm.

Another study [248] redesigned the AE model for CF, incorporating optimization techniques such as pre-training with RBM and stacking AEs together to increase the recommendation performance.

[31] presented a collaborative adversarial autoencoder (CAAE) to unify training across generative and discriminative models, combining AEs with GAN. This method extended conventional recommendation models like IGRAN [322] and GraphGAN [317]. Both IRGAN and GraphGAN are constructed using MF, and their overall objective functions can be written as:

$$\min_{\theta, \phi} \sum_{m=1}^N E_{i \sim p(i|u_m)} \log(\sigma(f(i|u_m))) + E_{i \sim p_{\theta}(i|u_m)} \log(1 - \sigma(f(i|u_m))) \quad (10)$$

The term $p_\theta(i|u_m)$ and $\sigma(f(i|u_m))$ represents generative probability distribution of positive items and discriminative probability distribution of users' preferences u_m over item i , respectively. The discriminative model is typically solved using stochastic gradient descent method with θ and ϕ representing model parameters of the generative and discriminative processes in particular. CAAE employed AE networks as positive item generators, Bayesian personalized ranking (BPR)-based MF as discriminators and another generator model (negative item generator) for sampling negative items. The objective functions for different networks in CAAE are as follows:

- *Discriminative Network:*

$$\min_{\theta, \phi} \sum_{m=1}^N -E_{i \sim p(i|u_m), j \sim p_\theta(j|u_m)} [\log(\sigma(f(i|u_m)) + b_i - b_j)] + \frac{\beta}{2} \|\phi\|^2 \quad (11)$$

- *Generative (Positive Item) Network:*

$$\min_{\theta, \phi} \sum_{m=1}^N \lambda \cdot -E_{i \sim p(i|u_m), j \sim p_\theta(j|u_m)} [\log(1 - (\sigma(f(i|u_m))) + (1 - \lambda) \cdot \Sigma \|(r_u - \hat{r}_u) \cdot e_u\|^2 + \frac{\beta}{2} \|\theta\|^2 \quad (12)$$

- *Generative (Negative Item) Network:*

$$\min_{\theta, \phi} \sum_{m=1}^N -E_{i \sim p(i|u_m), j \sim p_\theta(j|u_m)} [\log(1 - (\sigma(f(i|u_m))) + \frac{\beta}{2} \|\theta\|^2 \quad (13)$$

These equations are similar to IRGAN and GraphGAN, with slight variations. The last term $\frac{\beta}{2} \|\theta\|^2$ represents the regularization term as CAAE followed L2 regularization. The first term indicates the loss from the GAN network. The second term $\Sigma \|(r_u - \hat{r}_u) \cdot e_u\|^2$ in (11) indicates the loss from the AE networks, where r_u is the initial input of the encoder network and $\hat{r}_u$ is the reconstruction of initial input in decoder network. CAAE has proved successful in predicting users' preference scores for plausible items.

Like CAAE, [237] adopted a generative concept with CNN for movie rating prediction. [318] utilized Bayesian stacked denoising autoencoder (SDAE), while [177] employed marginalized SDAE, which computes parameters in closed form and enhances the scalability of the recommender system. Next, [291] used the SDAE architecture to point out the complex relationship between users/items' representations and additional constraints on the top of the network to handle sparse inputs. Another approach

incorporated side information of users and items to every layer input of the SDAE network [290].

An advanced work by [339] proposed deep recommendation models to handle complete and incomplete cold start problems, in which item features were learned from an SDAE network using online item descriptions. These features were then integrated into the time singular value decomposition (timeSVD++) model [162] to track temporal dynamics of user preferences and item features. [401] introduced AutoSVD and AutoSVD++ based on contractive AE to effectively capture item content within a low-dimensional embedding space. This approach tightly coupled AE-based deep learning frameworks with MF to make accurate recommendations. To advance the health-care domain, a research study proposed a food recommender system that combines k-means clustering with a denoising autoencoders (DAEs) [9]. This system groups users with similar dietary behaviors to train the AE model, which learns from corrupted versions of the item–user interaction matrix, ensuring that users with different dietary habits are excluded from the recommendation process.

AE approaches for top-N ranking. A paper presented a hierarchical Bayesian collaborative deep learning (CDL) model by incorporating SDAE network into collaborative MF for top-N recommendations [318]. The authors also integrated item information into SDAE to learn deep representations. In each layer l of the SDAE network, the model aimed for three things. The first was to find the weight matrix $W_l \sim \mathcal{N}(0, \lambda_w^{-1} I_{D_l})$ for each column n . The second was to find a bias vector $b_l \sim \mathcal{N}(0, \lambda_b^{-1} I_{D_l})$. Third, it aimed to find $\text{layer}_{l,i*} \sim \mathcal{N}(\sigma(\text{layer}_{l-1,i*} W_l + b_l), \lambda_l^{-1} I_{D_l})$ for each row i of layer_l . For each item i , it cleaned the input as $\text{layer}_{c,i*} \sim \mathcal{N}(\text{layer}_{l,i*}, \lambda_c^{-1} I_i)$; designed the latent vector as $\epsilon_i \sim \mathcal{N}(0, \lambda_v^{-1} I_D)$, and set the item latent vector as $V_i = \epsilon_i + \text{layer}_{\frac{T}{2},i*}$. Further the network obtained the user latent vector for each user u as $U_u \sim \mathcal{N}(0, \lambda_u^{-1} I_D)$, and finally, the rating for each user–item pair is obtained $r_{u,i} \sim \mathcal{N}(U_u^T V_i, C_{ui}^{-1})$. Here $\lambda_w, \lambda_l, \lambda_c$, and λ_v are hyperparameters and C_{ui} denotes confidence parameter. The model adopted a point-wise approach to the loss function and trained the parameters using the expectation maximization algorithm.

In another instantiations of CDL, [378] utilized a pair-wise approach to alleviate sparsity reduction and [71] integrated side information to improve the CDL model by capturing the implicit relationships between users and items. Next, [167] applied the CDL approach to identify long noncoding ribonucleic acid (lncRNA) disease association. Using the SDAE network, the proposed model

Table 4 Summary of existing traditional deep learning techniques for recommender system

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
RBM	[268]	Undirected graphical RBM network	Handling large dataset	Netflix	RMSE	• RBM outperforms the traditional SVD model by a very small margin • A linear combination of multiple RBM and SVD model versions produces an error rate over 6% better than the Netflix's own baseline score
	[374]	N-CRBM	Data sparsity and cold start	Kaggle dataset	RMSE, accuracy, sensitivity	• N-CRBM outperforms the cluster-based RBM model in RMSE, accuracy, and sensitivity because of its use of popularity and similarity scores. For instance, on the Kaggle dataset, N-CRBM has an accuracy of 78.5% compared to 76% for the cluster-based RBM • The choice of similarity scores impacts learning outcomes, and poor selection can lead to lower performance
	[751]	Item-based RBM model	Data sparsity	MovieLens	MAE, RMSE	• The multilayer model outperforms both two-layer user-based and item-based RBMs, providing recommendations more aligned with the user's current behavior. The item-based multilayer RBM achieves the best results with MAE of 0.6424 and RMSE of 0.78
	[253]	Preference relation-based RBM and CRBM models	User behavior representation learning, data sparsity, scalability	MovieLens (ML), BookCrossing (BC)	nDCG, MAP	• However, the deep structure takes more time due to its complex model • The model consistently outperforms others, achieving the highest nDCG (up to 0.5259) and MAP (up to 0.4770), especially in the ML-20 M dataset • It remains the top performer even with sparse data, achieving the highest nDCG (0.4263 on ML-1 M, 0.4508 on ML-20 M, 0.4010 on BC) and MAP (0.3657 on ML-1 M, 0.3992 on ML-20 M, 0.3457 on BC) with only 20% data, showing it handles sparsity effectively
DBN	[44]	CRBM	Top-N recommendation problem	MovieLens, Netflix	Precision, recall, F-score, nDCG and 1-call	• It converges faster than other NeuMF- and AE-based models including traditional RBM models • The model effectively addresses item recommendation challenges with best performance of 1-call@5 of 0.4830 ± 0.0036 on MovieLens against traditional MF and popularity-based ranking methods • Although the RBM's capacity to learn explicit feedback may be somewhat restricted compared to more intricate models
	[38]	DBN and WordNet sentiment analysis	Recommendation accuracy	Yelp, Amazon, Tripadvisor	RMSE, MAE, accuracy, precision, recall, F-score	• The model performs well in the Amazon and Yelp with 72.83% and 75.85% accuracy, respectively, but it does not do as well with the Tripadvisor, achieving only 49.23% accuracy compared to libSVM's 50.02%
	[39]	Deep learning ensemble model	Computational complexity	Yelp, Amazon, Tripadvisor	Accuracy	• Moreover, the method is not suitable for real-time testing due to slow computational speed • Unlike traditional DBNs, the proposed method does not struggle to achieve high accuracy because the model is trained to fit the minority class • The new method achieves improvement of 10% on Tripadvisor and Yelp, and over 20% on Amazon against traditional DBN, and does much better than linear models. This is a significant improvement in predicting bad ratings from an imbalanced dataset
	[335]	DBN and probabilistic graphical model	Warm start (WS) and Cold start (CS)	The Echo Nest Taste Profile Subset	RMSE	• This method achieves RMSE scores of 0.323 on WS valid, 0.323 on WS test, 0.477 on CS valid, and 0.478 on CS test, and works much better than standard MF and content-based methods • It shows that the features it learns automatically are more effective than the traditional ones for music recommendation

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
AE	[377]	POI-based DBN	Sparsity in POI matrix and cold start	Yelp, Foursquare	Accuracy, MAE	- The model demonstrates the usefulness of multimodal features learning by beating top recommendation methods, especially for out-of-town (with improvements of 20.13%) and cold start situations - On the Yelp dataset, it achieves the lowest prediction error with MAE of 0.976 for out-of-town and 0.891 for home-town recommendations - It can also recommend POIs to cold start users with limited check-ins using popular features
	[415]	Deep DBN model	Recommendation diversity	ask.39.net	MAP, precision	- Word embedding models perform better than translation-based models with a marginal improvement of 1.88% in MAP and similar results in precision - However, these methods may have trouble capturing the subtle meanings of complex or ambiguous words or phrases, which could lead to inaccuracies in similarity assessments - Also, word embedding strategies may not effectively handle out-of-vocabulary words or rare terms, which can limit their usefulness in certain situations
	[404]	Hashing-based DBN and CF method	Recommendation efficiency, cold start, and data sparsity	Amazon	Accuracy, MRR	- Hashing-based recommendation has a clear advantage over real-valued recommendation because hashing codes take less time and memory compared to real-valued features - The method performs similarly or slightly worse than the CTR model (the best of the real-valued models). However, in sparse and cold start settings, it outperforms the competing baselines
	[67]	AE and social trust ensemble learning method	Data sparsity, cold start, and Trustworthiness	Epinions, Flixter	RMSE, Coverage, F-measure	- Results show improved RMSE, F-measure and coverage compared to baseline methods, and the model effectively handles cold start users. For example, it shows RMSE improvement of 1.84% to 2.45% with new users - This method costs around 20% more time than the others on average
	[164]	Six-layer deep AE-based on I-AutoRec approach	Computational speed and recommendation performance	Netflix	RMSE	- Additionally, user ratings are time-sensitive, and outdated ratings may be unreliable for recommendations - Results show that the model has better generalization than the shallow ones. Also, the training algorithm uses iterative output re-feeding that helps to improve the RMSE from 0.9167 to 0.9100; however, the model brings some difficulty to interpret [238]
	[270]	Discriminative CF model based on AEs (AutoRec)	User behavior representation learning	MovieLens, Netflix	RMSE	AutoRec offers representational and computational advantages over existing neural approaches in CF. And the deep extension of AutoRec further enhances its performance by reducing RMSE from 0.831 to 0.827

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[401]	Contractive AE + MF	Scalability	MovieLens, Movie tweetings	RMSE	- It achieves RMSE improvement of up to 1.36% and 1.50%, on both the ML-100 K and ML-1 M, respectively - It can easily scale up to larger datasets without compromising performance or increasing computational costs
	[318]	Hierarchical CDL model from SDAE + a Bayesian formulation	User behavior representation learning, data sparsity	CiteULike, Netflix	Recall, MAP	For CiteULike, the two-layer CDL outperforms CTR [312] by a margin of 4.2% to 13.1% in the sparse setting and 3.3% to 7.6% in the dense setting. For Netflix, the two-layer CDL outperforms CTR by a margin of 1.9% to 5.9% in the sparse setting and 1.5% to 2.0% in the dense setting
	[177]	Marginalized SDAE + PMF	User behavior representation learning, scalability	MovieLens, BookCrossing, Advertising	RMSE, MAE, AUC	The approach outperforms PMF, Biased PMF and Bayesian MF with side information, achieving lowest RMSE values (e.g., 0.8628 at 50% sparsity on ML) and demonstrating the effectiveness of side information and robust latent factors representation learning through marginalized SDAE
	[167]	AE + MF	Time and space complexity	lncRNA disease–gene associations and related data collected from different sources	Precision, recall, AUC	- It has nearly linear time complexity and owns a good scalability - In tenfold cross-validation, the model shows a 4.75% improvement in AUC over the MF baselines
	[354]	CDAE model from AE network	Top-N recommendation problem	MovieLens, Netflix, Yelp	Precision, recall, MAP	- The combination of SDAE and MF outperforms than using either method independently - CDAE outperforms in MAP@10 and recall@10 across both Yelp and Netflix, outperforming MF by 15% on Yelp and Item-CF by 10% on Netflix - However, the model's ability to address the cold start problem remains unproved
	[370]	CDL-I model from SDAE network	Scalability	CiteULike, Netflix	Recall, MAP	- The initial trials with stacked CDAE did not yield notable improvements over shallow CDAE and requires further investigation - CDL-I achieves the highest MAP scores and faster convergence in N recommendation tasks, with 0.514 for CiteULike-a, 0.453 for CiteULike-t, and 0.312 for Netflix, compared to CDL and SDAE
	[23]	Stacked sparse AE + fuzzy clustering	Leverage CF performance	MovieLens	Hit ratio, nDCG	- The model's performance nearly stabilizes when the hidden layer reaches two layers - Neural models, including the proposed one, outperform traditional CF models, showing that AE effectively improves recommendation accuracy despite data sparsity - The model improves by 2.7% in hit ratio and 4.6% in nDCG on ML-100 K, and by 2.6% and 2.3% on ML-1 M compared to the fuzzy user-based CF model

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
GAN	[407]	AE + k-means	Recommendation accuracy and diversity	MovieLens	Precision, coverage, MAP	The model outperforms user-based CF, k-means, and CDAE, achieving a 9% increase in precision on ML-100 k and a 16% increase on ML-1 m compared to user-based CF results
	[187]	Multi-VAE multinomial distribution model	User behavior representation learning	MovieLens, Netflix, Million Song Dataset (MSD), Pinterest	Recall, nDCG	- Multi-VAE surpasses MF and CDAE baselines, with standard errors of 0.002 for ML-20 M and 0.001 for Netflix and MSD - Multi-VAE performs better than NCF, with standard errors of 0.00106 for ML-1 M and 0.0053 for Pinterest - On the ML-20 M, Multi-VAE with a multinomial likelihood function performs better than common CF likelihood functions, with a standard error of 0.002
	[276]	RecVAE composite prior distribution model	Focused on enhancing AE-based recommendation architecture	Netflix, Million song dataset	Recall, nDCG	- RecVAE performs significantly better than all baselines, particularly compared to Multi-VAE, achieves improvements in Recall@20, Recall@50, and nDCG@100 of around 0.019, 0.016, and 0.016 in ML; 0.010, 0.008, and 0.008 in Netflix Prize; and 0.010, 0.010, and 0.010 in Million Songs - Each new feature improves the results, with all proposed additions achieving the highest nDCG@100 scores across all three datasets. However, adding these features slows down model training and requires more memory - The model faces challenges in integrating the proposed composite prior into it
	[9]	DAE + k-means clustering	User behavior representation learning	Food.com recipes, Interactions	RMSE	The proposed model has an RMSE score of 0.1927, which is better than other AE models without clustering, which have RMSE scores of 0.4358 and 0.4354
GAN	[322]	Generative and discriminative frameworks (IRGAN) with minimax game optimization	Web search ranking, item recommendation, and question answering	Learning TO Rank (LETOR), MovieLens, Netflix, InsuranceQA	Precision, MAP, nDCG, MRR	- Results demonstrate significant performance gains as much as 23.96% on Precision@5 and 15.50% on MAP over traditional ranking baselines in a variety of applications including web search, item recommendation, and question answering - In web search, IRGAN point-wise focuses on individual document relevance, while IRGAN pair-wise assesses overall document ranking - In item recommendation, IRGAN offers more learning flexibility despite not explicitly optimizing final ranking measures - For question answering, IRGAN surpasses both basic CNN models with random sampling and enhanced CNN models with dynamic negative sampling

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
MLP	[317]	GraphGAN framework from adversarial learning + proximity-aware graph softmax	User behavior representation learning	MovieLens	Precision, recall	- GraphGAN outperforms baselines in link prediction task, with accuracy improvements ranging from 1.18% to 4.27% and 0.59% to 11.13% - In node classification, it achieves gains of 1.75% to 13.17% and 0.95% to 21.71% on two datasets - For recommendation tasks, GraphGAN consistently outperforms baselines, showing significant improvements, such as Precision@20 gains from 38.56% to 156.85% - Graph softmax satisfies desirable properties such as normalization, awareness of graph structure, and greatly reduces computation complexity - CAAE shows average relative improvements over IRGAN of 65.1% in top-5 and 59.4% in top-20 recommendations, and over GraphGAN of 149.2% and 114.4%, respectively. In particular, CAAE improves by 262% over GraphGAN in terms of M@5 - CAAE with BPR improves top-5 recommendations by 27.9% and top-20 recommendations by 20.6% compared to CAAE with MF - The model outperforms a series of baselines with an improvement of 9.3% MSE and 7.6% hit ratio - This versatile model can be extended to handle other prediction tasks even with a small amount of training data
	[37]	MLP + hierarchical attention layer	Data sparsity	Douban, Yelp	RMSE, MAE	- The model consistently outperforms SemRec, NeuMF, and FM-based methods across all training data settings on the Douban dataset, with RMSE improvements of 1.7% to 19.9% and MAE improvements of 1.7% to 15.9%, as well as on the Yelp dataset, with RMSE improvements of 3.0% to 34.3% and MAE improvements of 1.5% to 34.1% - Furthermore, it is faster than MF-based models because it directly learns user and item features for predictions
	[185]	Ties CF and content-based filtering into multiview neural network	Data sparsity	Douban, MovieLens	RMSE	While the shallow network performs better than traditional MF-based methods, it does not fully utilize domain interaction information and neglects explicit consideration of content features for recommendations
	[53]	MLP-based candidate generation network + ranking network	Scalability and accuracy	YouTube	A/B testing—expected watch time per video impression	This deep CF model outperforms MF approaches. However, the fixed-length representation of embedding features may lead to information loss

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
CNN	[265]	Sequence-oriented DNN + k-means clustering	Cold start	Korean e-learning platform (ubob.com)	Precision	- The precision rates of model are 0.626 for Top-1 and 0.492 for Top-5 courses, with cold start errors of 0.618 and 0.697 for 25 and 50 new learners, respectively - However, it struggles with the “gray sheep” problem, where users who do not fit into existing clusters cannot receive relevant course recommendations
	[119]	NCF neural-based collaborative approach	Leverage CF performance	MovieLens, Pinterest	Hit ratio, nDCG	The model outperforms state-of-the-art methods eALS and BPR, showing an average relative improvement of 4.5% and 4.9%, respectively, across datasets. Additionally, the model with pre-training achieves relative improvements of 2.2% for MovieLens and 1.1% for Pinterest
	[40, 41]	J-NCF approach tightly integrating deep feature (DF) learning and user-item deep interaction modeling (DI) within a single neural network	Cold start, data sparsity	MovieLens, Amazon	Hit ratio, nDCG	- This method delivers substantial improvements over traditional methods, with Hit@10 improvements of up to 10.81% and nDCG@10 improvements of up to 15.06% across various datasets - It shows greater benefits for “inactive” users, with Hit@10 improvements of 11.08% at 50% interaction to 7.85% at 90% interaction and nDCG@10 improvements of 9.57% to 7.32% on ML-100K - The method takes slightly longer for one training epoch than deep machine learning models but has a shorter total training time - The model achieves a 1.85% loss improvement compared to NCF and shows statistically significant gains of 0.37% to 3.98% on DepaulMovies, and 1.6% to 2.16% on Tripadvisor - Overall, it outperforms baselines with an average gain of 2.35% and achieves roughly a 6.2% average gain compared to KNN
	[11]	CA-NCF framework incorporates a pre-filtering contextual information into NCF method	Personalized recommendations	MovieLens, DepaulMovie, Tripadvisor	MAE, Top1Accuracy	- The model achieves roughly 4% and 90% improvements compared to context-free plain NCF and BiasedMF, respectively - The improvement of model over NeuMF (best from baselines) ranges from 0.6% to 13.7% in terms of Hit@10 and from 0.7% to 16.6% in terms of nDCG@10 across various datasets - The proposed iMusic system achieves a data classification accuracy of approximately 88.27% (0.379 RMSE) and outperforms traditional machine learning methods
	[10]	A retrofitted version of a user-splitting approach with NCF	Social recommender System	MovieLens, Tripadvisor	MAE, precision	
	[68]	CF + vanilla MLP model	User behavior representation learning	MovieLens, LastFM, Amazon	Hit ratio, nDCG	
	[264]	DNN + k-means clustering	Recommendation reliability, representation learning	iMusic (Indian Classical Music)	RMSE	
CNN	[410]	MLP investigation for automatic behavior modeling and interaction exploration	User behavior representation learning, architectural design	Beauty, Steam, Avazu, Criteo, Alimama datasets	AUC, hit ratio, nDCG	The AUC improvement of the model over DNN models ranges from 0.0037 to 0.0083, and over attention models ranges from 0.0031 to 0.0042, with the highest improvement observed at 0.0083 over DeepFM
	[419]	Two parallel CNN networks + MF	User behavior representation learning, data sparsity	Yelp, Amazon, Beer	MSE	The CNN model delivers an average improvement of 8.3%, including 8.5% on Yelp, 7.6% on Beer, and 8.7% on Amazon, demonstrating its effective in joint user modeling and leveraging reviews semantics
	[157]	Context-aware ConvMF model uses CNN + PMF	Data sparsity	MovieLens, Amazon	RMSE	ConvMF demonstrates consistent improvement over the best competitor, CDL, with gains of 3.92% on the ML-1M and 2.79% on the ML-10M. On the Amazon dataset, which is highly sparse and skewed, ConvMF outperforms CDL by 16.60%

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[158]	Rating-based context-aware R-ConvMF model uses CNN + PMF	User behavior representation learning, data sparsity	Movielens, Amazon	RMSE	- R-ConvMF, which considers Gaussian noise differently, outperforms ConvMF across all datasets. The improvements over ConvMF are 0.71% on ML-1M, 1.43% on ML-10M, and 2.87% on Amazon. Moreover, R-ConvMF shows greater resilience to extremely sparse datasets compared to ConvMF - Additionally, the improvement of R-ConvMF over traditional PMF consistently increases from 5.57% to 8.83% - The method performs well, achieving the highest accuracy values (e.g., 0.9048 on TTL) and surpassing approaches that do not account for users' tasks relationships - Its performance also highlights its potential for adoption across various web platforms with user behavioral data - ConvMF surpasses PMF by at least 15% in terms of MAE and RMSE, while this dual CNN approach improves ConvMF by 3.6% to 5.5% across all datasets in both metrics
	[160]	Temporal CNN network	User behavior representation learning	Movielens, Tencent TL news and video dataset, NAVER shopping dataset	MRR, accuracy	- Prediction-based review simulation outperforms cluster-based review simulation - Overall, the model performs well, improving over MF by 5.53% and over hierarchical network [373] by 2.7%
	[346]	Dual CNNs + PMF	User behavior representation learning	Amazon	RMSE, MAE	- Results show that splitting the problem of song recommendation at artist and song levels improves the recommendation quality - Learning semantic features and tags embedding enhances results compared to using only text or audio
	[48]	CNN + attention mechanism	User behavior representation learning	Yelp	MAE	
	[245]	Multimodal deep CNN	Cold start	MSD	MAP	
	[171]	CNN + PMF for online CF	Data sparsity, cold start	Movielens, HetRec2011	RMSE	- The method outperforms baselines, showing that CNN and bias factors improve online CF prediction accuracy. It achieves lower average RMSEs of 1.008583, 0.956433, and 0.871783 on the ML-100K, ML-1M, and HetRec datasets, respectively, compared to the low-rank online CF model RMSEs of 1.08238, 1.0581, and 0.907233, and ConvMF model RMSEs of 1.0253, 0.995683, and 0.8983 - Although it converges more slowly than ConvMF, it achieves better results, providing the most accurate predictions at a faster convergence rate
	[77]	POLAR method uses an attention-based CNN	Personalized recommendation	AMiner, Patent Full-Text Databases, Related-Article Recommendation Dataset	nDCG	- POLAR performs better than traditional ranking models as well as interaction-based models - It highlights the significance of local and global attention weights in extracting useful terms
	[78]	Adds active learning to the POLAR framework using a Bayesian neural network	Personalized recommendation	AMiner, Patent Full-Text Databases, Related-Article Recommendation Dataset	nDCG	- POLAR outperforms baselines, including the most competitive TF-IDF, by 6.9%–13.2% on nDCG @3 and 3.3%–20.3% on nDCG @5. It achieves average nDCG improvements of 3.8%, 8.1%, and 6.4% across datasets - Even the model is effective compared to other advanced neural methods and has been successfully employed as a recommendation engine in AMiner platform

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
RNNs	[231]	Connects a CNN image feature extractor with an MLP	Personalized recommendation	NUS-WIDE, Flickr-PTR	F1-measure	Results suggest that personalized models outperform non-personalized ones in prediction quality
	[329]	CNN-based probabilistic model for POI recommendations	POI recommendation performance and cold start	New York City (NYC), Chicago (CHI) obtained from Instagram API	Precision, recall	- Particularly, the model with CNN supervised features captures more information than using low-level features, resulting in an approximate 2% improvement in performance - VPOI shows significant improvements over VBPR, PMF, and NMF across NYC and CHI datasets, with performance gains ranging from 4.8% to 19.5% in precision@10 and recall@10
	[22]	Residual CNN model	Recommendation accuracy	MovieLens	Hit ratio, nDCG	- The method is more robust to cold start users by integrating visual content, leading to improvements of 2.71% to 3.43% in precision@10 and recall@10 across both datasets - The study finds that the outer product-based interaction method outperforms concatenation and Hadamard product approaches, achieving Hit@10 and nDCG@10 values of 0.6401 and 0.3668, respectively
	[387]	Content-aware CNN approach	Cold start and improving item ranking	Jester	RMSE, MSE, MAE, precision, recall, F1-measure, nDCG	- Additionally, the CNN approaches converge slightly faster than the MLP approaches - Combining both rating and semantic similarities in the model yields the best results across metrics: RMSE of 1.08, MSE of 1.06, MAE of 0.84, precision of 63.74, recall of 66.33, F1 of 64.75, and nDCG@10 of 55.39 - However, the evaluation is limited to model cases and does not compare with other recommendation baselines
	[308]	RNN	User behavior representation learning, improving top-N recommendations	ALLEGRO e-commerce user actions, AVITO ad click prediction	Recall, MRR	The RNN representation approach improves by up to 34.1% in recall and 53.6% in MRR compared to the BPR method, and by 75.5% to 126.3% compared to the traditional Top-N recommendation method. However, its performance benefits from not using session modeling data
	[282]	Contextual GRU network	User behavior representation learning	YooChoose dataset from RecSys Challenge, Internal e-commerce dataset	Recall	The method significantly outperforms non-contextual models in recommending relevant non-historical products and predicting sold items, with notable improvements in Recall@10 for sale events, achieving up to +12% on the YooChoose dataset and +6% on the Internal dataset
	[342]	RNN model with a recurrent part and nonrecurrent part	Time heterogeneous feedback recommendation	MovieLens, Taobao, CiaoDVD, LastFM	F1-measure	The model exhibits superior performance by achieving leading F1@10 scores of 0.1049 (MovieLens), 0.0247 (Taobao), and 0.0280 (CiaoDVD), as well as top F1@20 scores of 0.1303 (MovieLens), 0.0260 (Taobao), and 0.0275 (CiaoDVD), consistently surpassing the nearest competitors, including KNN- and MF-based models in all test scenarios
	[341]	LSTM networks	Recommendation accuracy and cold start	IMDb, Netflix	RMSE	- The flexible LSTM-based architecture outperforms PMF, AutoRec, and Time SVD methods, achieving lower RMSE scores of 1.9703 on IMDb and 0.9224 on Netflix - Even in dense scenarios, RNN maintains a smaller size while achieving comparable accuracy
	[432]	PMF + LSTM attention network	Temporal rating prediction	Netflix, MovieLens	RMSE	This approach leverages both static and temporal features, resulting in superior performance compared to methods that focus solely on static or dynamic features. Consequently, it surpasses PMF, NCF, and the RNN from (C.-Y. [341, 346]), achieving lower RMSE of 0.9197 on IMDb and 0.8847 on Netflix
	[192]	Rating-RNN + Category-RNN	Prediction accuracy	MovieLens	SPS classification accuracy	- Rating-RNN and Category-RNN outperform the basic RNN, with Category-RNN reaching 79.4% and 80.8% accuracy at SPS@3 - The evaluation shows the RNN models' effectiveness but lacks comparison with recommendation baselines

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[124]	Classic GRU	Session-based recommendation	RecSys Challenge, OTT video service platform	Recall, MRR	- This approach has substantial gain over ItemKNN on both datasets, with the maximum accuracy gain of 20–30% with TOP1 loss. However, adding extra layers to GRU results in worse performance - This study utilizes two pair-wise ranking losses, BPR and Top1, which produce superior results and emphasize the instability of point-wise losses such as cross-entropy for session-based recommendation tasks
	[285]	LSTM temporal model	Temporal-based recommendation	Commercial news portal	Precision, AUC, MAP, MRR	- Learning user interests at different rates enhances the model's performance, achieving a higher P@1 of 0.245 compared to MF-based method (0.211) and DSSM (0.229) - However, focusing on past user interests and neglecting new user information may hinder its ability to perform well for cold start users
	[15]	GRUs into CF	Sparsity and cold start	CiteULike	Recall	- For cold start scenario, the GRU model without multitask learning performs 18.36% to 32.74% better than CTR and Embed models. With multitask learning, the model's performance increases to 19.30% to 34.22% - In warm start scenario, the model without multitask learning shows an improvement of 8.5%, while the performance increases to 15.8% with multitask learning
	[72]	Gated RNNs + user-specific GRUs	Personalized recommendation	MovieLens, LastFM	MRR, recall	On MovieLens, GRU-based models surpass item-based KNN with 12% higher MRR@20 and 23% higher Recall@20. On LastFM, they achieve a 157% increase in MRR@20 and 21% in Recall@20. The best GRU variants, attentional on MovieLens and rectified linear on LastFM, offer a 32–33% increase in MRR@20 and 28–30% in Recall@20
	[433]	Time-LSTM model	Cold start	LastFM, CiteULike	Recall, MRR	In LastFM, the Time-LSTM model improves Recall@10 by approximately 10.84% over BPR and 4.72% over Session-RNN, and MRR@10 improvement of 32.27% over BPR and 17.83% over Session-RNN
	[228]	BRNN model with Linked Open Data cloud (LOD)	Recommendation accuracy	MovieLens	F1-measure	- Similarly, in CiteULike, the model enhances Recall@10 by approximately 6.82% over BPR and 4.20% over Session-RNN and improves MRR@10 by 9.58% over BPR and 6.08% over Session-RNN - BRNN significantly improves performance over other deep learning approaches, achieving maximum increases of 0.024 in F1@5 and 0.014 in F1@10 compared to AE, whereas CNN offers no advantage - Additionally, the BRNNs and LOD-based features model outperforms all baselines on ML with F1@5 and F1@10 scores of 0.567 and 0.654, respectively
	[20]	Latent cross RNN model	User behavior representation learning	YouTube videos	Precision, MAP	The latent cross approach, which uses multiplicative representations in RNNs, shows significant improvements with a precision of 0.1621 over concatenated RNN models (0.1465) and plain RNN models (0.1345)
	[47]	RNN-based hierarchical structure	Temporal-based recommendation	MovieLens, Steam	Hit ratio	- Additionally, latent cross is calculated linearly and does not increase computational complexity - The hierarchical RNN achieves a high Hit@10 of 78%, outperforming other models like popularity-based, BPR, and RNN-based approaches like GRU4Rec [124] and emphasizing the importance of temporal properties in long-term sequences
	[417]	Hierarchical collaborative method integrates RNN with PMF	User behavior representation learning	CiteULike, Yahoo	Recall, MAP	The method improves R@10 by an average of 4.09% over CDL (best baseline) in the Citeulike and outperforms CDL in MAP by 3.6% and 4.2% on the Citeulike and Yahoo
	[33]	LDA topic modeling + LSTM	User behavior representation learning	Scopus, custom collected research paper data	Accuracy, RMSE	The proposed LSTM model learns static representations effectively, achieving the highest accuracy of 0.75 and the lowest RMSE of 0.248, outperforming models like CNN. However, it struggles with capturing changes in user intentions

Table 4 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[92]	Dynamic RNN network	Personalized recommendation	Google +	Precision, recall, F1-measure	• The proposed method outperforms others in both long and short modes, achieving F-score improvements of 26.9% and 75.3%, respectively • It also experiences a smaller F-score reduction (12.4%) when historical behavior data is limited compared to LFM and KNN, which are reduced by 44.4%, and 29.7%
Methods: {SVD singular value decomposition, PMF probabilistic matrix factorization, BPR Bayesian personalized ranking, VBPR visual content BPR, POI point of interest, VPOI visual content POI, LFM latent factor model, LDA latent Dirichlet allocation, eALS element-wise alternating least squares, KNN K-nearest neighbor, FM factorization machines, DNN deep neural networks, BRNN bidirectional recurrent neural networks, LSTM long short-term memory, GRU gated recurrent unit, CTR collaborative topic regression, CDL collaborative deep learning, RNA ribonucleic acid, CAAE collaborative adversarial autoencoder, DAE denoising autoencoder, CDAE collaborative denoising autoencoder, SDAE stacked denoising autoencoder, VAE variational autoencoder, NCF neural collaborative filtering, CA-NCF context-aware NCF, TF-IDF term frequency-inverse document frequency, API application programming interface} Evaluation Metrics: {RMSE root mean squared error, MSE mean squared error, MAE mean absolute error, nDCG normalized discounted cumulative gain, MAP mean average precision, MRR mean reciprocal rank, AUC area under curve}						

extracted lncRNA features and lncRNA disease feature to make disease prediction with MF. Another study, where [370] improved CDL by considering the differences in items. The authors added private bias to each item, allowing the network to learn item parameters more effectively. The model optimized each part in parallel to improve system scalability. Next, the authors utilized the SDAE network to learn dense latent features [298]. The main idea was to provide diverse recommendations by considering surface and latent similarities among users. Different SDAE layers were utilized to obtain these similarities, which were then used to rank top-N recommendations. This approach resulted in a diversity of recommendation lists without a significant loss in accuracy. Another paper introduced tag-aware recommender system by adopting the SDAE network to learn more abstract and representative latent features from varied tag information [435].

To improve CF predictions, a model utilized deep learning methodology that learns a shared feature space from heterogeneous data, such as item descriptions, product tags, and online purchase history [179]. The authors employed the SDAE network to extract latent features from non-sequential data and the RNN encoder–decoder to extract features from sequential data that capture user preferences and potential interest shifts over time. The paper proposed a deep clustering model for user-based CF that uses a stacked deep AE to learn user features from the rating matrix and then applies fuzzy clustering to these learned representations to create effective user clusters and improve recommendation accuracy [23]. Similarly, Zhao and team presented a deep CF-based recommendation model that used AE network to analyze user attributes and k-means for classification, enhancing recommendation speed by generating lists based on user classifications [407]. Another paper proposed a collaborative denoising autoencoder (CDAE), a single-hidden-layer model to learn correlations between a user's item preferences on a known preference set [354]. The preference set is represented with binary values (0 means user likes the item and 1 means user dislikes the item). The encoder phase of the network mapped input to a latent representation, obtained in the hidden layer using $h = \sigma(W_1 \tilde{x}_u + V_u + b_1)$. The standard DAE adds corrupted input to extract highly representational features. Similarly, the CDAE model generated corrupted user feedback ($\tilde{x}_u$) using a conditional Gaussian distribution $p(\tilde{x}_u|x_u)$ process. Then in the decoder phase, the model reconstructed the derived latent representations back to the input vector space. The representations computed at the final layer are given as $z = f(W_2 h_u + b_2)$. CDAE used stochastic gradient descent in backpropagation learning and minimized the reconstruction error using formula as follows:

$$\operatorname{argmin}_{w_1 w_2 V_u b_1 b_2} \frac{1}{N} \sum_{u=1}^N E_{p(\tilde{x}_u|x_u)} L(\tilde{x}_u|z) + R.\lambda \quad (14)$$

$L(\tilde{x}_u|z)$ denotes the loss function of the model. $E_{p(\tilde{x}_u|x_u)}$ represents the expectation of a random Gaussian distribution. R denotes the regularization term—squared L2 norm as used in paper. This model consistently gained success in addressing the top-N recommendation problem as claimed in the paper.

A variational AE for top-N recommendation (RecVAE) model adopted *Mult*-VAE [187] to construct the network that takes implicit user feedback as input [276]. The *Mult*-VAE ideology is like normal VAE, except that *Mult*-VAE obtains multinomial distribution while VAE follows Gaussian and Bernoulli distributions. With this, the equation during the encoder and decoder phases are given as $z_u \sim N(0, I); \pi(z_u), \operatorname{softmax}(f_\theta(h_z)); x_u \sim \operatorname{Mult}(n_u, \pi(z_u))$. Here z_u denotes the latent representation and $\pi(z_u)$ is a nonlinear transformation of z_u . f_θ is a neural network with parameters θ and softmax is the activation function. x_u is the input feedback history consisting of interactions n_u such as clicks and purchases. For the normal VAE, the evidence of lower bound also known as the objective function is defined as:

$$E_{q_\phi(z_u|x_u)} \log p_\theta(x_u|z_u) - \beta KL(q_\phi(z_u|x_u) || p(z_u)) \quad (15)$$

RecVAE introduced new novelties to Eq. (15) and expressed $p_\theta(x_u|z_u)$ as a multinomial distribution of encoder representation. It shifted to the denoising concept by adding corrupted signals $E_{q_\phi(\tilde{x}_u|x_u)}$. Then it introduced a composite prior distribution in the network. The composite prior

$$p(z_u|\emptyset_{old}, x_u) = \alpha N(z_u|0, I) + (1 - \alpha) q_{\phi_{old}}(z_u|x_u)$$

is a mixture of the standard Gaussian prior and the latent representation of the previous iteration, which improved training stability and performance. Followed by an inclusion of setting the hyperparameter $\beta = \beta(x_u)$ (independent of implicit feedback) for the Kullback–Leiber divergence (KL). These changes leveraged the recommendation system performances. As per the RecVAE model, Eq. (15) can be rewritten as:

$$E_{q_\phi(z_u|\tilde{x}_u)} E_{q_\phi(\tilde{x}_u|x_u)} [\log p_\theta(x_u|z_u) - \beta KL(q_\phi(z_u|\tilde{x}_u) || p(z_u|\emptyset_{old}, x_u))] \quad (16)$$

3.1.4 Multilayer perceptron (MLP)

MLPs are typical feedforward neural networks that map input data to the input layer, process it through one or more hidden layers, and classify the output in the output layer. These layers are composed of neurons. Each neuron in the

hidden layer performs a weighted linear summation using input from the previous layer, followed by an affine transformation on this weighted sum using an activation function. The flexibility of MLPs allows them to improve recommendations by increasing the model's capacity with more data.

MLP approaches for rating prediction. A research study employed an attention and MLP module to extract implicit information and user preferences for rating recommendations [255]. In another research, [37] utilized MLP to learn user and item features from all meta-paths and then introduced a hierarchical attention layer to combine this learned information, described as user preferences and item properties. In the end, the method applied factorization machine (FM) to predict ratings. [88] introduced a novel rating prediction model that leverages MLP network to endow high level of nonlinearities for representing each user–product preference interaction. Following, the model employed a CNN network with multiple filters to capture local semantics of text features. These representations are then input into an MF projection layer to generate rating scores. Next, the iMusic system proposed a representation learning-based approach to recommend Indian classical music by analyzing session-sensitive features and note structures with DNNs and a context-aware k-means clustering algorithm for data filtering [264]. All these studies employed MLP for user–item representation learning, while other studies use MLP for the final prediction. For example, some methods designed compound architectures using different deep neural models, such as AE, CNN, or deep attention mechanisms to extract user–item latent features and then utilized MLP to find interactions between users and items for rating prediction task [325]. Based on this concept, an existing work employed an attention mechanism to learn user, item and social latent spaces from graph data, which were then passed to the MLP layer for prediction [84]. Another study proposed a unified framework to support recommendations in a cross-domain system and put forward a hybrid content collaborative deep framework to propagate content embeddings into neural-based CF model [185].

MLP approaches for top-N ranking. A paper combined the benefits of memorization (frequent co-occurrence of features) and generalization (transitivity of correlation and exploration of new feature combinations that have rarely or never occurred in the past) for recommender systems [45]. The next article introduced a vanilla MLP for recommending YouTube videos with the classic two-stage retrieval dichotomy [53]. In the first stage, the candidate generation network collected events from the user's YouTube activity history as input and retrieved a small subset (hundreds) of videos from a large corpus by generalizing high-order information. The second ranking stage accomplished this task by assigning a score to each video according to the desired objective function. A well-known

neural collaborative filtering (NCF) model interpreted MLP as MF to perform the inner product of user–item latent features [119]. Figure 3 illustrates the general concept of combining linear and nonlinear learning from latent features. Based on this concept, the NCF model defined two instantiations for learning linearities and nonlinearities from the user latent vectors (p_u) and item latent vectors (q_v) using generalized matrix factorization (GMF) and MLP.

In the GMF part, the model formulates latent vectors as $\alpha^{(g)} = (p_u^{(g)} \odot q_i^{(g)})$. In the MLP network, these latent vectors serve as inputs to the initial layer, computed as $x_1 = \begin{bmatrix} p_u^{(m)} \\ q_i^{(m)} \end{bmatrix}$. These values enable computations at hidden layers and the final formulation in at the MLP network is expressed as

$$\alpha^{(m)} = f_l(w_l^T h_{l-1}(\dots f_1(w_1^T \begin{bmatrix} p_u^{(m)} \\ q_i^{(m)} \end{bmatrix} + b_1)\dots) + b_l)$$

. The model fuses the output from GMF and MLP at the neural MF (NeuMF) layer to generate projected scores as $\hat{y}_{ui} = \sigma(W^T \begin{bmatrix} \alpha^{(g)} \\ \alpha^{(m)} \end{bmatrix})$, which are then used for items ranking.

NCF has opened new perspectives for researchers, leading to various modified versions of the same concept. Joint neural collaborative filtering (J-NCF) retrieved user characteristics and item features by mapping high-dimensional to low-dimensional features using MLP networks [40, 41]. This model combines these features either by element-wise multiplication or concatenation method and input them into another MLP to achieve a deep collaborative process. J-NCF also introduced a hybrid loss function that combines point-wise loss (typically used for rating prediction) and pair-wise loss (generally used in top-N tasks).

A new dimension in which the contextual information is incorporated into the NCF recommender [11]. The authors incorporated the context features vector into the item vector using a pre-filtering technique. The model redefined the NCF-based prediction at the output layer as $\hat{y}_{uic} = f_l(f_{l-1}(\dots f_1(P^T U_u, Q^T V_{ic})\dots) + b_l)$. Where, P^T and Q^T are users and artificial items latent matrices. Then U_u highlights feature vector of user u and V_{ic} denotes feature vector of combined item i and context c . The seamless integration of context-aware features into NCF gained a competitive advantage by reducing computational complexity.

Similarly, the potential of incorporating context into NCF for social recommender systems is demonstrated in the work by [10]. They aimed to improve contextual incorporation to achieve more accurate personalized

predictions efficiently. Another paper proposed a new ConvNCF model by integrating the interaction mapping function into the NCF [118]. The interaction function facilitated CNN in learning high-order correlations among user and item embeddings.

[68] proposed a novel model based on the vanilla MLP to understand the complex matching function while efficiently learning low-rank relations between users and items. Another approach involved a three-stage search space—behavior modeling [426], interaction exploration [45] and MLP investigation [53] to improve the accuracy and personalization of given recommender tasks [410]. Further, a study adopted a hybrid approach for e-learning platforms and developed a novel DNN that uses heterogeneous features and integrates k-means clustering to generate top-N course recommendations [265].

3.1.5 Convolutional neural network (CNN)

CNNs are specialized neural networks that have become prepotent in various computer vision tasks, and their influence is growing across a variety of other domains such as NLP, time series analysis, and visual recommendations. The architecture of CNNs resembles the animal visual cortex and is designed to automatically learn nonlinear hierarchies of features. A typical CNN has three layers: the convolutional layer, which extracts important features from input data; the pooling layer, which reduces the dimensionality of these features while keeping the most important information; and the fully connected layer, which maps these features to the final recommendations. This structure enables CNNs to handle complex patterns and relationships in data, enhancing the accuracy of recommendations.

CNN approaches for rating prediction. In recent years, CNN-based recommendation methods have gained attention to capture meaningful nonlinear features for modeling users (or items) [155]. The study proposed DeepCoNN model with two CNN networks: one for modeling user latent representations through reviews written by the users and the other for modeling item latent representations through reviews written for the items [419]. Another method introduced latent variables to capture user biases and item preferences, adopting embeddings to regularize the MF [346]. Building upon DeepCoNN, a method added dual local and global attention mechanisms to the CNNs to endow the recommender systems with interpretability regarding the reviews for modeling users and items [272]. A different research considered merge-CNN with an attention mechanism for rating predictions by extracting concise attention reviews with crucial words and sentences [48]. An article combined text and audio in a multimodal

approach to attain better performance in music recommendation problems [245].

Next research article proposed two variants of the model called deep bias probabilistic matrix factorization (DBPMF) and deep constrain bias PMF (DCBPMF) with additional constraints to improve the system performance [171]. Both variants featured CNNs to extract latent user and item features and were added to the PMF technique to classify predictions as biases. With user bias, the model learned the users' rating patterns and with item bias it tracked item popularity. Typically, the CNN-based recommendation architecture consisted of four layers: embedding, convolution, pooling, and output, as illustrated in Fig. 4. The embedding layer of DBPMF employed one-hot encoding method to encode user and item attributes into E_{user} (user matrix) and E_{item} (item matrix). The convolution layer then extracted latent user ($c_{user_i}^j$) and item ($c_{item_i}^j$) features using the formulas shown in (17) and (18), respectively. Next, the pooling layer reduced the dimensionality of extracted features to a fixed vector length as shown in (19) and (20). Finally, the output layer transformed the fixed vectors from pooling layer to k -dimensional space as given by (21) and (22). The model integrated CNN extracted features with CF algorithm to make rating prediction as $\hat{r}^{(i,j)} = \sum_{i=1}^m \sum_{j=1}^n (U_i^T V_j + f_i + g_j)$. Here, U_i represents user latent vectors and V_j represents item latent vectors. f_i is the user bias, while g_j is the item bias.

$$c_{user_i}^j = f(W^j * E_{user}(:, i : (i + ws - 1)) + b^j) \quad (17)$$

$$c_{item_i}^j = f(W^j * E_{item}(:, i : (i + ws - 1)) + b^j) \quad (18)$$

$$P_{user} = [\max(c_{user}^1), \max(c_{user}^2), \max(c_{user}^3), \dots, \max(c_{user}^{n_{convolution}})] \quad (19)$$

$$P_{item} = [\max(c_{item}^1), \max(c_{item}^2), \max(c_{item}^3), \dots, \max(c_{item}^{n_{convolution}})] \quad (20)$$

$$U = \tanh(W_1 \{ \tanh(W_2 P_{user} + b_1) \} + b_2) \quad (21)$$

$$V = \tanh(W_3 \{ \tanh(W_4 P_{item} + b_3) \} + b_4) \quad (22)$$

Like DBPMF, another model ConvMF predicts users' ratings using CNN and PMF [157]. However, the key distinction lies in ConvMF considering only item latent features, while DBPMF extracts both user and item latent features. An advanced work exploited Gaussian noise (number of ratings given to items) with document information and showed robustness in the accurate modeling of item factors [158].

CNN approaches for top-N ranking. A content-based approach focused on leveraging contents of the image, such as colors, shapes, visual aspects to make users' tagging decisions in social media [231]. This approach employed

CNN to find interactions between users and visual features. The resulting interaction data is then fed into a MLP network to compute tag scores. In their later study, the authors incorporated an adaptive factorization layer to predict the top-N image tags from the image tagging history and CNN extracted features [232]. User modeling involves creating user profiles for personalized recommendations by learning from user interactions with items. TERACon is a novel continual user representation learning method that retains learning capability throughout training and captures task relationships to improve training efficiency [160]. It uses a temporal convolutional network as its backbone for models' universal user representation learning. One research study demonstrated the potential to enhance POI recommendation performance using visual features of images [329]. Another research effort demonstrated the effectiveness of visual information for restaurant recommendations [51]. The visual features including restaurant attributes and user preference, extracted through CNN, were integrated into content-based filtering and CF to prove the effectiveness of visual information. A recent method implemented CNN network with CF and outer product-based function [22]. This model takes a sparse rating matrix as its input and, through an outer product function with a residual connection, learns high-order user-item correlations, subsequently employing CNN for further processing.

The model POLAR++ (PersOnaLized Article Recommendation Framework++) introduced a Bayesian neural network-based system to capture the uncertainty of user choice while actively learning a new user's preference through user interactions [78]. This active learning method was well exemplified in an earlier model, POLAR [77], which is an attention-based CNN model designed for ranking tasks. A different research primarily intended to improve the item cold start problem and item ranking by first conducting the required preprocessing, then using CNN to extract item vector representations, and finally estimating the content similarity of the items to use for predictions [387].

3.1.6 Recurrent neural network (RNN)

RNNs are among the most convictive deep learning algorithms for solving sequential and temporal issues including speech recognition and time series analysis. Because of their architecture, which uses internal memory to evaluate the next input in sequences, RNNs have a significant impact in the field of recommendation. Traditional feed-forward neural networks produce outputs that are independent of one another. RNNs, on the other hand, can infer new outputs from the information of past outputs. RNNs use a consistent input size to ensure a smooth transition

from one procedure (state) to the next. Each processing unit in the network has a time-varying real-valued activation and weight that can be adjusted, making RNNs well suited for modeling user behavior over time. Long short-term memory (LSTM), gated recurrent units (GRUs), and bidirectional RNNs are some variants of RNN architectures.

RNN approaches for rating prediction. RNNs are suitable for modeling users' temporal interests. For example, in a study by [341], two LSTM networks were employed to extract users and items temporal information, with an objective of predicting ratings using the temporal information recorded by the LSTMs. The model utilized dynamic and stationary states to make predictions as $\hat{r}_{ij}^{(t)} = f(u_i^{(t)}, v_j^{(t)}, u_i, v_j)$. Here, $u_i^{(t)} = W_u u_i^{(t)} + b_u$ and $v_j^{(t)} = W_v v_j^{(t)} + b_v$ represents users and items dynamic states over time index t . And u_i and v_j represents static latent attributes for user i and movie (item) j , respectively. The prediction process repeated by taking latest observations as input, updating the states, and predicting future trajectories in the unified manner. Another study leveraged the PMF model to learn both static and temporal user interests by feeding them to RNN at each time step [432]. Differently, a research developed a unique Mixing-RNN model to captures time and context change for item recommendation [192]. Unlike standard RNNs, the authors included a forget gate and an input gate in the model, with the forget gate deciding what information to keep and the input gate providing rating data to the model.

Subsequent research improved the GRU network to better fit the task by introducing session-parallel mini-batches, mini-batch-based output sampling, and a ranking loss function [124]. In a later research, [210] applied bidirectional RNN-based attention to learn feature representations from user-generated reviews. The proposed model comprised of four components. Word embeddings for mapping words to vectors, sequence encoding with bidirectional GRU for context annotations, topical attention to capture topic-related information, and feature projection for obtaining user and item latent features through non-linear transformations. For recommendations, they applied PMF on the extracted user features $\tilde{U}$ and item features $\tilde{V}$ that finds conditional distribution over the observed ratings as in (23).

$$L = \sum_{i=1}^m \sum_{j=1}^n I_{ij} (r^{(i,j)} - \hat{r}^{(i,j)})^2 + \lambda_u \sum_{i=1}^m \|U_i - \tilde{U}\|_F^2 + \lambda_v \sum_{j=1}^n \|V_j - \tilde{V}\|_F^2 \quad (23)$$

RNN approaches for top-N ranking. Although user feedback is valuable for the recommendations, considering

time stamps will improve recommendation results. In this context, a study focused on learning heterogeneous time feedback in their designed RNN, which consists of a nonrecurrent component representing the underlying user preference and a recurrent part that retains the impact of past feedback. [342]. The nonrecurrent part computes $R^{(n)} = \sigma(Du)$, where u is user vector and D represents weight matrix. And the recurrent part computes $R(t) = \sigma(I \cdot i(t) + W \cdot a(t) + X \cdot R(t-1))$, where $i(t)$, $a(t)$, and $R(t-1)$ denote item vector, feedback activity vector and previous hidden layer state, respectively. I , W , and X represent associated weight matrices. The model combines the results of the nonrecurrent and recurrent parts in the output layer as $O(t) = f(YR(t) + ZR^{(n)})$ to produce top product recommendations.

The research introduces a model for learning user preferences and predicting user intentions by combining latent Dirichlet allocation (LDA) and Word2Vec to analyze paper topics and contextual word relationships [33]. It extracts user preferences from click data and utilizes LSTM network with temporal context to model dynamic user behavior over time. Similarly, another research incorporated the temporal properties of user history in their RNN model [47]. The authors added short-term (s) and long-term (l) elements to show the influence of sequence length. The short-term layer processes the short-term sequences, which is a common consideration in every RNN structure as in (24). The long-term layer memorizes previous short sequences and uses them for long-term consideration as shown in (25).

$$h_s^{(t)} = f(X_s x^{(t)} + W_s h_s^{(t-1)} + W_l h_l^{(t-1)} + b_s) \quad (24)$$

$$h_l^{(t)} = f(X_l h_s^{(t)} + W_l h_l^{(t-1)} + b_l) \quad (25)$$

Another recommendation study utilized deep network Gated RNN to generate a personalized next item recommendation list [72]. The system created temporal dynamics of item consumption sequences and then integrated user-based information into the model. In addition, it derived an attention-based user integration method to exclude outliers. Recently, a model called Time-LSTM demonstrated the importance of time interval information for dynamic user preference modeling [433]. Other study applied RNNs to learn users' dynamic interests and items over time to leverage the recommendations [285]. [228] adopted a content-based bidirectional RNN to discover items, textual descriptions, and features from the Linked Open Data. The average mean of these learned representations is concatenated where logistic regression exploits these vectors to find top-N items. Another representation learning study utilized an RNN-GRU network to model context-dependent item representations and RNN dynamics [282]. By

adapting item representations to context and making RNN transitions context-dependent, the method captured how user behavior and item similarities change with context. Moreover, a research article explored the use of contextual data in an RNN-based recommender system, incorporating features such as request and watch time, device type, and page, and applied it at YouTube [20]. It critiques the modeling of context as direct features, especially in MLP networks, and introduces “latent cross,” a technique that embeds contextual data and combines it with the RNN’s hidden states. The latent cross method demonstrated performance improvements across top-N recommendation settings. [417] constructed a hierarchical RNN to extract semantic meaning from textual content by embedding words in a latent semantic space [417]. Differently another research used multitask learning concept in deep RNN network to extract text semantics to produce tag prediction [15], while [92] employed RNNs to uncover the semantic embedding of videos from textual features.

3.2 Modern deep learning models

The emergence of modern deep learning models such as attention models, LLMs, and GNNs has revolutionized the fields of AI and NLP. Additionally, the rapid progress of modern deep learning models in recommender systems has substantially advanced the field. This section provides a literature analysis of modern deep learning models for recommendation tasks and presents detailed summaries of these approaches in Table 5.

3.2.1 Deep attention network (DAN).

DANs leverage attention mechanisms to focus on the most relevant parts of input data, making them highly effective for tasks including sequential or spatial data. In recommender systems, attention mechanisms allow these networks to dynamically prioritize important user interactions and contextual information, giving more attention to relevant items and less to irrelevant ones. By assigning different weights to different items, the attention mechanism can concentrate on contextual items, making user preferences more personalized. Additionally, DANs can extract rich and heterogeneous user behavioral patterns to accurately predict future user intentions.

DAN approaches for rating prediction. [255] leveraged the concept of dual learning approach as depicted in Fig. 5 and the self-attention mechanism to create a recommendation model that enhances the capture of preference information for ratings. The framework includes user self-attention for user influence as $a_u^{(l)} = f(W_u^{(l-1)} a_u^{(l-1)} + b_u^{(l-1)})$, item self-attention for item-

specific features $a_i^{(l)} = f(W_i^{(l-1)} a_i^{(l-1)} + b_i^{(l-1)})$, user-item self-attention for user attention in different items $a_{(u,i)}^{(l)} = f(W_{(u,i)}^{(l-1)} a_{(u,i)}^{(l-1)} + b_{(u,i)}^{(l-1)})$, and item-user self-attention for item attention across different users $a_{(i,u)}^{(l)} = f(W_{(i,u)}^{(l-1)} a_{(i,u)}^{(l-1)} + b_{(i,u)}^{(l-1)})$. The output of each attention module is passed to the MLP layer to obtain vector with user feature as $mlp_{a_u} = (a_u a_u)_{mlp}^{(l)} = f(W_{(mlp)}^{(l-1)} (a_u a_u)_{mlp}^{(l-1)} + b_{mlp}^{(l-1)})$. Similarly, the model obtained other features’ vectors as mlp_{a_i} , $mlp_{a_{ui}}$ and $mlp_{a_{iu}}$. All these vectors are then pooled and reduced to one-dimensional output in the form of pooling vectors as $p(mlp_{a_u})$, $p(mlp_{a_i})$, $p(mlp_{a_{ui}})$, $p(mlp_{a_{iu}})$. Additionally, two more pooling vectors $p(mlp_{ui})$ obtained through dot product of user and item features; and $p(mlp_{ui}^a)$ obtained through dot product of a_u and a_i . These pooling vectors are combined into a single layer for final ratings prediction: $\hat{r}_{ui} = p(mlp_{ui}) + p(mlp_{ui}^a) + p(mlp_{a_u}) + p(mlp_{a_i}) + p(mlp_{a_{ui}}) + p(mlp_{a_{iu}}) + b_u + b_i + b$.

To enhance representation learning in recommender systems, a study introduced a sentiment-aware deep recommender system utilizing a neural attention network to integrate product aspects and user sentiments [57]. This approach employed a semi-supervised topic model, LSTM encoder, and co-attention mechanism to enhance prediction performance. Later, the authors developed a multilevel attentive model combining review-based and interaction-based latent factors, employing a Bi-LSTM sequence encoder and a co-attention network to capture contextual and semantic details, along with an attentive FM to further improve both prediction and ranking performances [55]. Next, CARL representation learning model estimates personalized ratings by integrating two feature learning components: review-based and interaction-based [350, 351]. It uses convolution and attention mechanisms to extract semantic information from user reviews and combines this with interaction data to create a joint representation for rating prediction. Clustering is a widely used technique for managing large-scale datasets. However, traditional clustering algorithms often fail to uncover deep-level relationships between clients [324]. To address this, the authors introduced a personalized e-commerce recommendation system that integrates KNN clustering with LSTM and attention mechanisms, enabling the system to capture deep relationships and sequence features in user data for more accurate recommendations.

[287] incorporated graph learning into the proposed multiattention network. The network employed a dual-branch residual graph attention module to extract features from graphs, and multiscale latent matrices to mitigate the issue of dimension selection. Additionally, a fusion graph

attention module captured personalized preferences from interaction graph. The complete framework was optimized using a geometric factorized regularization loss. Another study in recommendation presented a general self-attention framework for personalized and explainable recommendation [36]. [43] proposed a time-aware model that provides adaptive recommendation explanations by integrating GRU with an attention mechanism. The authors believed that GRU is effective in modeling users' dynamic preferences, while the attention mechanism facilitates explainable recommendations. Another attention-based personalized recommendation developed by [273]. The attention layer aimed to select informative keywords contributing to users' preferences for e-commerce websites. Some researchers leveraged the attention mechanism to extract valuable item reviews for explaining the rating prediction process [34]. Next, a research paper proposed an attention-based deep learning method to solve the data sparsity and cold start problems [123]. The approach combined a language model and deep AE to represent and reduce the dimensionality of contextual information. This information was used as regularization terms in the MF framework to form an objective function. An iterative algorithm was developed to solve this function, facilitation the prediction of unknown rating values.

DAN approaches for top-N ranking. A co-attention approach presented powerful interaction functions by combining meta-path learning with user and item representations [130]. A meta-path is a relation sequence that connects object pairs in a heterogeneous graph network. Each instance p from the meta-path is embedded into the embedding layer of the CNN as $E(p) = \text{CNN}(X(p); \theta)$, where θ is CNN parameters, and $X(p)$ denotes the embedding matrix. Next, the max-pooling layer selects N instances of p embeddings to capture important features as $M(p) = \text{max-pooling}(\{E(p)\}_{p=1}^N)$. Additionally, the method applies average pooling to aggregate the meta-path. Further, it adopts a two-layer attention architecture as in (26, 27, 28 and 29): one for the meta-path context $M(p)$ and another for users U_u and items V_i to improve the interactions.

$$\alpha_{u,i,p}^1 = f(W_u^1 U_u + W_i^1 V_i + W_p^1 M(p) + b^1) \quad (26)$$

$$\alpha_{u,i,p}^2 = f(W^{2T} \alpha_{u,i,p}^1 + b^2) \quad (27)$$

$$\alpha_u = f(W_u U_u + W_{u(i)} C_{u(i)} + b_u) \quad (28)$$

$$\alpha_i = f(W_i V_i + W_{u(i)} C_{u(i)} + b_i) \quad (29)$$

The final meta-path attention obtained score using $\alpha_{u,i,p} = \frac{\exp(\alpha_{u,i,p}^2)}{\sum_{p' \in mp(i)} \exp(\alpha_{u,i,p'}^2)}$. This interpreted in aggregation

form $C_{u(i)} = \sum_{p \in mp(i)} \alpha_{u,i,p} M(p)$ to learn new meta-path embedding for each user interaction $u(i)$. Here, $mp(i)$ is the set of the considered meta-path for the current interaction. Similarly final representations of user and item are computed using element-wise product as $\tilde{U}_u = \alpha_u \odot U_u$ and $\tilde{V}_u = \alpha_i \odot V_i$. At last, all the three embeddings are concatenated as $\tilde{X}_{u,i} = \tilde{U}_u \oplus C_{u(i)} \oplus \tilde{V}_i$ to feed into MLP for modeling complicated interactions $\hat{R}_{ui} = \text{MLP}(\tilde{X}_{u,i})$. This co-attention model showed effectiveness in alleviating cold start problem and provided interpretable recommendation results.

MARS: Memory attention-aware recommender system, a CNN-based attention model presented by [418]. MARS utilizes a parallel CNNs architecture that takes text reviews as input and extract visual features using squared filters. The system adopts a factorization method that learns these visual properties to make top-N prediction. Another method conducted explainable recommendation by reasoning over knowledge graph embeddings that establishes explanation routes between user and item to generate knowledge-enhanced explanations [7]. Based on sequential recommendation, a study employed an attention network to differentiate the importance of items in long- and short-term user preferences [367]. The methodology comprises three blocks: an embedding block to transform raw interactions into low-dimensional spaces, an attention block to weigh items based on their relevance, and a neural network to combine these weights with user embeddings, generating a mixture of long- and short-term user preferences for personalized recommendations. To handle the drifting interests of users, a research article introduced an MLP-based session recommendation algorithm that prioritizes short-term memory and uses an attention mechanism to balance the influence of users' long-term and current interests [180]. This approach is then combined with the clustering analysis of a user tagging system to generate personalized recommendations.

Different from the conventional methods, [234, 355], and [369] studied user profile learning from multiple interests. For example, [234] introduced the deep user perception network, which uses RNNs, attention mechanisms, and multitask learning to improve user representations based on behavior sequences. The proposed attention layer integrates content, behavior, and temporal information for better user modeling in e-commerce tasks. Another research by [355] integrated diverse user information, including social graphs along with attributes and behaviors into a pre-training recommendation framework. It introduced two user-aware pre-training tasks: user attribute prediction to inject attribute knowledge from users' representations and social relation detection to make the representations between socially connected users or detect

social relations for different users. These tasks are versatile and can be easily applied in different recommendation scenarios. Further, [369] developed a user tag profiling model that used multiple user information fields and a multihead attention mechanism to combine key attributes. It also featured an FM-based cross-layer and a joint loss function to improve performance on user tag preferences by learning from multiple tags simultaneously.

Next, the paper aimed to improve recommendation interpretability and introduced a novel deep attentional aggregative interaction network to effectively learn high-order cross-features [115]. It utilized a cross-attention unit to learn sum combinations of features and a compress layer to filter relevant cross-feature combinations. The approach was then combined with a classic DNN to help model high-order features. A recent study applied the graph-based attention mechanism to improve preference mining, and a relation-aware graph convolutional neural network (GCN) to capture deep multirelation features for improving recommendation performance [93]. Another study deployed a knowledge-aware attention mechanism that explicitly integrated the varying weights assigned by users to neighboring entities in the knowledge graph for personalized recommendations [80]. Differently, research by [35] focused on a multilevel attention mechanism in the multimedia recommendation. The authors introduced the item- and component-level attention model to assign attentive weights for inferring user preferences encoded in the implicit user feedback. Following research, the authors

claimed the first real-time attention-based look-alike model for recommender systems [204]. This similarity-based approach featured the attention merge layer for user representation learning and modeled global and local representations of seeds using global and local attention units to optimize seeds representation in look-alike learning. Additionally, the method used asynchronous online seeds clustering to reduce seed scale and minimize the time complexity of online prediction. Later, a research study implemented hierarchical attention mask mechanism to solve the cold start issues for users and items by randomly masking some nodes in the item domain and in the social domain during the training process [411].

3.2.2 Large language model (LLM)

LLMs are advanced neural networks trained on vast amounts of text data to comprehend and generate human language. LLMs offer innovative approaches to enhancing recommendation systems by leveraging their extensive understanding of natural language, context, and user-generated content. Their ability to understand context and nuances allows LLMs to accurately derive user preferences. For example, LLMs can craft personalized recommendations by detecting subtle relationships between elements in a sequence, leading to more precise and relevant suggestions for users.

LLM approaches for rating predictions. The paper addressed the limitation of LLMs in understanding collaborative information by combining traditional recommendation systems with LLMs through an adaptive merging module [211]. This approach utilized the strengths of both systems, employing diverse prompt ensemble techniques and averaging multiple answers from LLMs to enhance accuracy. Additionally, the method leveraged LLMs to supply valuable side information, further improving the accuracy and efficacy of the recommendation system. The authors extended their ideology of augmenting traditional and LLM-based models for both rating prediction and ranking tasks [212]. They leveraged LLMs for data augmentation to address data sparsity and long-tail issues in conventional models, while traditional models provided prompt augmentation for LLMs. Their approach effectively combined predictions from both models through an adaptive module to refine recommendations. Differently [338] leveraged the concept of multiagent collaboration by incorporating specialized agents such as a Manager, User/Item Analyst, Reflector, Searcher, and Task Interpreter to collaboratively tackle recommendation tasks. The model was applied to various recommendation tasks including rating prediction, sequential recommendation, conversational recommendation, and explanation generation. To address the explainability issue in traditional

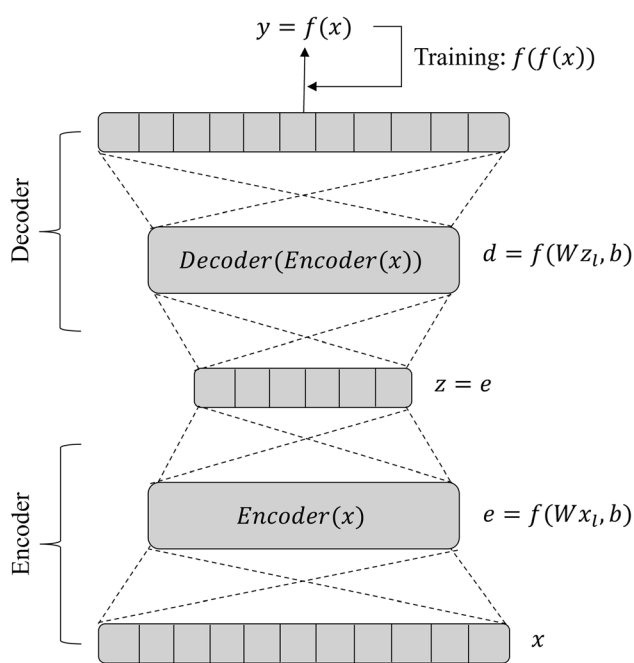


Fig. 2 Deep AE recommendation network

recommender systems, the study introduced Chat Generative Pre-Trained Transformers (ChatGPT) augmented recommender system called as Chat-Rec, a novel approach that utilized LLMs for conversational recommender systems [96]. Chat-Rec relied solely on in-context learning without requiring training, enabling the system to learn and update users' preferences during conversations and facilitate cross-domain product recommendations.

Meng and Juan implemented the R-Transformer financial product recommendation algorithm to generate state vectors for users and financial products from historical behavior sequences [186]. They calculated the inner product of user and financial product states to obtain rating scores. Another study on the rating prediction task processed the transformer encoder separately for item features and user information, combining the two outputs in a final transformer decoder [149]. The attention mechanism allowed for predicting the rating of a new item based on the features of the previous vector item, while attention to user features helped avoid scenarios where a customer never rated any items. Next a research article explored the potential of incorporating LLM technology, particularly ChatGPT, into book resource understanding and recommendation [183]. It introduced a ChatGPT-like book recommendation system (BookGPT) that handled tasks such as book rating recommendation, user rating recommendation, and book summary recommendation, demonstrating the viability of LLM technology in these areas. Furthermore, [197] used ChatGPT and developed task-specific descriptions for various recommendation tasks, such as rating prediction, top-N recommendation, and explanation generation to conduct in-context learning based on the relevant input–output examples for each recommendation task.

With the goal of improving recommendation explainability, [172] proposed a personalized Transformer and utilized IDs to predict words in target explanations, giving IDs linguistic meanings and enabling personalized learning.

LLM approaches for top-N ranking. Building on the BERT4Rec model [294], the study employed a deep bidirectional self-attention model (BERT—Bidirectional Encoder Representations from Transformers) to learn contextual relationships using user's interaction history with songs and their listen counts for music recommendation systems [366]. Using LLM directly to handle semantic information in recommendation scenarios proved to be unreliable and less effective [213]. Therefore, the study [213] proposed the knowledge-enhanced LLMRec (KELLMRec) framework, which incorporates external knowledge to enhance the accuracy and reliability of LLM outputs. It also used a contrastive learning module to leverage additional external knowledge in recommendation

training. Next, a pioneering research study employed LLMs for graph augmentation in recommendation systems by enriching user–item interactions, item node attributes, and user node profiles [340]. It addressed the scarcity of implicit feedback signals by enabling LLMs to analyze user–item interaction patterns explicitly and improved recommendation quality through a denoised augmentation mechanism. Another article featured the LLM-based GANs approach for job recommendation that combines users' explicit attributes and inferred implicit characteristics to improve resume quality [76]. To tackle the few-shot problem and enhance recommendations, the approach aligned low-quality resumes with high-quality generated ones using GANs refining resume representations for more accurate recommendations.

The paper [159] presented an LLM-based recommender system that excels in both cold and warm scenarios. The main concept was to allow an LLM to leverage collaborative knowledge from a pre-trained state-of-the-art CF-RecSys by aligning item embeddings of the CF-RecSys with the LLM's token space. This alignment transferred collaborative knowledge to the LLM, enabling it to understand and utilize the knowledge for downstream recommendation tasks. Furthermore, a study proposed an intention-aware re-ranking model that used an intention discovering module to mine latent user intentions from text reviews and employed a dynamic transformer encoder to capture inter-item relationships based on these intentions [190]. It also applied an individual attention network over the self-attention layer to efficiently perform user-specific predictions. Further, a research study argued that the text description generated from LLMs could replace traditionally obtained descriptions to achieve competitive results and introduced a two-step approach to enhance recommendation systems [2]. First, it generated item descriptions using the Alpaca-Lora LLM with few-shot prompting for accuracy, and second, it trained the recommendation model by combining item ID embeddings with BERT embeddings from these descriptions, which were processed through a GRU layer to predict the next items. Recently, a study introduced a novel ID representation paradigm that integrated pre-trained ID embeddings learned from user–item interactions with LLMs for improved recommendations [383]. It combined ID embeddings with a novel alignment module and an efficient tuning method tailored for better performance across various LLM architectures and ID-based methods. In a different research, the authors conducted extensive experiments to review the use of LLMs, particularly ChatGPT-based systems, in recommender Systems [65]. They focused on the differences in behavior between generative models and traditional CF methods in movie recommendations, examining how specific system roles and prompt design strategies impact recommendation

quality in terms of accuracy, fairness, diversity, stability, genre dominance, and temporal freshness. In their subsequent research [66], the authors assessed the fairness of RecLLMs by assessing how recommendations vary with the inclusion of sensitive attributes like gender and age. They also explored how different strategies for constructing user profiles impact recommendation alignment, highlighting bias mechanisms within RecLLM. The recommendation study from [60] uncovered the capabilities and limitations of LLMs in recommendations and aimed to enhance ChatGPT's recommendation abilities by aligning it with traditional ranking methods such as point-wise, pair-wise, and list-wise ranking. Similarly, [70] explored the potential of LLMs in recommender systems to integrate LLMs for conversational AI and leverage LLMs at the lower level to improve LLM-based recommenders' capabilities.

The work by [361] explored how LLMs can be used as recommenders through the prompting LLMs for recommendation (ProLLM4Rec) framework, providing a systematic analysis on leveraging LLMs for recommendation. It assessed the impact of natural language prompts and factors such as model architecture, tuning strategies, and parameter scale on recommendation quality, and proposed a general framework to summarize existing work on using LLMs as foundation models for recommendation across various scenarios and tasks. Moreover, a research study examined the use of open- and closed-source LLMs to improve content-based recommendation [199]. Open-source LLMs were used as content encoders to enhance content representation, while closed-source LLMs utilized prompting techniques to enrich training data.

Inspired by the success of the Transformer model in machine translation, researchers proposed a new self-attention-based sequential recommendation model called SASRec [150]. This model effectively addressed sequential recommendation problems by drawing context from past actions and framing predictions with a small number of actions. SASRec modeled the entire user sequence without recurrent or convolutional operations and adaptively assigned weights to previous items for prediction. Additionally, the core self-attention block of SASRec enabled parallel acceleration, making it significantly faster than CNN/RNN-based alternatives. During training, the model transformed input user sequence s into fixed-length sequences o through truncation or padding of items. It used binary cross-entropy loss as the objective function (30) to guide learning.

$$\sum_{s(u) \in s} \sum_{t \in [1, 2, \dots, n]} \left[\log(\sigma(r_o, t)) + \sum_{j \notin s(u)} \log(1 - \sigma(r_j, t)) \right] \quad (30)$$

The authors [348, 349] argued that the SASRec model is a standard Transformer that is inherently impersonal. Inspired by SASRec's state-of-the-art performance, they enhanced interpretability and emphasized recent engagement patterns for each user. They employed a new regularization technique called stochastic shared embeddings [350, 351], which played a crucial role in the success of personalization. Moreover, a variant of the model effectively managed long sequences and surpassed SASRec in ranking results while maintaining a similar training speed. At time t , the predicted probability of user u engaging with item i was expressed as $p_{uit} = \sigma(r_{uit})$, using the predicted score r_{uit} at that point. The model applied binary cross-entropy loss between the predicted probability for a positive item and one uniformly sampled negative item, resulting in an objective function for learning defined as:

$$\sum_u \sum_{t=1}^{T-1} \sum_{k \in I} -[\log(p_{uit}) + \log(1 - p_{utk})] \quad (31)$$

A recent work introduced a hierarchical architecture that separates user behavior representations from behavior interactions [98]. Basically, it utilized LLM's shallow layers to extract and store granular user behavior embeddings and further intricate complex inter-behavior interactions to generate comprehensive user embeddings through deep layers of LLMs. This approach refined user embeddings which were then combined with item embeddings in the collaborative topic regression (CTR) model to calculate the CTR score. Another recent research examined the use of LLMs as zero-shot ranking models in recommender systems [129]. It designed natural language prompts and specific strategies to leverage LLMs' understanding of sequential behavior. Additionally, bootstrapping and prompting techniques were implemented to mitigate position and popularity biases in LLM-based ranking models. [86] developed a novel pre-training model by integrating a product knowledge graph (PKG) into LLMs to enhance item features. The approach introduced a unique paradigm that fine-tunes the PKG-pre-trained model for zero-shot item-based recommendation, which involves learning user and item embeddings through various pre-training tasks based on PKG. In the e-commerce domain, a research article demonstrated the effectiveness of ShopperBERT, a pre-trained model with general user representations, for various recommender system tasks [279]. Another study in literature introduced pre-trained user models (PTUM) that executes two pre-training tasks—masked behavior prediction and next N behavior prediction—to model user behaviors in recommender systems [344]. PTUM masked single-user behaviors, rather than spans of tokens, and used other behaviors in the user's interaction sequence to predict the masked behavior.

A novel study used LLMs to enhance text-rich sequential recommendations by summarizing user historical behaviors with LLM-based summarizers, including hierarchical and recurrent summarization techniques [424]. It combined user preference summaries, recent interactions, and candidate item information in an LLM-based recommender, then fine-tuned the model with supervised fine-tuning and low-rank adaptation for efficient model performance. Next to handle massive sequential data [181, 182] designed Transformer-based model for sequential recommendation that implements a Transformer encoder layer to represent item embeddings in sessions and employs multihead attention for optimization. The model incorporated rating information into self-attention weights to better capture user preferences and balances global and local preferences for accurate top-N recommendations. Specific research works [235], [236] investigated BERT models to perform multimodal representation learning in expert recommender systems from text and graph information. Building on the concept of multitask learning, one study developed the solution for multiobjective E-commerce recommendation problem by modeling users' various behavior sequences with Diverse Multi-Transformers [103]. The model utilized multiple transformers to represent diverse user behavior sequences, leveraged multitask mixture-of-experts to jointly optimize multiple objectives,

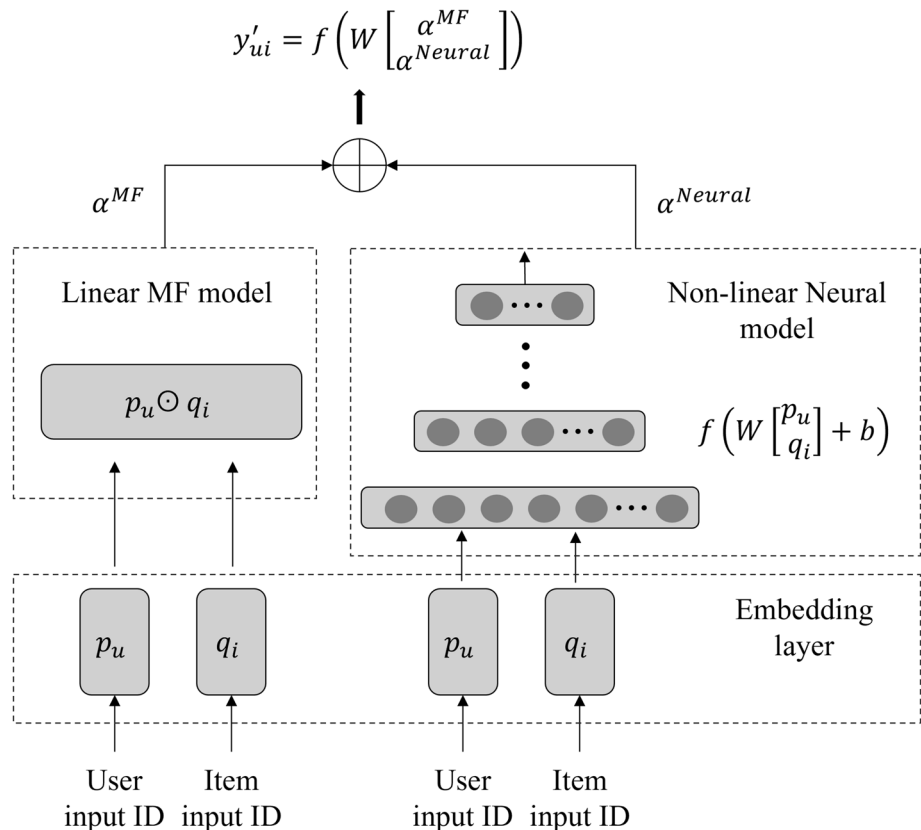
and applied a bias DNN to minimize bias in E-commerce recommender systems.

3.2.3 Graph neural network (GNN)

GNNs are designed to work with graph-structured data, where users and items are represented as nodes, and the relationships between them are represented as edges. GNNs excel at modeling complex relationships, such as capturing social interactions and dependencies between similar users, which helps overcome data sparsity and improves the accuracy of recommendations. By effectively analyzing the interconnectedness of user behaviors and preferences, GNNs enhance the ability to recommend items that are more likely to match user interests.

GNN approaches for rating predictions. The research [141] reinterprets traditional CF methods like k-nearest neighbors (KNN) and linear latent factor models (LFMs) as specific band-stop graph filters and bandlimited interpolation algorithms, respectively, based on networks of user and item similarities. This approach proposed new methods for improved rating prediction, such as the novel predictor mirror filtering (MiFi) that filters across user and item networks. Additionally, it focused on implementing a low-complexity method for linear latent factors by leveraging the eigenvectors of correlation matrices from known

Fig. 3 Illustration of hybrid linear and nonlinear latent features learning



ratings. The novel method, GraphRec, employed GNN to model social recommendations effectively by capturing interactions and opinions within the user–item graph [84]. It also accounted for a mathematical approach to consider heterogeneous strengths of social relationships. Extending this work, the authors developed GraphRec + for social recommendations, which coherently modeled graph data to learn improved user and item representations [85]. They introduced a principled approach to jointly capture interactions and opinions in the user–item graph and proposed an attention mechanism to differentiate the diverse strengths of social relationships. Next, with the aim of leveraging both rating scores and review information, the research [405] designed a multiattributed graph to model heterogeneous user–item interaction features. It implemented GCN to learn user and item features and used an attention mechanism to fuse features and dynamically predict ratings.

Another method [94] innovatively employed GNN to add and model the degree of contextual features, which not only discovers users' short-term preferences in time but also captures complex interactions as $\sum_{n_i \rightarrow n_j} A[n_i, n_j] Wh_i + b$ between feature nodes (n_*) and using initial node state (h_i) to enhance recommendation predictions.

An article [107] explored a social recommendation application for future Internet of Things (IoT) users and used the GNN method to capture complex relationships. The approach focused on encoding item attribute correlations into a vector $\sum_{\alpha=1}^z \sum_{\beta=1}^z \beta = 1, A(\alpha, \beta).M$, using $\beta \neq \alpha$

transformation matrix (M) and weight connections $A(\alpha, \beta)$ between different attributes. This integration improved the granularity of the overall feature space and ultimately helped predict unknown preference ratings. Another article developed a recommender system for social recommendations, aiming to tackle challenges such as heterogeneity, personalization, and privacy protection [206]. The system employed relational attention and aggregation to manage heterogeneity, inferred user embeddings from local data to maintain personalization, and utilized pseudo-labeling with item sampling to protect privacy and improve training. [189] focused on social recommendation and employed GNN to combine dynamic and static representations of users and items. This method featured two main components: interaction aggregation, which used RNNs for short-term dynamic modeling and graph attention mechanisms for long-term static modeling of user–item interactions; and relational graph aggregation, which used graph attention mechanisms to integrate influences from user–user and item–item relational graphs. The method then combined

interaction and relational representations through an MLP for rating prediction.

A different approach [356] used GNN and multiview learning to predict users' ratings for unrated mobile apps. It designed semantic meta-graphs using user reviews, app genres, and rating information to create a similarity matrix. The method then implemented GNN with attention to gather feature information from neighboring nodes, forming representations for users and apps. Different semantic relationships were treated as separate views for multiview learning, predicting user ratings for each view. Finally, the method evaluated the importance of each view and combined the ratings to predict the user's final score for an unknown app. Likewise, [343] adopted a multiview learning framework to leverage both review content and user–item graphs. It incorporated a three-level attention network to analyze words, sentences, and reviews for informative user/item representations. Additionally, the method used a hierarchical GNN to capture user–item, user–user, and item–item interactions with attention to prioritize these relationships, resulting in a comprehensive understanding of users and items.

Unlike previous methods that focus on exploiting interaction graphs, the novel approach [256] emphasizes the importance of utilizing the attribute graph. The authors created a new eVAE structure to infer preference embeddings from attribute distributions and developed a gated-GNN structure to better combine various attributes in a neighborhood. The proposed model significantly improves strict cold start recommendations and either matches or exceeds baseline performance in warm start scenarios. A unique approach emphasized an aspect-GNN for rating prediction [138, 139]. It defined aspects as noun words from user reviews and product descriptions, focusing on three types: user reviews aspect, item reviews aspect, and item description aspect. An aspect graph was built with nodes representing aspects and edges representing relations such as co-occurrence and similarity. This graph enhanced semantic representation learning and addressed sparsity issues. Finally, a graph attention network (GAT) was used to learn the association between aspects and integrate the aspect graph with the neural network. Additionally, the graph diffusion network [348, 349] and graph attention mechanism [352] were implemented to capture user influence and integrate social relationships into recommendations to further address data sparsity problem.

In scenarios where only the matrix to complete is available without any additional information, the paper [397] presented an innovative inductive matrix completion method. This approach did not rely on content or the entire rating matrix; instead, it inferred ratings for individual user–item pairs using local graph patterns and converted unknown rating predictions into labeled link predictions in

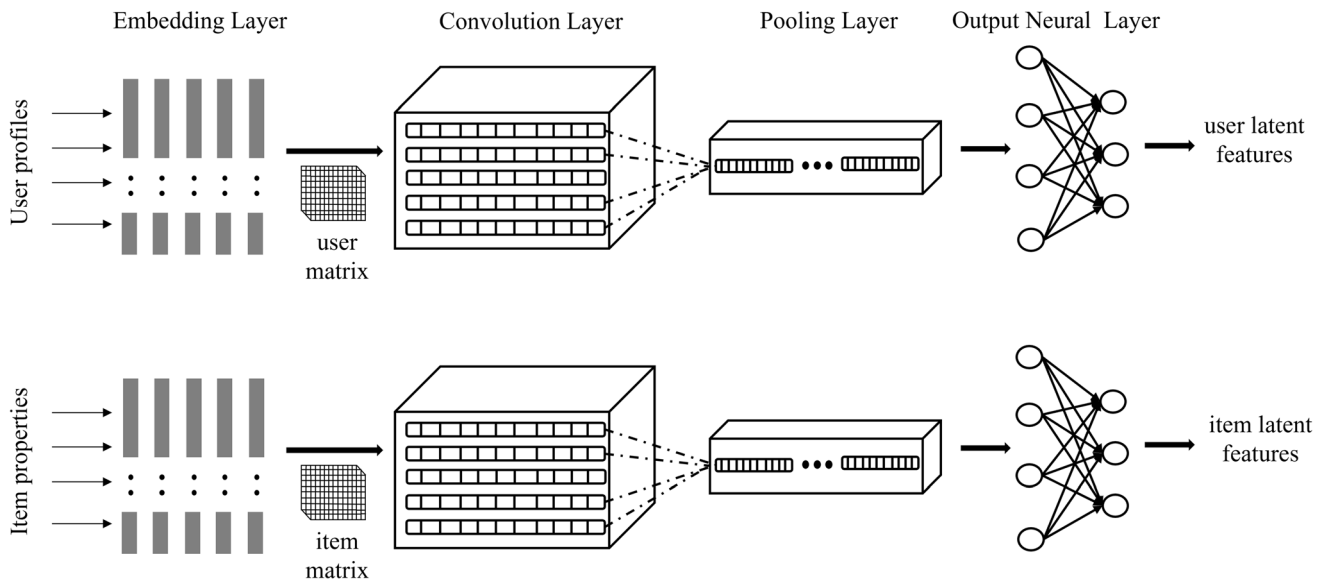


Fig. 4 CNN-based recommendation architecture design

a bipartite graph. Another study treated user–item interactions as a bipartite graph and used a GNN to improve the representations of each node [18]. It focused on modeling each node with its adjacent neighbors using an aggregation module for rating prediction.

GNN approaches for top-*N* ranking. One of the challenges of CF method is efficiently extracting negative samples from implicit feedback. To address this issue, [140] proposed generating samples instead of directly extracting them from data, which are then used for training GNN-based CF recommendation models. Methodologically, the framework incorporates positive mixing (interpolation) and hoping mixing (pooling) strategies to synthesize negative samples by combining information from various local graphs. Besides this, a few other studies have focused on negative sampling using graph information. For instance, [379] sampled nodes by ranking items in a graph according to their Personalized PageRank scores. [372] studied how negative sampling affects recommendations and used self-contrast approximation to estimate the positive node pairs. Next recommendation method utilized high-order graph convolution within subgraphs of users with similar interests, using an unsupervised module to identify these users and prevent negative information propagation [193]. Furthermore, in research, a study addressed the challenge of category diversification in recommendation systems through GCN [420]. It introduced a rebalanced neighbor discovery method to improve accessibility to underrepresented categories and adjusted the negative sampling process to increase the likelihood of sampling similar, yet negative items. Additionally, it employed adversarial learning to extract implicit category preferences, resulting in the development of an end-to-end

model. In a separate study, the authors highlighted the significance of the price factor in recommending economically costly items and proposed a graph-based approach to integrate item price and category influence for effectively learning user preferences [421].

A study on recommendation ranking aimed to identify the set of feature interactions that would optimize recommendation performance [292]. It implemented an automated approach to identify advantageous feature interactions, selectively preserving only the most beneficial edges. This led to the creation of a learned graph, which was subsequently utilized in a graph classification model to estimate the ranking score $y' = f(G_n(X_n, E'_n); \theta)$, where X denotes pair-wise statistical interactions between variables, G is the graph with the predicted edge set E' , and θ represents model parameters. In previous research, NGCF method [333, 334] effectively utilized multihop community information by employing recursive message propagation and integrating representations from various layers to improve model performance. Conversely, LightGCN model [117] simplified the design of NGCF by eliminating nonlinear feature transformations, thus streamlining the aggregation module for a more efficient approach.

Like [9], a research study combined deep AE network and clustering algorithm to develop a food recommender system [263]. The system operated in two phases: first, it predicted user ratings by analyzing user similarities and trust networks, and second, it predicted food ratings using deep learning embeddings of food ingredients. The final recommendations combined both user and food predictions to recommend the top food items to users. Another hybrid approach introduced a novel, multistage recommendation system that integrated clustering, graph modeling, and deep

learning to effectively manage and recommend scientific articles from the extensive AMiner Citation-network dataset [289]. A different study proposed a personalized ranking method that bridges the gap by integrating embedding learning, which maps user and item IDs to embeddings through user–item graph aggregation, with interaction modeling that uses the learned embeddings to match user–item interactions [137]. The research implemented personalized recommendation using GNN in MOOC [321]. It incorporated co-learning and co-category relationships into item embeddings to represent users' general preferences. To capture current preferences, it modified sequences for CNN input to enable high-order connectivity in item embeddings. Then, it implemented CNN to extract key features of sequential patterns in users' current item choices. A recent method introduced a novel knowledge-enhanced GNN for explainable recommendation by using semantic knowledge from an external knowledge base to create enhanced embeddings for users, items, and user–item interactions [216]. It constructed a user behavior graph based on user–item interactions and initialized it with the knowledge-enhanced semantic embedding. The approach implemented a GNN-based model to understand user behaviors and perform multihop reasoning over the user behavior graph. Furthermore, hierarchical CF layers were developed for rating prediction, incorporating a GRU-based generator and copy mechanism to produce humanlike semantic explanations. Next, a sequential recommendation literature utilized a dynamic graph concept to model user–item interactions and predict future items for users [398]. It first converted user sequences into dynamic graphs, then extracted relevant subgraphs, and further used dynamic GNNs to learn user preferences and item characteristics. The method employed attention modules and RNN to capture both long-term and short-term user preferences and transformed the next item prediction task into a link prediction task. An earlier sequential recommendation work involved using GCN and dynamic graph pooling to effectively capture user preferences by transforming loose item sequences into tight item–item graphs [32]. This approach gathered weak signals to strengthen them for accurate user preference reflection and used dynamic graph pooling to adaptively reserve core preferences for predicting future user behavior. Next research developed a heterogeneous graph neural CF framework that integrated sequential patterns of user–item interactions with high-order collaborative signals [168]. It enhanced user–item interactions by incorporating additional item attributes in a heterogeneous information network and used attention mechanisms to capture user preferences, dynamic interests, and important neighboring nodes for more effective recommendations. To address the uncertainty in user behavior, a study developed the

GRASER method [108]. It constructed a directed graph for each session, using GNN to learn item representations and capture both non-sequential and sequential transitions. GRASER combined local session embeddings from the GNN output with global session representations from an attention mechanism, and then used these combined representations to recommend the top-N items. Few other works [138, 139, 181, 182, 358] focused on modeling historical session sequences using GNN model. Another investigation focused on context-aware recommendation systems and proposed a model that transformed user, item, and context features into an attributed graph [347]. The approach used GNN to capture interactions among user behaviors and employed FMs to model feature interactions within individual behaviors. Additionally, another study introduced the attribute-aware attentive GCN, which enhanced representation learning by leveraging available item attributes and establishing connections between items and attributes [194]. This approach allowed item embeddings to effectively capture both user behaviors and attribute information. Advancing this approach, subsequent work accounted for the different effects of attributes and utilized attribute information within a graph structure to enhance the learning of user and item representations [386]. Unlike existing approaches, a study automatically discovered explanatory paths from items in a user's history to potential recommendations [319]. It enhanced recommendation performance by combining embedding-based and path-based methods through the propagation of user preferences using a knowledge graph and the aggregation of entity information. Data in a single social network is often limited and sparse. Therefore, an article designed a multiview user representation learning model to enhance user modeling by integrating knowledge from various heterogeneous social networks [331]. It aimed at addressing the challenges of fusing diverse information sources and focuses on user account linkage and user attribute prediction.

A study developed a model based on GraphSAGE to model design attributes [6]. The initial GraphSAGE method [111] represented graph structures in lower-dimensional vectors and used the vectors as input for downstream classification. [6] introduced a permutation-based technique to examine feature importance and assist in design decisions. [388] investigated a structured representation learning approach using a gated-GNN to capture feature representations and developed a causality-based CTR prediction model. It built user and ad graphs based on statistical analysis and used GraphSAGE to generate embeddings. The model combined graph representations of features, users, and ads with a multihead attention mechanism, and an attention-aware predictor leveraged weighted representations to forecast clicking probability. Another

study used graphs to provide clear model explanations for CTR prediction [184]. It created a weighted graph of all input features using combined GRU and GAT in the encoder, and attention networks in the predictors. In addition, a study presented a dual graph method for CTR prediction that enhances embeddings by utilizing an attribute graph and a collaborative graph [106]. The model leveraged divide-and-conquer and curriculum-learning-inspired strategies to optimize graph representation.

3.3 Hybrid models

Hybrid models leverage ensemble methods or meta-learning techniques to maximize recommendation performance and robustness. This section provides a literature analysis of hybrid deep learning models for recommendation tasks and presents detailed summaries of these approaches in Table 6.

Hybrid approaches for rating prediction. A method employed DAE and MLP to extract intensive user–item interactions for rating prediction tasks [325]. It is a hybrid technique designed of two parallel DAE networks: AutoRec-U for extracting user latent features and AutoRec-I for extracting item latent features. These features are combined and fed into MLP to represent user–item interactions and produce expected ratings. In [345], the authors enhanced their previous research [346] by proposing a method for both rating prediction and top-N recommendation. They used a multilayer neural network with CNN and bidirectional GRU to enhance text latent features. [381] proposed a novel hybrid network to anticipate and recommend the tags of news article to users. They used CNN for text features, VGG-16 for image features, and combined them in an AE network before classifying news articles using MLP. Another hybrid methodology by [267] exploited the RBM-CNN combination to implement an effective health recommender engine. Another novel approach DeepHCF utilized the MLP network for rating data and a CNN network for item reviews data [12]. SARWAS, a revolutionary recommendation architecture, employed deep ensemble learning techniques that combines reviews and ratings in a weighted way to create an association score for sentiment-based recommendations [49].

Inspired by RBM-CF and neural autoregressive distribution estimator (NADE) architectures, [423] proposed CF-NADE, a neural autoregressive model for CF recommendation task. In later study, the authors conducted experiments with a variant of CF-NADE that incorporated implicit feedback [422]. [73] used a neural co-autoregressive model for CF tasks, which directly and simultaneously utilizes structural correlation in both users and items.

Hybrid approaches for top-N ranking. A novel study integrated CNN-RNN model to create an online recommendation system for medical pre-diagnosis [430]. The CNN algorithm obtained text information from input phrases, whereas the RNN classifier learned probable patterns and correlations from input dialogues. Additionally, the model used K-means clustering to improve the training efficiency of these networks. To address data sparsity problem, [126] proposed a novel neural CF-based DeepLSGR hybrid model endowed with two hidden layers with LSTM and GRU for improved recommendations. An article introduced a method known as CHAMELEON deep learning meta-architecture for news recommender systems [63]. The authors implemented CHAMELEON meta-architecture as a concrete design with CNN to learn textual properties from news articles and with LSTM to model the sequence of clicks from users' sessions [64]. Later in next year, a pipelined hybrid recommender system was introduced by [241] that considers two inter-session and intra-session contexts simultaneously via weighted summation of loss functions. The model designed using the combination of a RNN, which captures the user's intention from the embedded session and a CNN, which captures the context of items on the displayed page in service from the embedded display. [390] proposed a collaborative knowledge-based embedding recommendation model that employed a hybridization framework combining CNN and AE, specifically SDAE and stacked convolutional AEs, to extract textual and visual representations of items, respectively. In contrast, [316] developed a collaborative recurrent AE that sophisticatedly models the generation of item sequences while extracting the implicit relationship between items and users.

A research article proposed streaming recommender model based on neural memory networks with external memories to capture and store both long-term stable and short-term dynamic interests in a unified manner [326]. For effective training of the streaming recommender system, the authors used GAN with the negative sampling method. Recently, other paper demonstrated a great potential for applying deep reinforcement learning networks in recommendation systems [136].

4 Pre-modeling and post-modeling of recommender systems

In addition to the methodology, it is crucial to explore other domains that contribute equally to the research development of recommender systems. This section discusses the fundamental elements of recommender systems, covering pre-modeling aspects (dataset and feature learning), and post-modeling (evaluation metrics).

Table 5 Summary of exiting modern deep learning techniques for recommender system

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
DAN	[36]	Attention-driven gated model	Explainability	MovieLens, Douban	RMSE, MAE	- The attention-driven model achieves an average MAE improvement of 2.42% and RMSE improvement of 2.25% over LibFM, SVDFeature, CDL, and NeuMF in ML data - Similarly, it achieves average MAE and RMSE improvements of 4.06% and 3.58% in the Douban Music dataset, respectively, and improvements of 4.56% for MAE and 5.13% for RMSE in the Douban Movie dataset - The range of improvement performance of this GRU attention model over PMF is from 0.110 to 0.264, over HFT is from 0.102 to 0.112, over CTR is from 0.100 to 0.137, and over ConvMF is from 0.042 to 0.094
	[210]	Attention GRU + MF	Recommendation transparency and explainability	Yelp, Amazon	MSE	NARRE achieves the lowest error rates in RMSE compared to PMF, NMF, SVD, HFT, and DeepCoNN (Zheng, Noroozi, et al., 2017). Specifically, NARRE's RMSEs are 0.7783 to 0.9965 in Amazon, and 1.1559 in Yelp
	[34]	Neural attentional model (NARRE)	User behavior representation learning and explainability	Amazon, Yelp	RMSE	The model performs best across all datasets, achieving the lowest error rates in RMSE compared to other models such as PMF, GRU4Rec, Time-LSTM, and NARRE. The resulting RMSEs for the model are 0.8981 to 0.9678 in Amazon, and 1.2314 in Yelp
	[43]	Attention time-aware GRU method	Explainability	Amazon, Yelp	RMSE	The model's self-attention and dual learning components lead to improved performance, resulting in at least 0.3% improvement in MAE and 0.2% improvement in RMSE on ML-100K, as well as at least 0.3% improvement in MAE on ML-1M
	[255]	Self-attention network	User behavior representation learning	MovieLens	MAE, RMSE	Overall, the model achieves an average relative improvement of 4.46% over the best baseline. In the sparsest dataset, it improves by 5.18%, and even with limited review data, it still offers a 3.30% relative gain
	[351]	Attention-based CNN network	User behavior representation learning	Amazon, Beer, Yelp	MSE	It surpasses most Siamese architectures, such as DeepCoNN, TransNet [30], D-attn [272], and, though it incurs higher computational costs compared to DeepCoNN and TransNet
	[273]	Attention-based CNN network	User behavior representation learning	Yelp, Amazon	MSE	CNNs with attention achieves RMSE of 1.212 on Yelp and 0.928 on Amazon, outperforming HFT, ConvMF, and CNN-only methods with significant improvement on both datasets
	[272]	Dual local and global attention-based CNNs + MF	User behavior representation learning	Yelp, Amazon	MSE	CNNs with dual local and global attention achieves RMSE of 1.191 on Yelp and 0.855 on Amazon, outperforming HFT, ConvMF, CNN-only, local-attn only, and global-attn only methods with significant improvement on both datasets
	[35]	Attention mechanism in CF	Multimedia recommendation	Pinterest, Vine	Hit ratio, nDCG	The method excels on both datasets, significantly outperforming state-of-the-art MF and hybrid methods, with an average relative improvement of 5.19% over the best baseline (SVD). However, the approach is most effective when strong user-item patterns exist in data

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[369]	Multithread attention network + cross-feature FM layer	User behavior representation learning	WeChat, MovieLens	Precision	• With the attention mechanism, the proposed model shows improvements ranging from 1.1% to 9.3% on the industrial and 0.8% to 1.3% on ML • With the cross-feature implementation and joint loss functionality, the proposed model demonstrates improvements ranging from 2.6% to 11.7% on the industrial and 1.2% to 1.7% on ML • The model surpasses traditional clustering algorithms by 3.8% to 11.6% in all tested categories • It also exceeds context-aware, trust-aware, knowledge-based, CF, and hybrid recommenders by 8.8% to 16.7%
	[324]	KNN clustering + LSTM + attention mechanism	Personalized recommendation	User-product e-commerce data	MSE	• The proposed model outperforms baseline sequentialRec models, achieving 68.81 Hit@10, 39.63 nDCG @10, and 32.31 MRR on Yelp, and 79.32 Hit@10, 52.92 nDCG @10, and 45.75 MRR on WeChat. This shows that self-attention is better for sequence modeling and that incorporating diverse user information into pre-trained models results in better user representations and improved performance
	[355]	Multithread self-attention BERT framework	User behavior representation learning	Yelp, WeChat	Hit ratio, nDCG, MRR	• The proposed method outperforms traditional methods (KNN, SVD, Pearson similarity, DNNRec), especially for sparse datasets and cold items. For example, in the BookCrossing comparatively sparse dataset, the method's improvement ratio is higher with MAE and RMSE improvements of 19.58% and 17.28%, respectively • In the MovieLens 100K dataset, the method achieves an impressive MAE improvement rate of about 15.09% for cold items
	[123]	Self-attention + MF with BERT and AE as embedding methods	Data sparsity, cold start, scalability	BookCrossing, MovieLens	RMSE, MAE, hit ratio	• The model's performance is compared among embedding methods such as TF-IDF, Word2Vec, and BERT, and results indicate BERT performs well than other methods • Experiment results demonstrate superior performance of this approach with certain level of interpretability across both synthetic and real-world datasets, outperforming MF and GNN-based deep learning baselines with a minimum of 0.68% relative improvement
	[287]	Multigraph attention fusion method with dual-branch residual graph attention module, fusion graph attention module and geometric factorization	User behavior representation learning	Netflix, MovieLens, Flixter	RMSE	• The method highlights how the co-attention mechanism effectively utilizes meta-path-based context, resulting in superior accuracy and interpretability. For example, on ML, the method achieves the highest performance improvements over models (ItemKNN, BPR, MF, NeuMF, and SVD), ranging from 4.81% to 33.87% in precision, 4.20% to 46.97% in recall, and 3.27% to 21.22% in nDCG
	[130]	Co-attention mechanism	Heterogeneous information network-based recommendation	MovieLens, LastFM, Yelp	Precision, recall, nDCG	

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[418]	Memory attention-aware system	User behavior representation learning and explainability	Yahoo, Amazon	Recall, MAP	- The method shows average improvement over the best baseline (Wide & Deep [45]) by 20.8% in recall@50 and 13.0% in MAP across three datasets - It also allows for interpreting recommendation results based on items purchased or liked, making it more applicable to real-world scenarios
	[234]	Gated LSTM + attention mechanism	User behavior representation learning	Taobao	AUC, precision	- The multitask learning feature converges faster and performs about 1% better than single-task learning, with precision increasing from 40 to 44%. Using information from multiple tasks helps user representations better reflect their interests - Moreover, the pre-trained version of model outperforms its word-2-vec variant by 2.0% to 3.6% in AUC and by 6.0% in precision
	[411]	Hierarchical self-attention mechanism + masked GCN	Cold start	Cora, CiteSeer, and PubMed	Hit ratio, nDCG	It outperforms all baselines across different output dimensions with accuracy on three citation network datasets is 1.9%, 2%, and 2.1% higher than the GAT model
	[367]	Long- and short-term users' preferences attention networks + fully connected network	User behavior representation learning	Gowalla, Tmall	Recall, precision, AUC	Approaches that combine users' sequential behaviors with their general tastes typically outperform traditional methods that ignore short-term preferences, showing the value of using sequential data
	[57]	LDA + LSTM encoder + neural attention network	User behavior representation learning	Amazon, Yelp	MSE	The proposed sequential network improves Recall@20 by 21.5% and Precision@10 by 35.5% on the Tmall dataset and achieves 37.5% improvement in Recall@20 and 35.4% in Precision@10 on the Gowalla dataset, compared to the best baseline
	[55]	Multilevel attentive + co-attention-based model	User behavior representation learning	Amazon, Yelp	RMSE, MAE, precision	- The model performs better in rating prediction, showing up to 86% improvement over MF, up to 55% over topic modeling methods, and up to 6% over other deep learning models like DeepCoNN, TransNet, and D-Attn - The model significantly improves rating and ranking performance compared to other models. It outperforms CNN-based DeepCoNN by 10.1% in RMSE and 14.2% in MAE for rating performance. For ranking performance, it shows improvements of 4.1% to 12% in Precision@10 compared to traditional deep learning models
	[93]	Knowledge graph model attention mechanism + relation-aware GCN	User behavior representation learning, data sparsity	MovieLens, BookCrossing, LastFM, and Dianping-Food	AUC	- The method improves AUC by 26.8% on ML, 12.4% on BookCrossing, and 20.7% on LastFM compared to baselines, proving effective for CTR prediction, especially on large-scale sparse datasets like Dianping-Food with a 39% improvement - It performs well in Top-N recommendation scenarios but slightly underperforms on LastFM against one baseline method - Although its complex strategy increases space complexity, it provides significant advantages in recommendation accuracy, stability, and time efficiency

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
LLM	[80]	Knowledge-aware and user-specific attention networks	Personalized recommendation	MovieLens, LastFM, BookCrossing	Recall, nDCG, AUC, F1-score	- This approach improves performance compared to the collaborative attentive model with gains of 1.2%, 2.7%, and 1.9% in AUC on ML, LastFM, and BookCrossing - For nDCG @N, the method outperforms by 2.1% in ML, 5.1% in LastFM, and 2.9% in BookCrossing, highlighting the value of a relation-aware attention mechanism - The model's attention units achieve the best performance, surpassing the YouTube DNN model with 0.0109 AUC gain and 0.0078 precision@10 gain - Furthermore, the attention merge layer uncovers more user interest factors, boosting category and tag diversity among users - The method is effectively implemented in WeChat's "Top Stories" recommender system, indicating great improvement on diversity and quality of recommendations
	[204]	Attention merge layer + self-attention-based local and global attention units + k-means clustering	Recommendation quality, diversity and scalability	Wechat "TopStories"	AUC, precision	- The method shows an AUC improvement of 0.74% on ML-1M against Wide & Deep [45] and 0.73% against DeepFM, and an improvement of 3.89% on Avazu against Wide & Deep and 3.82% against DeepFM - The model improves recall by 0.25% to 4.53% and MRR by 0.45% to 5.33% over the neural attentive recommendation machine (J. [169, 175]), with training time roughly one-third to one-half of that
	[115]	Cross-attention unit + DNN	Recommendation interpretability	MovieLens, Avazu	AUC	The use of LAMAR significantly improves the performance of base models across different datasets. For example, DeepFM improves RMSE from 0.9873 to 0.9582 and MAE from 0.7765 to 0.7505 in the Kaggle Movie dataset. In the ML data, neural FM's RMSE drops from 1.0581 to 1.0345 and MAE drops from 0.8323 to 0.8157 with LAMAR. These improvements showcase LAMAR's effectiveness
	[180]	MLP + attention network + Tag clustering	Recommendation accuracy, User behavior representation learning	Yoochoose from RecSys Challenge, Diginetica from CIKM cup	Recall, MRR	- Integrating LLMs with traditional models improved rating prediction. For instance, on ML-100K dataset, it achieves up to 4.58% improvement in RMSE and 4% in MAE against DeepFM. For ML-1M, it gains up to 1.24% in RMSE and MAE against DeepFM. In the BookCrossing dataset, Llama4Rec excels, with up to 9.86% RMSE and 9.89% in MAE against DeepFM, and similar gains over neural FM - Llama4Rec also performs well in top-k recommendations, particularly on the BookCrossing dataset due to its fine-grained ratings - LLM-based recommenders require longer training and inference times, indicating limitations and high computational resource demands
	[211]	Traditional recommenders + LLM-based recommenders. (LAMAR)	Augmenting LLMs with conventional recommenders	MovieLens, Kaggle movie	RMSE, MAE	- The proposed method excels in interactive resume completion using LLMs. Only LLM methods perform poorly, suggesting that relying solely on textual descriptions is ineffective - It effectively tackles few-shot scenarios, improving the quality of resume generation - It compares against traditional, GNN-based, and LLM-based models, achieving an average gain of 7.19% MAP, 7.81% nDCG, and 5.38% MRR
LLM	[212]	mutual augmentation between traditional recommenders + LLMs through data and prompt augmentation. (Llama4Rec)	Augmenting LLMs with conventional recommenders	MovieLens, BookCrossing	RMSE, MAE, hit ratio, nDCG	
	[76]	LLM-based GAN method	LLMs challenges: fabricated generation and few-shot problem	Designs, sales and tech collected from real online logs	MAP, nDCG, MRR	

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[186]	Financial product-based transformer method	Computational complexity	Custom dataset	RMSE	- The method obtains significant gains in RMSE compared to CF across all datasets, with gains of around 30.86% to 35.56%. It also performs slightly better than R-RNN, with gains of around 1.08% to 7.04% - The average processing time of this approach is faster than RNN and traditional CF methods
	[172]	Transformer-based language modeling	Explainable recommendation	Yelp, Amazon, Tripadvisor	RMSE, MAE	- This method matches other advanced models in performance but falls slightly short of SVD and other neural attention models such as NRT (P. [169, 175]) in some areas. In the final scenario, its 0.81 RMSE and 0.63 MAE indicate slightly higher error rates - On large-scale datasets, this approach performs better, but it may underfit on smaller datasets due to having more parameters to learn
	[149]	Attention layer + self-attention mechanism	Cold start	MovieLens	RMSE	- The consideration of both user attention layer and the movie attention layer gives a positive impact on the prediction performance - Yet the model fails to produce reasonable predictions for users who do not have existing profiles in the system
	[183]	Reasoning LLMs framework for book recommendation tasks	Book recommendation	Douban, Goodbook	RMSE, MAE, MAPE, nDCG	- BookGPT shows promising results in classic book recommendation tasks, particularly in scenarios with limited information such as zero-shot or one-shot learning - The model achieves substantial improvements in the 20-shot sub-task, with error reductions of 21.67% in MAE, 16.53% in MAPE, and 16.84% in RMSE
	[383]	SentenceBERT-based LLM model	User behavior representation learning	Amazon	Hit ratio, nDCG	- The model improves Hit@100 and nDCG@10 by 25.9% and 7.9% on Amazon Clothing, and 21.7% and 15.1% on Amazon Books, compared to methods like MLP4Rec, GRU4Rec, SasRec, Bert4Rec, and GPT4Rec - It also outperforms baselines with up to a 3.0% absolute Hit@100 improvement while using less than $10 \times$ the training data
	[65]	LLM-based framework, particularly uses ChatGPT systems	Accuracy, fairness, diversity, stability, and recency	MovieLens	Recall, nDCG, Gini Index (GI), Herfindahl–Hirschman Index (HHI), Entropy	- Adoption of the fair recommender role in GPT models results in enhancements in fairness and diversity metrics, effectively addressing popularity bias - While certain GPT models display higher popularity bias than some CF models, this is not consistent across all models. Models like NGCF outperform certain GPT models in fairness indicators. However, excluding ItemKNN's outlier performance, GPT models generally match CF models in fairness and diversity - GPT models, when equipped with the fair recommender role, exhibit flexibility in improving recommendation diversity and fairness. For instance, the average GI coefficient of the GPT models under fair recommender surpasses baseline models by approximately 1.5%, and the average HHI is about 120% higher
	[66]	Leveraging fairness in RecLLMs models	Fairness of RecLLMs	MovieLens	Jaccard similarity and PRAG	- The study emphasizes the importance of RecLLMs being sensitive to various user identities to avoid unequal system quality across demographic groups. It uncovers significant differences in recommendation outcomes based on user profile sampling methods such as random, top-rated, and recent. For instance, in the Age group category (teen, young, adult), similarity scores show improvements from 0.1669, 0.1847, and 0.1421, respectively, to 0.5773, 0.6548, and 0.6565, respectively - However, the study's evaluation relies primarily on a single dataset

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[199]	Open-source and closed-source LLMs	Content-based recommendation	MIND, Goodreads	AUC, MRR, nDCG	- Open-source LLMs, particularly LLaMA [306], exhibit strong performance by providing a robust understanding of content and semantic knowledge - Closed source LLMs also perform effectively, demonstrating the benefits of enriching data to enhance semantic features - This dual LLM-based approach combines GPT-3.5 and LLaMA for better performance, but its impact is less pronounced than open-source fine-tuning due to different levels of closed-source LLMs representations. It improves AUC on the MIND dataset by 7.89% to 11.39% and on the Good Read dataset by 9.56% to 16.79%
	[70]	LLaMA, GPT, and PaLM [50]	Robustness and adaptability of LLM-based recommenders	MovieLens, Facebook books	nDCG, hit ratio, MAP	ChatGPT performs well across on Facebook Books, but its performance is similar to other models on ML. The study throws an idea of improving results with advanced prompt engineering techniques
	[150]	Self-attention-based sequential framework (SASRec)	Sequential recommendation	Amazon, Steam, MovieLens	Hit ratio, nDCG	- It examines limitations of LLMs in recommendation tasks, emphasizing the need to tackle popularity bias and explore out-of-distribution items for better diversity and serendipity - The model outperforms traditional (popularity-based, BPR) and sequential (GRU4Rec) recommendation methods across both sparse and dense datasets, with average improvements of 6.9% in hit rate and 9.6% in nDCG compared to the strongest baseline
	[424]	Hierarchical LLM-based and recurrent LLM-based summarizers	Overlength limitations, computational efficiency, recommendation accuracy	Amazon, MIND	Recall, MRR	- LLM-based recommendation approach outperforms non-sequential and sequential models. For example, hierarchical LLM and recurrent LLM improve recall and MRR by 0.14% to 1.37% in Amazon, while they show gains of 1.67% to 10.33% on MIND dataset - The recurrent method excels on the Amazon dataset, while the hierarchical method does better on the MIND dataset, indicating that different user preference summarization paradigms may suit different scenarios
	[348, 349]	Leverages personalization concept into self-attentive network	Personalized recommendation	Amazon, Steam, MovieLens	Recall, nDCG	The model surpasses SASRec by nearly 5% in nDCG@10 on real-world datasets, offering better ranking performance in the same training time
	[181, 182]	Transformer encoder with multihead attention	Sequential recommendation	MovieLens, LastFM	Hit ratio, MRR, nDCG	The model shows significant improvements over traditional and neural models. It improves hit ratio by 8.34%, MRR by 1.54%, and nDCG by 4.49% in ML, and hit ratio by 5.75%, MRR by 10.21%, and nDCG by 9.07% in LastFM
	[159]	LLMs to leverage collaborative knowledge from pre-trained CF-RecSys	Cold and warm start problem	Amazon	Hit ratio	It consistently outperforms CF models like NCF, SASRec, modality-aware models, and other LLM-based models like MLP-LLM across different domains, with improvements ranging from 20 to 70% in nDCG scores
	[96]	Chat-Rec algorithm links traditional recommender systems with conversational AI such as ChatGPT	Explainability and cold start	MovieLens	RMSE, MAE, precision, recall, nDCG	- In top-N ranking, all three GPT-3.5 models surpass LightGCN in nDCG scores, with text-davinci-003 achieving the highest precision (0.3240, 6.93% higher than LightGCN) and nDCG (0.3802, 11.01% higher). However, its recall rate (0.1404) is 3.51% lower than LightGCN's - In rating prediction, text-davinci-003 performs best among GPT-3.5 models with an RMSE of 0.785 (15.86% better than ItemKNN) and an MAE of 0.593 (19.21% better). These results indicate that GPT-3.5 models can effectively predict user preferences for movies without relying on traditional recommender systems

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[60]	ChatGPT and other LLMs models	Explainability and cold start	MovieLens, Amazon, MIND	nDCG, MRR	- LLMs outperformed random and popularity-based approaches (pop) in zero-shot recommendation settings. For instance, in the music domain, ChatGPT (gpt-3.5-turbo) achieves an improvement of 0.1305 in nDCG@3 over random and 0.0944 over pop. For MRR@3, ChatGPT achieves an improvement of 0.1283 over random and 0.0920 over pop - LLMs excel in list-wise and pair-wise ranking but are less effective in point-wise ranking in most cases - ChatGPT exhibits potential for mitigating the cold start problem and improving explainable recommendations - MLPs with pre-trained user representations outperform transformers in most tasks - Pre-training significantly improves performance on cold-start tasks, offering 4–5% better results compared to models trained from scratch. Also, the model performance improves with scaled-up pre-training data - The analysis shows that LLM-generated descriptions perform similarly to IMDB movie plots, with Hit@10 values of 0.707 for IMDB and 0.700 for LLM, and nDCG@10 values of 0.484 for IMDB and 0.480 for LLM - The cosine similarity between LLM and IMDB descriptions is 0.4398, indicating good similarity between the two sources - The method greatly improves efficiency and CTR performance, introducing new strategies for using LLM-based CTR models. At text length 2048, it achieves an AUC improvement of 0.0133 over DNN and 0.0029 over LLM-CTR - In industrial CTR prediction, it successfully reduces training time and memory usage, making LLMs practical for industrial applications - KELLMRec achieves a significant improvement over LLMRec, with AUC scores of 0.7797 (versus 0.7785 for LLMRec) and LogLoss scores of 0.5559 (versus 0.5586 for LLMRec) when using DIEN (two layers of GRU) as the backbone model. KELLMRec also outperforms LLMRec in AUC with DeepFM and DIN (attention network), proving its independence from the choice of recommendation algorithm - The model outperforms other sequential models (RNN, LSTM, GRU, Bi-LSTM) by 13% in MAP and 0.5% in recall on the Listenbrainz dataset - However, the results for The Echo Nest dataset are less reliable due to bias in listening counts; most users tend to listen only once, which leads to lower precision and MAP values
	[279]	ShopperBERT for e-commerce recommender systems	Generalizability, cold start, representation learning, scalability	Purchased logs from e-commerce platform	AUROC, F1-score, AUC, hit ratio, MRR, nDCG	
	[2]	Alpaca—open-source LLM model + BERT embeddings + GRU	User behavior representation learning	MovieLens, Goodreads	Hit ratio, MRR, and nDCG	
	[98]	LLM's hierarchical encoding technique	User behavior representation learning, computational efficiency	Industrial dataset with CTR records	AUC	
	[213]	Knowledge-enhanced LLMRec framework (KELLMRec)	Augmenting or adapting LLMs to recommendation tasks	MovieLens, WeChat-ArticleRec	AUC, nDCG, MAP	
	[366]	Bidirectional transformer layers consisting of multihead self-attention and position-wise feedforward network layers	Sequential recommendation	Echo Nest, Listenbrainz	Precision, recall, MAP, RMSE	

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
GNN	[103]	Deep multifaceted transformers framework for multiobjective ranking in e-commerce recommender systems	Multitask learning	Recsys dataset	AUC	The approach is highly effective for click and order prediction, outperforming DNN, DIN, and DIEN models. It achieves substantial gain over DIEN in both click (0.0126 AUC gain, 7.4% relative improvement) and order prediction (0.0192 AUC gain, 11.54% relative improvement). Overall, the model performance is substantial for industrial applications
	[190]	Re-ranking transformer model	Personalized recommendation	Amazon	Precision, MAP, nDCG	The self-attention layer provides stronger representational capability and achieves up to 13.95%, 9.60%, and 13.03% relative improvements as measured by Precision@5, MAP@5, and nDCG@5, respectively, against GMF and other ranking models
	[340]	Augments interaction graphs using an LLM-based algorithm	Recommendation accuracy and data sparsity	Netflix, MovieLens	Precision, recall, nDCG	The LLM-based algorithm outperforms various baseline categories, including CF, recommenders with side information, data augmentation methods, and self-supervised methods. On Netflix, it achieves approx. improvements in recall (5.82–13.95%), nDCG (19.06–21.43%), and precision (7.89%). On MovieLens, it improves recall (4.88–7.82%), nDCG (9.06–10.52%), and precision (6.29%)
	[129]	LLMs as zero-shot ranking models	Recommendation performance	MovieLens, Amazon	nDCG	LLMs demonstrate strong zero-shot ranking abilities. GPT-3.5 excels in ML with N@1 of 23.33 and N@20 of 53.73, while GPT-4 leads in Amazon with N@1 of 39.50 and N@20 of 65.25
	[84]	GraphRec coherently models graphs and heterogeneous strengths	GNN challenges: graph complexity and heterogeneity	Ciao, Epinions	RMSE, MAE	GraphRec integrates rating and social network data using advanced components, resulting in improved performance compared to other GNN models. For example, In the Ciao with 60% training data, it achieves the lowest MAE of 0.7540 and RMSE of 1.0093, with improvements of 2.2–20.1% in MAE and 4.2–15.6% in RMSE
	[85]	Building on GraphRec, GraphRec + explicitly integrates relationships between items	GNN challenges: graph complexity and heterogeneity	Ciao, Epinions, Flixster	RMSE, MAE	- In the Epinions dataset, it records an MAE of 0.8441 and an RMSE of 1.0878, improving MAE by 1.2–20.4% and RMSE by 1.4–18.4% - GraphRec + consistently outperforms other algorithms (including GraphRec) across these datasets, achieving the lowest RMSE and MAE metrics with improvements ranging from 7 to 19%
	[343]	Hierarchical GNN + attention mechanisms	User behavior representation learning	Amazon, Yelp	RMSE	It outperforms all the baselines, including PMF, NMF, SVD, DeepCoNN, Attn + CNN, and NARRE, with the highest improvement against PMF. It achieves improvement of 29.20% (average) RMSE in the Amazon and 16.67% RMSE in Yelp
	[141]	Applies graph recommendation techniques such as graph band-stop filters, mirror filtering (MiFi) on user and item networks, and eigenvectors from similarity matrices to handle graph-based challenges	Graph signal sampling and reconstruction problem	MovieLens, Yahoo music	RMSE	- The proposed design reduces RMSE by up to 4.56% for user-based graph filters, 2.78% for item-based graph filters, and 6.85% for MiFi compared to the CF approach on ML data - User-based graph filters and MiFi slightly outperform graph-based RNN methods - The model significantly reduces computational complexity and is much simpler than matrix completion by separating the estimation of singular vectors and singular values

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[331]	Multiview attention method	User behavior representation learning, Data sparsity	Twitter, Foursquare, Facebook	Hit ratio, MRR	- The multiview graph method improves Hit@20 by 27.19% on Foursquare, 54.27% on Twitter, and 44.63% on Facebook, and MRR@20 by 31.08% on Foursquare, 20.89% on Twitter, and 48.11% on Facebook, compared to best in single-view graph methods - However, it assumes that all social networks share the same semantic space, meaning content words are represented similarly across different networks. This assumption may not always align with the unique characteristics or contexts of each network
	[319]	Knowledge graph-aware networks	User behavior representation learning, explainability	Movielens, BookCrossing, Bing news	AUC, accuracy, precision, recall, F1-score	- The model achieves average AUC gains of 9.1% in movie, 11.5% in book, and 10.3% in news recommendations compared to baselines like LibFM and Wide & Deep - It also excels in top-K recommendations, although the performance improvement is lower for news data due to its larger size
	[108]	GCN + Gated GNN + attention mechanism	User behavior representation learning	Yoochoose from RecSys Challenge, Diginetica from CIKM cup	Precision, MRR	The model significantly improves MRR@20 by 10–11.6% on Yoochoose and Diginetica datasets, outperforming GRU4Rec and other session-based models. However, there is a slight drop observe in Precision@20 (1.9% on Yoochoose and 0.9% on Diginetica)
	[189]	Graph attention mechanism	User behavior representation learning	Ciao, Epinions, Delicious	RMSE, MAE, MRR, nDCG	The model shows improvements in RMSE by 5.84% to 7.14% and MAE by 6.74–10.54% across Ciao and Epinions compared to other models (such as SocialMF and GraphRec). In the Delicious dataset, it also outperforms base models with MRR improvements of up to 15.16% and nDCG improvements of up to 22.32%
	[206]	Graph-based federated learning framework	Heterogeneity, personalization, and privacy protection	Ciao, Epinions, Filmtrust	RMSE, MAE	The model provides average relative improvements of 2.99% in RMSE and 4.03% in MAE compared to MF and graph-based baselines
	[321]	GNN for MOOC recommendation	User behavior representation learning	MOOC dataset	Recall, nDCG	- Its sensitivity analysis highlights the ability to effectively preserve user privacy data locally - The method improves Rec@20 by 2.1% and nDCG@10 by 4.6% compared to PinSage (Z. [379, 380]), and also minimizes the impact of users enrolling, a common occurrence in MOOCs with high dropout rates, compared to Session-GCN and GRU4Rec
	[6]	A GNN-based product relationship network	User behavior representation learning	The survey dataset includes car attributes and customer information	AUROC, F1-score	The analysis using 29 features across the three years yields an average F1-score of 0.68, with consistent AUC scores of 0.84 for training and test sets
	[388]	Causal-GNN + GraphSAGE	CTR prediction	Criteo, Avazu, MovieLens	AUROC	Survey data restrictions lead to a limited sample size in the study
	[216]	Semantic knowledge-based graph structure + hierarchical neural layers	Explainable recommendation	Amazon	RMSE, MAE, ROUGE	- The method significantly improves CTR prediction performance in AUC at the 0.01 level and reduced Logloss at the 0.001 level compared to low-order, high-order, and graph models - The method's success in cross-domain CTR tasks suggests strong generalization abilities
	[405]	Attributed-GCN method	Prediction accuracy	Amazon, Epinions	RMSE, MAE	In the Electronic category, the method improves RMSE by 9.74% and MAE by 4.02% over NRT (one neural network-based rating method). In the Home & Kitchen category, it surpasses NRT with 2.59% MAE and 0.88% RMSE. In the Music Instrument category, the method outperforms NRT with a 7.38% RMSE and GCMC (simple GNN-based approach) with a 1.63% MAE
	[256]	eVAE + gated-GNN method	Cold start	Movielens, Yelp	RMSE, MAE	The result analysis shows that using both rating and review information improves prediction performance compared to using only reviews or only ratings. It obtains increasing prediction accuracy by over 3%
	[107]	Deep GNN-based model	Recommendation efficiency and scalability	Epinions, Amazon, Yelp, Flixster	RMSE, MAE, nDCG	- The method excels in strict cold-start recommendations and ranks first or second in warm-start scenarios. It improves RMSE on ML by -0.32–2.62% and on Yelp by 1.10–2.47%. MAE improvement ranges from -0.31% to 3.19% on ML and 2.40–3.12% on Yelp - Their large number of experiments demonstrate improvement in most scenarios - Further, the results indicate the method's stability due to the comprehensive formulation of the feature space and its robustness across different parameters

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[94]	CA-GNN + attention layer	Interpretability of context-aware recommendation	Food, Yelp	RMSE, MAE	- The results show that context-aware methods outperform traditional MF methods, highlighting the positive impact of context features on score prediction - Overall, the method improves performance by 1.8–2.3% compared to top baselines that rely solely on attention networks
	[356]	GNN + multiview learning	Mobile application recommendation	Google Play	RMSE, MAE	- The model achieves a maximum gain of 17.59% in MAE and 12.87% in RMSE with 20% training samples compared to PMF, FM, and previous graph-based models - The embedding approach of the method may struggle to adapt to changing user behavior or app characteristics over time and could limit the model's ability to capture nuanced interactions due to its reliance on pre-constructed similarity matrices and fixed embedding vectors
	[7]	Heterogenous knowledge-based embedding	Explainability	Amazon	Precision, recall, hit ratio, nDCG	- It achieves at least 80% improvement over MF models and at least 10% over deep learning models - It gains nDCG improvement over the best baseline JRL (Y. [392, 399, 401, 402]) on Amazon by 5.6% for CDs and Vinyl, 78.16% for Clothing, 23.05% for Cell Phones, and 45.56% for Beauty
	[137]	GNN propagation layer	Personalized recommendation	Amazon, CiteULike, Yelp	Hit ratio, nDCG	The method outperforms other models across all datasets. For instance, on Amazon, it improves Hit@5 by 1.67 times against GMF, 1.48 times against NCF, and 1.35 times against LightGCN. nDCG metrics also improve, with gains of 1.27–1.54 times compared to GMF, NCF, and LightGCN
	[138, 139]	Aspect-based graph model	User behavior representation learning, data sparsity	Amazon	MSE	The method achieves an average RMSE improvement across Amazon categories equal to 1.48% compared to NMF, 8.10% compared to SVD, 8.39% compared to HFT, 10.57% compared to DeepCoNN, and 12.38% compared to NARRE
	[398]	GNN encoder	Sequential recommendation	Amazon, MovieLens	Hit ratio, nDCG	- This GNN method outperforms the popular neural method GRU4Rec and demonstrates the effectiveness of modeling user sequences using a dynamic graph - It improves nDCG@10 on Amazon dataset by 6.87% for Beauty, 3.71% for Games, and 3.81% for CDs compared to the strongest baselines. Particularly notable for the Beauty dataset, which is sparse and has shorter interactions
	[32]	Item-item GNN method	Sequential recommendation	Taobao, Kuaishou	AUC, GAUC, MRR, nDCG	This graph-based method outperforms the popular neural model GRU4Rec, improving AUC by around 0.03 on the Taobao dataset and 0.04 on the Kuaishou dataset, both with statistically significant improvements (p-value < 0.001)
	[140]	GNN strategy with negative sampling	Capturing negative sampling	Alibaba, Yelp, Amazon	Recall, nDCG	- The method significantly brings improvement to GNN-based recommenders, showing a 26% improvement for NGCF and a 22% improvement for LightGCN in terms of nDCG@20 - It demonstrates comparable level of complexity to other sampling methods
	[263]	Stacked AE + BERT embedding + Graph-based clustering	Health awareness food recommendation, Cold start	user-food rating data from Allrecipes.com	Precision, recall, F1, AUC and nDCG	- The deep network shows improvements of 9.6% to 16.4% on used metrics compared to user-only and item-only recommendations - Additionally, the time-aware graph clustering algorithm outperforms the non-time-aware system, with improvement of 13.4% in precision and 12.5% in recall
	[289]	Graph-based DNN + k-means	Recommendation performance and scalability	CiteULike, AMiner Citation	Recall, nDCG	Clustering and graph-based approaches boost the deep learning component's performance, making the system outperform ItemKNN [191], especially with sparse data and user cold-start problems, achieving high recall with a median of 0.415

Table 5 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[397]	Inductive matrix completion based on GNN model	Scalability	Flixster, Douban, Yahoo, MovieLens	RMSE	The method shows a significant improvement in performance across all datasets, with the most notable improvement on the Yahoo Music, achieving a 16.59% RMSE than the PinSage

Method: {HFT hidden factors topics, GRU gated recurrent unit, LSTM long short-term memory, Bi-LSTM bidirectional LSTM, GCN graph convolutional neural network, GAT graph attention network, DNN deep neural network, GPT generative pre-trained transformers, BERT Bidirectional Encoder Representations from Transformers, LLaMA large language model meta-AI, PaLM pathways language model, KNN K-nearest neighbors, LDA latent Dirichlet allocation, SVD singular value decomposition, eVAE extended variational autoencoder, CDL collaborative deep learning, CTR click-through rate, NRT neural rating and tips generation, MOOC massive open online course, BPR: Bayesian personalized ranking, MC Markov chain, TF-IDF term frequency–inverse document frequency}

Evaluation Metrics: {RMSE root mean squared error, MSE mean squared error, MAE mean absolute error, nDCG normalized discounted cumulative gain, MAP mean average precision, MRR mean reciprocal rank, AUC area under curve, ROC receiver operating characteristic, AUROC area under the ROC curve, ROUGE recall-oriented understudy for gisting evaluation, GAUC group AUC; MAPE mean absolute percentage error, PRAG pair-wise ranking accuracy gap}

4.1 Pre-modeling: datasets and feature learning

4.1.1 Dataset

Big Data is an impelling cause behind recommender systems. Based on the recommendation filtering techniques, a system demands a different type of data:

- Content data: This includes data like item attributes, user profiles, and system information. For example, if an item is a “book” then associated attributes can have the “book name,” “author of the book,” “genre of the book,” and “published year.” Similarly, a user profile can have attributes such as “age,” “gender,” and “occupation.” Content-based filtering highly depends on content keywords to make recommendations.
- User Interaction data: These data are user-generated feedbacks such as implicit (clicks, check-ins, purchases) [191], [37] and explicit (ratings, feedbacks) [145], [325]. These data along with algorithms equally contribute to achieving the desired outcomes. Implicit data, for example, analyzes users’ activities and helps understand users’ preferences when providing recommendations. On the other hand, explicit data assists in properly modeling user–item latent variables.
- Contextual data: This type includes either static or dynamic context [3] as used in context-aware recommendation systems. [3] further classifies these context type into fully observable, partially observable, and unobservable taxonomies. Different combination of these two context taxonomies are well described in [165]. Context data can include hidden features such as user emotions extracted from user comments, and item reviews. Alternatively, these data can be external features such as social, tags obtained from various platforms or real-time event data such as temporal, weather, and location.

Researchers follow a common practice of using publicly available datasets to perform recommendation tasks. Exploring such literature datasets with associated elements may assist future improvements in the recommendation research area. To this end, we provide a detailed summary of existing benchmark datasets, including domain, respective publications, description, and data types given in Table 7.

4.1.2 Feature learning

Feature learning tasks such as data augmentation and feature representations are important components that significantly contribute toward improving recommendation performances. The manual crafting of rating matrix,

textual, visual, or graph-based data requires lots of time. Also, it limits the learning capabilities of extracting relationships between features. Deep learning is changing the way of representational learning by employing supervised or unsupervised strategies to enable automatic features crafting. The advantage of deep learning in extracting intricate relationships shows their dominance in various fields including recommendation. In this section, we discuss deep learning literature focused on feature learning in recommender system. Table 8 lists the summary of feature learning categories alongside deep learning publications.

Numerical Features. A research study introduced a self-supervised strategy to construct a nonlinear high-dimensional representation in the AE bottleneck layer using a user–item rating matrix with the goal of engineering features for recommender systems [260]. Another study employed representation learning via dual AE, simultaneously learning new hidden representations of users and items from a rating or check-in matrix [434]. A valuable contribution made to e-health and medical informatics by [227], where the authors applied a categorical/numeric feature selection process to diagnose multiple heart diseases. The model assessed the risk of predicted disease to make medical recommendations. In another work, the recommendation system used RBM network to describe the user’s explicit feedback, such as numerical ratings to build a top-N list for each user [44]. The next research technique developed a personalized recommendation model for analyzing sparse and dense category information [230].

Textual Features. In a study, a CNN-based model called deep cooperative neural networks was developed to utilize user reviews for modeling user behavior and item attributes [419]. In the following year, a similar method that used the same type of CNN network with an additional transform layer to approximate the user and item review texts [30]. Unlike prior representational techniques, an interaction-based matching matrix was created for a given pair of input texts and then fed into CNN to compute the matching score [69]. Furthermore, another method proposed a recommendation model using GAN [40, 41]. In the methodology, the generative network aimed at generating user profiles and item representations from rating and text reviews and the discriminative network aimed at distinguishing the predicted ratings from real ratings. Another publication utilized the RNN-based GRU model for encoding text sequences into latent vectors [15]. The proposed model improved the sparsity of the observed rating matrix by regularizing the CF model.

Topical Features. In the past few years, several studies have emphasized deep learning-based topic modeling approaches to learn various topical or domain-specific features for better user/item representation learning. For example, one method adopted a semi-supervised topic

model to extract product attributes and sentiment lexicons from user textual evaluations, which were then further fed into an LSTM encoder using an interactive neural attention mechanism to enhance the learning of user and item sentiment-aware representations [57]. In the following year, the authors devised a multichannel CNN architecture designed for aspect extraction tasks. They subsequently clustered the resulting aspects using LDA technique [56], and estimated aspect ratings using a lexical method. In their recent work, they utilized neural attention approaches to construct adaptive user/item representations and obtained fine-grained user–item interaction to improve item recommendation accuracy [59]. Another example involves the application of regularized latent Dirichlet allocation (rLDA), which estimates latent themes for each document in relation to other documents [209]. The authors used a collective inference approach to estimate the distribution of latent themes and utilized a deep network structure to examine the benefits of rLDA.

Semantic Features. A research method developed a hierarchical and symmetrical RNNs model called SEMA (Semantic mEanings and teMporal dynAmics) to capture semantic meanings from textual content and temporal dynamics from historical information [393]. Following, a novel RNN-LSTM network approach extracted meaningful semantic data from multisource supplemental reviews [336]. Another article proposed an attention-based bidirectional GRUs to capture semantic information in scientific papers for tag recommendation [114]. An approach used multitasking CNN to capture short and long semantic vectors for tag recommendation [394]. Next, a method analyzed the semantic gap to offer personalized recommendations using deep learning [275], whereas other method extracted group semantic features for the group recommendation model [143]. Furthermore, a research study coupled deep learning with LFM to tackle the problem of item cold start by learning items’ semantic representations [79].

Multimedia. An approach attempted to integrate multimedia content with the CF mechanism [362]. It employed two GCNs as encoders to model users and items latent factors. The model used the inner product of the resulting latent vectors to construct preference scores. Another novel exploration applied MLP to extract a rich set of features, including videos and user behavior data for YouTube recommendation [53]. Other reference focused on image tag recommendation and used deep learning to extract powerful visual features [231]. Yet, another method feature-engineered visual attributes, non-visual attributes, and time dynamics to build a large-scale recommender system [368]. Next, a novel attentive network (Ante-RNN) method jointly learned image representations with textual alignment and text representations with a topical attention

mechanism in parallel [198]. This approach aimed to make the recommendation process more transparent and explainable by combining whole latent visual–semantic alignments with a hybrid attention mechanism that includes topical and contextual attention. Finally, a different study developed a deep model to learn unified feature representations for both users and images by transforming heterogeneous user–image networks into homogeneous low-dimensional representations, facilitating a recommender to recommend images to users [99].

Spatial and Temporal. An article proposed the temporal-CNN recommendation (TCR) model to capture the evolving tendencies of user preferences and item attributes over time, aiming to better capture the interaction between users and items [220]. In the next article, LSTM networks were utilized to capture temporal dependencies for both users and movies [341]. This method incorporated past observations to predict future trajectories in an integrated manner. Recently, several approaches have emerged that incorporate both spatial and temporal factors using different ways to enhance the modeling of users' preferences. For example, a method leveraged spatial and temporal intervals between neighboring POIs and initialized a time window to take several POIs as input using the RNN network [200]. Another recommendation research developed a hierarchical model to directly incorporate spatial and temporal intervals into LSTM's existing multiplicative gates [161]. Other LSTM models focused on capturing short- and long-term user preferences. For example, [409] with the help of new time and distance-specific gates and [296] using a non-local network and a geo-dilated RNN. Additionally, a model was proposed to learn POI–POI relationships based on spatial, temporal, and preference factors using a GAT. [188]. It is worth noting that previous approaches often overlooked the importance of item usage sequences and time intervals of the time series data from users [47]. Therefore, in a study, the researchers designed an RNN model with a hierarchical structure (comprising two layers: one for a long time patterns and another for a short time patterns) to consider the sequence and temporal aspects of users' item usage history [47].

Graph Embeddings. A multitask, end-to-end deep recommendation framework utilized knowledge graph embedding (KGE) to design a cross and compress unit to link the KGE and recommendation tasks. This unit automatically learns high-order interactions of item and entity features and transfer knowledge between the two tasks [320]. Another method proposed a graph-enhanced representation learning architecture for news recommendation [97]. This methodology adopted a transformer architecture to build news semantic representations from textual content and user representations from their ID embeddings and historically clicked news. Users' and news representations

are further enhanced by aggregating their neighbor embeddings through a GAT. The final recommendations are generated by computing a dot product between the enhanced user and news representations. A different method proposed a novel graph-based method to facilitate a cross-domain recommendation system [406]. The model is comprised of two parts. The first part deployed graph convolution to capture the high-order interaction relationships across domains and a propagation module to capture how user preferences propagate in the graph. The second part defined the knowledge integration and prediction module, where the latent knowledge acquired is integrated and fed into the MLP network to get recommendation predictions. Another method focused on transferring cross-session information during individual session representation learning using a full GNN [258]. More recently, a technique was proposed for top-N personalized recommendation using a GNN to take advantage of co-learning relationship and sequential patterns [321]. In this approach, CNN is used to extract significant features from the sequential patterns with the current items.

4.2 Post-modeling: evaluation metrics

Evaluation metrics require measuring the performance of recommender systems to ensure their generalization on the future unseen data. The recommender system metrics can mainly be divided into accuracy metrics and beyond accuracy metrics. The accuracy metrics differ with the recommendation problem definition, which is typically classified into three types: First, the classification problem, in which the system recommends an item; Second, the regression problem based on rating prediction task; Third is the ranking problem based on top-N items ranking task. The other aspect beyond accuracy criteria measures the quality of the recommender system. A brief insight of popular recommender system evaluation metrics is discussed in Table 9.

5 New generation recommender system trend

Conventional recommender systems are mostly dependent on either rating matrix or user's behaviors (clicks or check-ins) to find user–item interactions. However, in real-world scenarios, this data tends to be highly sparse and even biased toward highly ranked items [45]. Eventually becomes harder for recommender systems to identify the underlying user–item interactions accurately. Therefore, traditional methods are unable to effectively recommend relevant items to users. New generation recommender systems use effective elements with innovative ways to

exploit intrinsic relationships between users and items. Consequently, these systems gain the ability to provide more accurate recommendations. Here in this section, we present brief discussion on new generation recommender systems, along with some insights into future research perspectives.

5.1 Joint representation learning (JRL)

The trending works have adopted JRL using different information sources [402]. For example, the article used MLP networks to learn the corresponding user and item representations from review text, product image, and numerical rating [402]. The model was trained using a pairwise learning algorithm to retrieve top-N recommendations. Another example is a model that jointly learns textual reviews representations and image representations using Attentive RNN networks [198]. Through RNN attention networks, the model extracted long-term and short-term behaviors. In addition to this, the model extracted historical behaviors and diverse temporal factors of the user's clicking sequence of items (time interval and the time of week) to further improve the recommendation performance. A different method architected two encoder models: a rating-based encoder for learning explicit features from user and item rating patterns and a review-based encoder for modeling users and items from text reviews [198]. The prediction model used these joint rating review representations to find item ratings for users. Next study combined user–item interaction graph and review text in the representation learning process [284]. The proposed model comprised of three networks. The first and second networks extracted user and items reviews representations using self-attention mechanism. And the third network extracted node representation of user–item interaction graph using graph feature extraction module. Because direct concatenation of different modalities is not possible in this process, the authors developed a mutual attention technique to correlate these modalities. The method combined these correlations along with the outputs of feature extraction modules in the prediction layer to obtain scores. In other work, MLP was used to capture user–product preference interaction, and CNN was used to capture local semantics of review text [88]. These representations were jointly projected into MF to yield rating scores.

One possible research avenue involves exploring JRL, which involves considering modalities of data to address recommendation quality in diverse recommendation scenarios including soundtracks, videos, and research articles. Furthermore, exploring the integration of external knowledge sources, such as semantic ontologies or domain-specific databases, can gain a deeper understanding of the content and context, leading to more relevant

recommendations and providing stronger explainability. To tackle the persistent challenge of cold start, enriching user profiles from multisources [198] can be beneficial.

5.2 Session-based recommender system (SBRS)

SBRS takes session data to capture short-term but dynamic user preferences, enabling more accurate and timely recommendations [328]. The study introduced a GRU-based RNN model that takes the current session event (user click) as an input and predicts the next item event [124]. To capture sequential dependencies, the model utilized concurrent mini-batch fragments and computed prediction scores within a small subset of output items. Another study focused on preprocessing techniques such as click sequence augmentation that handles noisy clicks, adds temporal threshold to discard older click sequences, and embeds information of items clicked by user to improve recommendation quality and enable deployment in real item settings [300]. Recently, a technique employed parallel RNNs to incorporate item attributes from images and text descriptions [125]. They used multiple training approaches (alternating, residual, and interleaving) rather than backpropagating errors in the neural network to increase recommendation accuracy. Similar item-based multimodal features are leveraged to improve top-N recommendation, combining a marginalized DAE model with RNN [54]. Next study utilized cross-session information in representation learning, employing a weighted graph RNN attention method to link different sessions [258]. A different method adopted hierarchical RNN to transfer user activities across sessions [259]. Another method modeled sequential patterns as local features of the embeddings of previous items using CNN [303].

One research direction involves refining SBRS representational learning using side factors. Past studies [207, 277] used contextual factors to improve the SBRS performance. However, building a context-aware framework in SBRS remains challenging, resulting in underutilization of side factors. Another opportunity exists to investigate SBRS that comprehend short-term and longer-term representations, as traditional long-tail recommendation methods have difficulty considering the sequential order of data, leading to decreased accuracy when predicting the next item in a session [201]. Additionally, leveraging complex user–item transition relationships within sessions could provide valuable insights. Another aspect is exploring SBRS in cross-network applications, where capturing users' global interests through cross-session dynamics can enhance recommendations. Furthermore, enhancing users' dynamic preference for accurate SBRS result. The continuous interactive process between users and items in session generation significantly

influences the precise learning of users' dynamic preference, yet it has been overlooked in existing SBRS studies.

5.3 Context-aware recommender system (CARS)

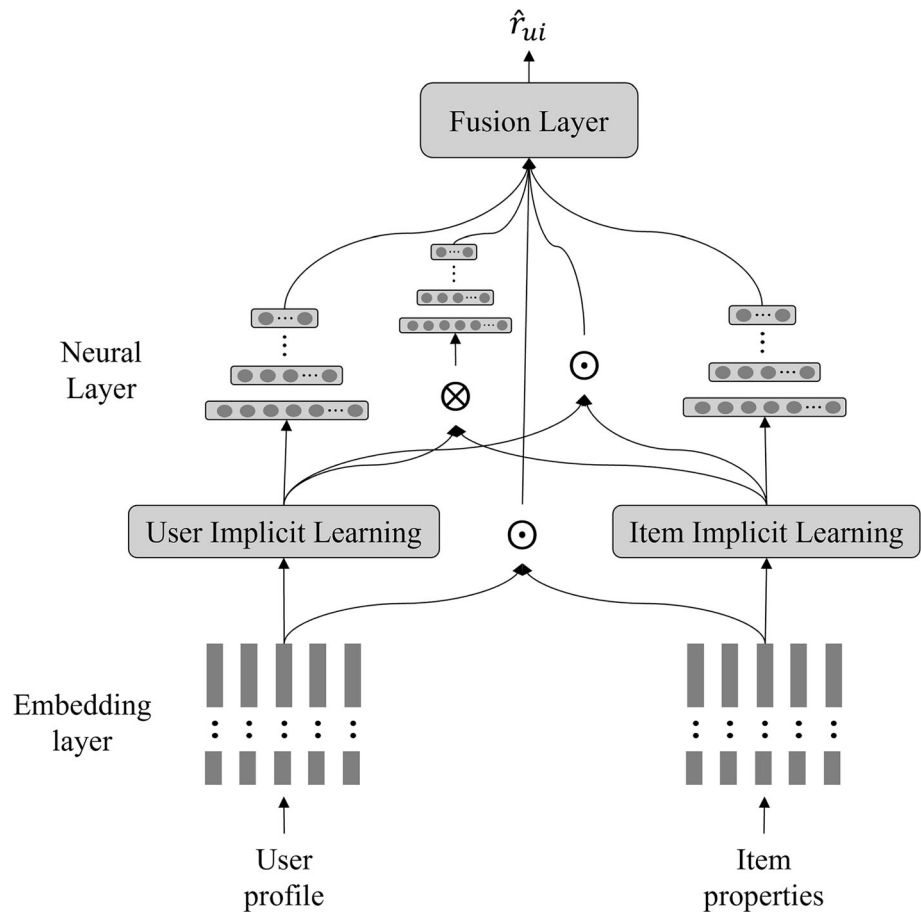
CARS adapts system applicable contextual information for effective recommendations. Contextual data can be static or dynamic. For instance, one method integrated static user contexts into RNN session model to learn high-order interactions between user contexts and previous items [286]. Another approach generated sequential latent context from dynamic contextual factors (time, location, mood) using two encoder–decoder models (AE and LSTM) [207]. Next, a study accounted spatial–temporal dependencies from heterogeneous contextual factors for POI recommendation [176]. They proposed an attention layer in LSTM encoder–decoder framework to obtain selective information. Another study extracted review context features that includes POI properties, user preferences, and sentiment indications using CNN for POI recommendation [357]. A recommendation method acquired contextual modeling and converted the joint effects of spatial and temporal context into knowledge graph, which further used as embeddings to capture successive user activities for POI suggestion [257]. On the contrary, a study defined pre-filtering contextual incorporation into NCF algorithm using item-splitting approach, which means item attribute values in the different contexts is split and handle as different context items [257]. On the contrary, other study defined pre-filtering contextual incorporation into NCF algorithm using item-splitting approach, which means item attribute values in the different contexts is split and handle as different context items [11]. For example, if a book read by user based on seasons (Christmas, Winter, Autumn) then it can be scaled into three representations for item book. Alternatively in their recent work, the authors designed user-splitting approach to perform pre-filtering contextual with NCF [10]. A popular method ConvMF by [157] and extension method by [158] extracted item latent factors from document (context) using CNN and integrated into PMF for recommendations.

CARS are vulnerable to shilling attacks, with users manipulating positive information for their products and negative information for competitors. Detecting these attacks and building trust in CARS usage requires further research. Future efforts can improve contextual learning by combining pre-filtering, post-filtering, and contextual modeling filtering methods. Moreover, current research mainly emphasizes the representational aspect of context and ignores the interactive nature of context, which considers users' dynamic preferences. Considering the interactional view insights to improve CARS accuracy. Additionally, there is a research demand in handling

heterogeneity of contextual data that can bring high significance and offer explainability for CARS models.

5.4 Cross-domain recommender system (CDRS)

CDRS is a promising research direction for tackling issues like cold start and data sparsity. Elkahky's team proposed a multiview DNN that maps users and items to a latent space, maximizing similarities between users and their item preferences [81]. It is content based that considers the mapping of users and items to a latent space. These mapping maximized the similarities between users and their preferences for items. Next research proposed an embedding and mapping CDRS [219]. The system is comprised of two phases. The first phase employed MF and BPR to learn latent factors of users and items in both source and target domains. The second phase mapped only popular entities across domains using linear and MLP-based non-linear functions. In contrast, a different approach used a Bayesian variational autoencoder (VAE) to develop a cross-domain recommendation model [278]. The method extracted the latent representations of users and items in many domains using input as the rating data and the model trained using the data of overlapping users. Predicting user preferences for items in the target domain based on user preferences for items in the source domain yielded the final recommendation set. An innovative approach took beneficiaries of both text content embeddings and selective knowledge transfer using a multilayer feedforward neural network (G. [131]). In the source domain, the approach focused on text words to match user preferences, and then transferred the adaptive weights to the target domain to learn user–item interactions. A similar approach fused side information into a CDRS [90]. It utilized two DNNs, SDAEs to extract latent factors from review texts and item information in both auxiliary and target domains and MLP network to transfer latent factors between domains. The networks are highly dependent on side information to enable feature transferring. A deep domain adaptation recommendation (DARec) model only relies on rating matrices from which users' rating patterns are extracted and transferred between domains [384]. At first, DARec used the famous AutoRec algorithm [270], to learn latent embeddings in source and target domains. These embedding sequences are interleaved and fed into a DNN to extract rating patterns. Further, these patterns are reconstructed in the source and target domain using a deep feedforward neural network. Also, DARec contained another DNN that takes these patterns to predict corresponding domain labels. With the dual learning concept in cross-domain, a proposed model extracted feature embeddings using an AE network and defined an orthogonal mapping function to enable the bidirectional transfer of

Fig. 5 Dual learning of user and item features

these latent representations [174]. It is an iterative process, which means learning repeatedly occurs in both domains until stability is achieved. Another dual learning approach conceptualized dual knowledge transfer and allowed dual mapping in the network's hidden layers (G. [130]). The recent work adopted unsupervised domain indistinguishable learning through discriminative adversarial networks [313]. The method addressed the cross-domain data imbalance issue with adversarial learning in the source domain and employed an adversarial loss function with a discriminative objective in the target domain.

CDRS relies on a large amount of data from multiple source domains; however, existing datasets contain single-domain information. For example, Netflix Prize holds information about Netflix users and movies and Amazon about the amazon users and their products. This brings difficulty in obtaining the interrelation among different domains. Also, efforts are required to standardize conditions and limits that should be considered when using existing datasets [154]. This calls for the development of new datasets that justify their association with the appropriate domain and CDRS recommendation scenario. Furthermore, research demands to study the influence of time-varying information, since the number of items is always

changing over time [134], and with that user interest for items can change. Although deep learning algorithms have proved effective in CDRS, their potential remains largely unexplored. Leveraging deep learning or deep hybrid methods can enhance recommendation performance in CDRS models.

5.5 Group recommender system

The goal of group recommendation is to capture the preferences of each group member into a single recommendation list which meets the needs of all group members [333, 334]. An article created probabilistic generative model for a geo-social group recommendation system that recommends POIs for groups [408]. The model considered different influential factors like personal preferences, group topics and social relationships for POI selection in a target group. Another generative model recommended new items especially satisfying different groups of users [311]. The approach used VAE to generate low-dimensional latent spaces and adopted a greedy maximum coverage technique that learns these vector spaces to find promising items. An attention-based method adopted the NCF framework to learn group-item interactions [29]. The network assigned

Table 6 Summary of existing hybrid deep learning techniques for recommender system

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
Hybrid	[88]	MLP + CNN	User behavior representation learning	IMDB, Yelp	RMSE	- The model achieves significant improvements over sentiment classification-based baselines. For example, it outperforms UPNN [301] by 11.4%, 13.6% and 14.1% on IMDB, Yelp 2014 and Yelp 2013, respectively - The model achieves significant improvements over CF-based baselines. For example, it outperforms TLFM [283] by 5.2%, 7.3% and 6% on IMDB, Yelp 2014 and Yelp 2013, respectively
	[325]	AE + MLP	User behavior representation learning	MovieLens	RMSE, MAE	- The model significantly outperforms SVD, AutoRec, and DAE with RMSE improvements of 0.03 on ML-1M and 0.03 on 10M, and MAE improvements of 0.04 and 0.06, respectively - Deeper model architecture improves accuracy but slow down the training process due to increased complexity
	[241]	CNN + RNN	Session-based recommendation	Recsys Challenge	MRR	The hybrid method achieves a final MRR score of 0.6702, securing 16th place in the RecSys Challenge 2019 and demonstrating the effectiveness of combining inter- and intra-session context to represent user interactions in session-based recommender systems
	[118]	Neural feedforward + CNN	Explainability	Yelp, Gowalla	Hit ratio, nDCG	The model improves by 3% over NeuMF, 3.9% over JRL, 9.2% over MLP, 9.5% over MF-BPR, and 227.6% over ItemPop, all with statistical significance ($p < 0.05$)
	[345]	CNN + gated RNN	User behavior representation learning	Amazon, Yelp	MAE, RMSE, precision, recall, nDCG	The hybrid model performs better than standalone CNN and neural CF models by using context to enhance CF methods, leading to average improvements of 3.7% on the Amazon Instant Video, 5.2% on the Amazon Apps for Android, and 2.1% on the Yelp
	[381]	CNN + AE + MLP	Recommendation accuracy	Adressa news	Accuracy	- Overall, this method shows 4% to 58.2% higher accuracy in the test set and 0.2–59% higher in the whole dataset compared to other methods - Specifically in the test set, the model achieves 16.1% higher accuracy than CNN and 58.2% higher than LSTM and reasonably higher than other methods such as SVM, NB
	[267]	RBM + CNN	Personalized recommendation and privacy-preserving	Healthcare [113]	RSME, MAE	- The proposed RBM-CNN outperforms all other models significantly, with an MAE of 0.03095 after 14 epochs compared to MF (0.06219), SVD (0.05239), WSVD (0.04718), and RBM (0.04093) - For $K = 10$, the model performs best, achieving an RMSE score of 2.58216, which is lower compared to MF (2.69592), SVD (2.67776), WSVD (2.66688), and RBM (2.62337)

Table 6 continued

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[27]	Bidirectional LSTM + MLP	User behavior representation learning, scalability, privacy-preserving and resource-efficient	Sticker from messaging app and Subreddit	AUC	- Federated user representations improve AUC by up to 50.51% on Subreddit and 8.39% on Sticker - The model produces similar user embeddings and clustering patterns as server-based models, with minimal accuracy drop (3.72% on Subreddit) and no AUC loss on Sticker, balancing privacy and performance
	[390]	SDAEs + stacked convolutional	Recommendation quality	MovieLens, IntentBooks from Microsoft engines	MAP, recall	The results show that using structural knowledge and network embedding greatly enhances recommendation performance, while traditional MF is the least effective due to ignoring structural knowledge. This method outperforms others by leveraging the heterogeneity of network embedding but lacks clear evidence of its impact on sparsity
	[316]	AE + RNN	User behavior representation learning	CiteULike, Netflix	Recall, MAP, BLEU	- The model offers significant gains over all baselines, with relative improvements of 2.0–16.7% in CiteULike and 5.7–22.5% in Netflix - It enhances MAP by 35% for CiteULike and 10% for Netflix compared to the strongest baseline (CDL) - The model also achieves higher BLEU scores of 46.60 for CiteULike and 48.69 for Netflix, exceeding CDL, CTR, and PMF
	[326]	GAN-based neural memory recommender networks from neural Turing machines [102] and memory network [293]	Streaming recommendations	MovieLens, Netflix	Hit ratio	The model addresses the limitations of classical negative sampling approaches, leading to better model parameter inference. Further, experimental results demonstrate the model's superiority in streaming recommendation scenarios
	[423]	RBM + NADE	Recommendation accuracy	MovieLens, Netflix	RMSE	The model with a single hidden layer achieves RMSE scores of 0.830 on ML-1M, 0.772 on ML-10M, and 0.803 on Netflix over baselines (MF and traditional deep models)
	[12]	CNN + MLP	Data sparsity	MovieLens, Amazon	RMSE, MAE	The model improves RMSE by 7.42% and MAE by 9.38% over the second most accurate method on dataset with over 99.9% sparsity
	[64]	CNN + LSTM	Session-based news articles recommendation	Globo.com	Hit ratio, MRR	- The proposed meta-architecture shows significant improvements in session-based recommendation methods, achieving a 10% increase in hit rate and 13% increase in MRR over the best benchmark (rule-based method) - It consistently maintains higher accuracy over time, with a median Hit@5 of 0.72 compared to 0.65 for the best benchmark, representing a relative improvement of about 10%
	[73]	Neural co-autoregressive models + CF	Improving top-N ranking	MovieLens, Netflix	RMSE, hit ratio, nDCG	- Our model achieves a substantial RMSE improvement of 0.006 over CF-NADE on the MovieLens dataset, and an RMSE of 0.799 on Netflix, significantly outperforming AutoRec (0.823) - Additionally, it delivers an nDCG@10 score of 0.492, outperforming the state-of-the-art method NeuMF by a large margin, with a relative improvement of 10.1% (0.045)

Table 6 (continued)

Models	Existing ref	Key techniques	Problem focus	Datasets	Evaluation measures	Remarks
	[430]	CNN + RNN	Intelligent recommendation	ask.39.net	Accuracy, precision, recall, F1-score	- The proposed model averages 0.96 accuracy and 0.86 F1 across medical departments, surpassing baselines (CNN, RNN, or combinations) with accuracies over 0.88 and F1 values over 0.72 - It also obtains high precision (0.985) and recall (0.940), highlighting its strong performance across these metrics
	[327]	Time-aware attention network + Deep MF + MLP	User behavior representation learning	MovieLens, Amazon	Hit ratio, nDCG, RMSE, MAE	The attention hybrid model achieves a maximum improvement of 4.43% on ML and 1.76% on Amazon compared to traditional integrated models (NCF, NeuMF, DeepCF [68])
	[49]	Ensemble model from fully connected layers, convolutional layers, and pooling layers	Recommendation performance	Amazon	Accuracy	It achieves an accuracy of 0.99, outperforming various ML algorithms, including LR, KNN, CART, SVM (all with 0.97 accuracy), and Linear Discriminant Analysis and NB classifiers (both with 0.96 accuracy)
	[126]	Bidirectional network with LSTM + GRU	Data sparsity	Amazon, OpinRank	RMSE, accuracy, recall	- This method shows strong performance on both datasets, achieving an accuracy of 97%, recall of 61%, and RMSE of 0.87 - On Amazon dataset, the PMF model performs poorly with an RMSE of 0.983, while CTR's bag-of-words approach results in an RMSE of 0.976. LSTM and GRU outperform PMF and CTR with RMSE of 0.943 and 0.932, respectively. Meanwhile, this method achieves an RMSE of 0.876, surpassing all baselines - Additionally, the method addresses the data sparsity issue, as the recall rate is higher in the sparser OpinRank dataset (37%) compared to Amazon (27%)

Method: {*NeuMF* neural MF, *SVD* singular value decomposition, *WSVD* weighted SVD, *SVM* support vector machine, *NB* naive Bayes, *PMF* probabilistic matrix factorization, *CDL* collaborative deep learning, *CTR* collaborative topic regression, *BPR* Bayesian personalized ranking, *DAE* denoising autoencoder, *SDAE* stacked denoising autoencoder, *JRL* joint representation learning, *LSTM* long short-term memory, *GRU* gated recurrent unit, *LR* logistic regression, *KNN* K-nearest neighbors, *CART* classification and regression trees, *NADE* neural autoregressive distribution estimator}

Evaluation Metrics: {*RMSE* root mean squared error, *MSE* mean squared error, *MAE* mean absolute error, *nDCG* normalized discounted cumulative gain, *MAP* mean average precision, *MRR* mean reciprocal rank, *AUC* area under curve}

different weights for each user of the group to enable dynamic learning of score aggregation strategy for the group from given data. Following, a method developed a novel type of attention mechanism in which each member of a group is represented by a separate sub-attention network [310]. Each sub-attention network captured the representative group member's preference in relation to the other members of the group. A method aimed to extract a high level of representation of group preferences in a group recommendation [133]. An article developed a DBN network to induce high-level representations from low-level features, and a dual-wing RBM network to learn representations of group preferences through joint modeling of individual and collective features. Research defined a latent group recommendation based on a dynamic PMF model

with CNN [314]. The authors from a study highlighted two aspects of the method to achieve accurate group recommendation: the first aspect utilized social attributes and converted them into a latent vector to learn the deep semantic features from it, and the second aspect performed preference learning to identify interactions between group and its members [144]. In advance, two attention-based approaches used social information to enhance the representation of an individual member in a group and group-item interactions that eventually leverage the performance of the group recommender systems [376].

One research interest is to explore dimensionality reduction methods for representing multidimensional data in two-dimensional space. This can provide compact representations of users and items for accurate item

Table 7 Summary of existing benchmark datasets

Domain	Dataset (publications)	Description		Content data			Contextual data
		User instances	Item instances	User profile	Item attributes	Explicit/implicit	
Books	BookCrossing [151, 260]	278,858	271,379	Not Available (NA)	✓	Both	Cover image, Uniform Resource Locator (URL), time (publication year)
	Library Thing [42, 44]	73,882	337,561	NA	NA	Both	Reviews
	Goodbooks [428, 430]	53,424	10,000	NA	✓	Both	Tags
Citation/ Research publications	AMiner [73, 77]	1,436,990	1,932,442	NA	✓	Implicit	Title, category
	Open Corpus [239]	8.3 million	6.9 million	NA	✓	Implicit	Documents, citations
Business/e-commerce	Yelp [38]	43,873	229,907	NA	NA	Implicit	Reviews, images, tips, location, user-to-user relationship
	Amazon Product Reviews [38]	20.98 Million	9.35 million	✓	✓	Both	Product description, reviews, images, review time stamps, item-to-item relationship
	Tripadvisor [38]	NA	20,491 unique reviews	NA	NA	Explicit	Item reviews
Jokes	Jester [387]	73,496	100	✓	NA	Explicit	Jokes
Movies	MovieLens (ML-1 M) [42, 44]	6,040	3,900	✓	✓	Explicit	Rating time stamps
	MovieLens (ML-10 M) [42, 44]	71,567	10,681	NA		Explicit	Tags, rating and tag Time stamps
	MovieLens (ML-20 M) [145]	138,493	27,278	NA	✓	Explicit	Tags, rating and tag times tamps, item-to-item links
	Netflix Prize [42, 44]	480,189	17,770	NA	✓	Explicit	Time stamps
Musics	Million Song [245]	1.2 million	380,000	✓	✓	Implicit	Tags
	LastFM [130–132, 136]	50,202	122,877	✓	✓	Implicit	Tags
News	Globo.com [64]	80 million	3 million	NA	✓	NA	News articles, user interaction time stamps
	Smart Media Adressa [389, 394, 396, 404]	561,733	11,207	✓	✓	Implicit	Item reference URL, title, category, device information, event time, article published time
	Yahoo news [243]	7,642	11,915	✓	NA	NA	News category, sub-category and reference URL
Social	Pinterest [379, 380]	1,000,000	1,456,540	NA	✓	Implicit	Tags (pins), Images

recommendations [62]. Investigating group-specific features, such as embedding group profiles and semantics, to enhance recommendation quality for both similar and diverse groups. Another aspect to consider is understanding the complexity of group preferences, requiring the

development of a qualitative deep model that can address the complexities of heterogeneous groups in real time for improved recommendations.

Table 8 List the summary of feature learning categories with respective deep learning publications

Categories	Types of features	Deep learning techniques	Deep learning publications
Numerical	rating matrix, check-ins, users' age, user id, item id	RBM, AE, MLP	[260, [434, [227, [230, [44]
Textual	reviews, tags, documents, pins, URLs	CNN, RNN, DAN, GAN	[419, 30, 69], 41, 15], 46, 284]
Topical	topics (word or a couple of words), aspects	AE, DNN, RNN, CNN, DAN	[58, 57, 59], 209, 429
Semantic	sentiments polarity, sentiments properties, usefulness	RNN, DAN, CNN, DNN	[393], 336, 143, 275, 114, 79, 394]
Multimedia	images, videos, audio	DBN, CNN, MLP, RNN, DAN	[335, 127, 362, 53, 231, 232, 368, 198, [99]
Spatial and temporal	location and time	CNN, RNN, DAN	[393, 341, 200, 161, 409, 296, 188, 47, 220]
Graph embeddings	Knowledge graph, network graph, entities graph	DAN, GCN	[320, 97, 406, 258, 321]

5.6 Social recommender system

An approach used social network user trust information to get accurate recommendations and used a deep AE to learn the initial values of users' and items' latent attributes [67]. Next a method presented dTrust for rating prediction that takes advantage of the trust–user–item network's architecture and employed a feedforward neural network to connect user relations with user–item ratings [61]. As several works focused on direct social trust to improve recommender system, while a method included both direct and indirect social trust with deep learning [16]. The model proposed a shared layer to incorporate direct and indirect trust values as well as rating information in AE network that retrieves correlated representations. This shared layer of representations significantly improved the recommendation results. A different method considered interactions among social influences [250]. The authors proposed an improved Sparse SDAE model to learn compact and efficient representations from social information. Recently an approach utilized a DNN to bridge the gap between recommender system and social relations [83]. Another recent study focused on GNNs to improve social recommendations by learning representations from various graphs [84]. The proposed method coherently modeled two graphs and heterogeneous strengths of social relations. The principled strategy to obtaining latent spaces from this graph data is to capture interactions and opinions. These user–item latent spaces are then concatenated and input into the MLP, which predicts ratings.

One research perspective is to explore effective social data, including investigating user interests and preferences, user relationships in social networks, and utilizing NLP and

sentiment analysis to enhance recommendations. Considering time-sensitive information is crucial for accurate recommendations in social networks as users' social relationships can change over time. Outdated information such as user ratings and connections may become noise. Social RSs rely on social relationships, and uncovering and incorporating either direct or indirect correlations among items within these relationships is a significant research challenge.

5.7 Co-clustering recommender system

The co-clustering technique is a subspace clustering paradigm that simultaneously clusters entities [166]. In CF-based recommender system, co-clustering clusters the user–item ratings matrix into smaller sub-matrices [24] that can capture local preferences to predict unknown ratings from similarities of users and items. Many CF algorithms assign users or items to a single subgroup based on implicit or explicit features. However, users may have multiple preferences and items may have diverse attributes, making it more reasonable to associate them with more than one groups [173]. Co-clustering effectively identifies groups of users who share local preferences for a particular group of items, hence these have been successfully applied to CF for nearly two decades [87, 91, 100]. To enhance CF performance, researchers used co-clustering with a new unipartite form of bipartite modularity and an optimized network representation, resulting in improved recommendation accuracy, reduced computation time, and addressing the cold start problem [87]. Additionally, nonnegative matrix tri-factorization-based recommender focused on clustering POIs and users simultaneously based on time and location

Table 9 Evaluation metrics for recommender system

Type	Problem measurement type	Evaluation metrics	Description
Accuracy metrics	Classification	Accuracy $A = \frac{TP+TN}{TP+TN+FP+FN}$	Accuracy computes a relation of all correct recommended products to all recommended products
		Precision $P = \frac{TP}{TP+FP}$	Precision determines the proportion of products that are relevant to the user among the recommended instances
		Recall $R = \frac{TP}{TP+FN}$	Recall determines the proportion of relevant products that are recommended to the user over the total number of relevant products
		F-measure $F1 - score = \frac{2 \times P \times R}{P+R}$	F-measure is a measure that considers both precision (relevant products) and recall (all possible relevant products) in the recommended list of the test to compute the score
		Specificity $S = \frac{TN}{FP+TN}$	Specificity measures the fallout scores of the recommended products
		Area under curve (AUC)	AUC computes the area under the ROC (Receiver Operating Characteristic) curve, whose value is obtained by plotting the true positive rate (recall) alongside the y-axis against the false positive rate (fallout) alongside the x-axis. A larger the AUC score better the recommender system
	Note: With terminologies of confusion metrics denote TP as True Positive, TN as True Negative, FP as False Positive, and FN as False Negative		
	Regression	Mean absolute error $MAE = \frac{ \sum_{i=1}^N \hat{r}_{(u,i)} - r_{(u,i)} }{N}$	One of the popular metrics for rating prediction tasks. The predicted ratings are compared with their corresponding actual ratings to obtain error values. MAE calculates the average of these error values
		Mean squared error $MSE = \frac{ \sum_{i=1}^N \hat{r}_{(u,i)} - r_{(u,i)} ^2}{N}$	Another popular metric for rating prediction task. MSE computes squared error values
		Root mean squared error $RMSE = \sqrt{\frac{ \sum_{i=1}^N \hat{r}_{(u,i)} - r_{(u,i)} ^2}{N}}$	RMSE is obtained by computing square root of MSE
Note: $\hat{r}_{(u,i)}$ and $r_{(u,i)}$ indicates predicated ratings and actual ratings. N is the total number of actual/predicated rating pairs. Lower the MAE, MSE and RMSE values means more accurate rating predictions			
Ranking	Mean reciprocal rank (MRR) $MRR = \frac{1}{ U } \sum_{u \in U} \frac{1}{R_u}$	The rank-based metrics decide the relative preference of the products in the recommendation list. The MRR metric follows three steps: • Generate the products recommendation list and obtain the products rank—the product which the user likes should have a higher rank and be at top of the list • Compute the users’ reciprocal rank (R_u). • Compute the mean reciprocal rank—By taking the average mean of all products’ reciprocal rank over all users	
	Normalized discounted cumulative gain (nDCG) $DCG = \sum_{i=1}^p \frac{rs_i}{\log(i+1)}$	DCG is a classical measure of the item gain over its position in the recommended list using graded relevance score. The values are discounted logarithmically and then accumulated at a particular rank position p . nDCG is based on the normalized form of DCG	
	Hit ratio (HR) $HR = \frac{No.ofhits}{ U }$	HR is a count of the hits or correct predictions a recommender system has achieved. It means the number of users who have interacted (liked, rated, watched). If HR is equal to one, then all the items are ranked as per users’ desirability	

Table 9 continued

Type	Problem measurement type	Evaluation metrics	Description
		Mean average precision (mAP) $AP@N = \frac{1}{r} \sum_{i=1}^N P(i) \cdot rel(i), \quad mAP = \frac{1}{m} \sum_{u=1}^m (AP@N)_u$ Here r is relevant recommendations, N is total number of recommended products, $P(i)$ is the precision value at the i^{th} recommendation, $rel(i)$ indicates whether i^{th} recommendation is relevant ($rel(i) = 1$) or not ($rel(k) = 0$) Precision@N, Recall@N, F1@N $Precision@N(u) = \frac{ P_u \cap G_u }{N}$ $Recall@N(u) = \frac{ P_u \cap G_u }{G_u}$ $F1@N = \frac{2 \times Precision@N(u) \times Recall@N(u)}{Precision@N(u) + Recall@N(u)}$	mAP computes the mean for average precision (AP) over all the users. The (AP) is measured by first subdividing the recommendation list using user relevant products to obtain sub-lists. Next calculating the precision on these sub-lists and finally averaging them for each user. mAP value can be obtained by calculating the AP for all say m users and taking mean of it Precision@N: It reflects how many users' interested products in the recommended list Recall@N: It represents the ability of coverage in the user's ground truth (F. [214, 215]) In the formulas given on the left, P_u is recommended set of products for user u and G_u is users' ground truth set
Beyond accuracy metrics	Quality	Diversity $Diversity(R_u) = \frac{1}{ R_u (R_u - 1)} \sum_{i,j \in R_u} simDist(i, j)$ Novelty $Novelty(R_u) = -\frac{1}{R_u} \sum_{i \in R_u} \log_2 P(i)$ Serendipity $S_{(u)} = \frac{1}{R_u} \sum_{i \in R_u} \max(P_u(i) - Prim_u(i), 0) \cdot rel_u(i)$ Coverage $Coverage(t) = \frac{n}{N} \times 100$ n is number of recommended products	Diversity measures the dissimilarity between the products in the recommended list R for a user u. The value can be determined by calculating pair-wise distance between products say i and j using similarity methods like cosine, Jaccard, etc. The result with high similarity distance relates low diversity Novelty defines not popular products (products unknown to user in the recommendation lists. The novelty of an individual u recommendations is aggregated into a single score for a list of recommendations R. This formula represents inverse ($-$) popularity (P) means self-information of a product i. In other words, high novelty value means less popular products are recommended Serendipity reflects unexpectedness (degree of surprise) and relevance for a user. It means the measure of estimation scores given to those items in the recommendation list R_u that are unlikely to users but put out to be favorable to them Where $P_u(i)$ and $Prim_u(i)$ in the Serendipity formula are the estimated scores of the item i for user u obtained from the predicted and primitive models, respectively. $rel_u(i)$ is the relevance score of the item i for user u Coverage represents the percentage of anything (user/product) t that are recommended by the recommender system over the total potential things N

similarities using a soft co-clustering approach, providing better semantic interpretation and handling clustering ambiguity without requiring labels or geographical data [135]. Other approaches such as fuzzy co-clustering [89], parallel co-clustering [100], and multiclass co-clustering [26, 358] contribute to improving recommendation efficiency.

Unlike simple clustering approach, co-clustering is a powerful tool that helps to uncover underlying hidden patterns and relationships in the in the feature space of input data [74, 121, 295, 403]. Once latent users' preferences are discovered, they are used to estimate missing ratings and rank recommendations. Although co-clustering and MF approaches represent an improvement against unidimensional clustering approaches to CF, they still

generally struggle to guarantee strong preference correlation in underlying subspaces [91].

While these statistical co-clustering approaches have their strengths, integrating deep learning with co-clustering can provide significant advantages [359], particularly in handling high-dimensional data, capturing complex patterns, and scaling to large datasets. Deep learning models, such as AEs and GNNs, can learn rich, low-dimensional representations of high-dimensional data, making the clustering process more effective. For example, DeepCC leverages a deep AE to generate low-dimensional representations for instances and features [359]. These representations are then fed into an inference neural network within the framework of a Gaussian mixture model for cluster assignment prediction. This integration advances the effectiveness of the clustering process.

In recommender systems, this process involves representing the data as a matrix or graph, using deep learning to learn feature representations, and then applying an appropriate co-clustering algorithm to these learned representations to identify clusters of users and items. The resulting clusters can be interpreted to uncover user interests. To model complex interaction data, a study proposed the co-cluster interactions via attentive hypergraph neural network, which constructs a hypergraph to represent interactions and uses an attentive hypergraph neural network for learning interaction representations with an attention mechanism for attribute selection [371]. Additionally, the co-clustering module jointly clusters entire interactions and their corresponding attention weights, leading to significant improvements in click-through rate prediction. Recently, an article focused on addressing the privacy-preserving problem in recommendation by introducing probabilistic preference distribution modeling, which uses federated learning to share user and item preferences across domains [202]. It further implemented a co-clustering method to group users based on their tastes or characteristics. This combination of distribution modeling and compactness clustering significantly improves model performance. These are a few examples that demonstrate how integrating co-clustering with deep learning can lead to more powerful and scalable co-clustering methods that can uncover deeper insights from the cluster data. Consequently, there is a high demand of extending co-clustering approach to a deep learning version [8, 403] to better handle the variability of user preferences and improve the effectiveness of clustering-based recommender systems.

6 Conclusion

AI technology-based recommender systems have shown success in reading people's minds by recommending relevant items to them. The preeminent nature of the recommender systems influenced us to thoroughly investigate the ideology, structure, working, challenges, and prospects of such systems. This article considered from beginners' viewpoint and conducted a complete study of the deep learning-based recommender system. Initially, we discussed the classification methods and general aspects of recommendation tasks in the research background. Next, we provided a comprehensive survey of deep learning-based recommender systems covering the deep state-of-the-art recommendation models for rating prediction and top-N ranking tasks. An exploration of this survey indicates incredible opportunities for deep learning methodologies in the recommender system. More importantly, deep learning methods have the potential to rise to a new level of algorithms for the reasoning behind the desired recommendations. We also investigated critiques of the deep learning recommender system and proposed a review analysis on pre-modeling aspects such as datasets and feature learning, as well as post-modeling evaluation metrics. These insights assist researchers and developers in understanding the building prototype behind the deep learning-based recommender engines. Finally, we discussed new generation recommender trends and outlined prospective research directions in each. The exploration of these trends is intended to guide researchers in developing advancements in recommender systems with deep learning techniques.

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Declarations

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